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High temperature piezoelectric ultrasonic transducers: Materials, design, and applications

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ABSTRACT

Since the advent of ultrasonic testing, piezoelectric ultrasonic transducers have been widely applied in nonintrusive inspection across aerospace, nuclear energy, and petrochemical industries, owing to their compact structure, low cost, and broad applicability. However, as service temperatures increase, issues such as the degradation of piezoelectric material properties and mechanical failure of transducer components become increasingly pronounced, highlighting the need for further optimization to enhance transducer performance under high-temperature conditions. This paper reviews the recent research progress on high-temperature piezoelectric ultrasonic transducers (HTPUTs) from the perspectives of materials, structures, and applications. Specifically, commonly used piezoelectric material systems for high-temperature environments and their performance characteristics are systematically summarized, with particular emphasis on degradation mechanisms at elevated temperatures. Design optimization strategies for transducers operating under high-temperature conditions are discussed in detail, including the selection of matching layers and backing materials, acoustic coupling optimization, and the development of novel structural configurations. Representative applications of HTPUTs in the energy and manufacturing sectors are also highlighted, together with the key challenges encountered. Finally, future development directions for HTPUT research are outlined.

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NOMENCLATURE

ABS	Acrylonitrile butadiene styrene
BFY-PZT	$x[(1-y)BiYO_3-yFe_2O_3]-(1-x)PZT$
BMN-BS-PT	$Bi(Mn_{2/3}Nb_{1/3})O_3$-BS-PT
BSPT	$BiScO_3$-$PbTiO_3$
BS–PT–BZT	$BS-PT-Bi(Zn_{1/2}Ti_{1/2})O_3$
BS–PT–xCTO	$BS-PT-xCaTiO_3$
BW	Bandwidth
c_p	Speed of sound
CBN	$CaBi_2Nb_2O_9$
CBN-xNa	$Ca_{1-x}Na_xBi_2Nb_2O_9$
CTE	Coefficient of thermal expansion
d_{33}	Piezoelectric coefficient
EMAT	Electromagnetic acoustic transducer
f_a	Anti-resonance frequency
f_c	Center frequency
f_r	Resonance frequency
FFT	Fast Fourier transform
HTPUA	High-temperature piezoelectric ultrasonic array
HTPUT	High-temperature piezoelectric ultrasonic transducer
k_{33}	Longitudinal electromechanical coupling coefficient
k_{eff}	Effective electromechanical coupling coefficient
k_p	Planar electromechanical coupling factor
k_t	Thickness electromechanical coupling factor
MEMS	Microelectromechanical systems
NDE	Nondestructive evaluation
NDT	Nondestructive testing
PBI	Polybenzimidazole
PI	Polyimide
PIN–BS–PT	$Pb(In_{1/2}Nb_{1/2})O_3-BS-PT$
PMN-PZ-PT	$0.15Pb(Mg_{1/3}Nb_{2/3})O_3$-$(0.85-x)PbZrO_{3-x}PbTiO_3$
PMUT	Piezoelectric micromachined ultrasonic transducer
PP	Polypropylene
PUT	Piezoelectric ultrasonic transducer
PYN–PZT	$Pb(Yb_{1/2}Nb_{1/2})O_3$-PZT
PZT	$Pb(Zr_xTi_{1-x})O_3$
Q_m	Mechanical quality factor
RMS	Root-mean-square
SEM	Scanning electron microscope
SH	Shear-horizontal
SNR	Signal-to-noise ratio
T_c	Curie temperature
T_d	Depolarization temperature
US	Ultrasound
V_{pp}	Peak-to-peak voltage
YSZ	Yttria-stabilized porous zirconia
Z	Acoustic impedance
α	Acoustic attenuation coefficient
ε_r	Relative dielectric permittivity

I. INTRODUCTION

Since the 20th century, ultrasonic testing has rapidly matured under the combined advances in physics, acoustics, materials science, and electronic technologies, and has become one of the most important techniques for ensuring equipment safety and life-cycle management in industries such as aerospace, nuclear energy, power generation, oil and gas, and chemical engineering.[1–7] In many engineering applications, service environments are often characterized by sustained temperatures of several hundred degrees Celsius or even higher. For example, in petrochemical refining facilities, the surface temperature of steel pipelines and their joints commonly approaches 400 °C; in polymer processing, materials must be heated to a molten state of approximately 220 °C prior to injection molding or extrusion; in the nuclear energy sector, many reactor systems employ opaque heavy metals such as sodium, lead–bismuth, and lead as coolants, with melting points of approximately 97.99 °C (sodium), 123.5 °C (lead–bismuth), and 327.5 °C (lead), respectively, and these systems operate for extended periods at temperatures well above their melting points.[8–13] Conventional high-temperature inspection methods largely rely on offline measurements performed after shutdown and cooling, or on intermittent testing within limited temperature ranges using techniques such as water immersion or couplant-assisted testing with local cooling.[14–16] Advanced gas turbines are moving toward higher combustion temperatures, nuclear and petrochemical installations are expected to operate for longer periods, and new processes such as additive manufacturing and intelligent manufacturing are emerging. These trends require high-temperature ultrasonic testing to operate at higher temperatures and under more severe conditions, while also offering long-term stability, miniaturized integration, and adaptability to complex geometries.[17–21]

At present, high-temperature ultrasonic testing techniques used in engineering applications can be broadly classified into four categories: piezoelectric ultrasonic transducers, electromagnetic acoustic transducers,[22–27] laser ultrasonic based on laser excitation and interferometric detection,[28–31] and optical ultrasonic technologies represented by fiber-optic sensing and optical microcavities.[32–35] Compared with the latter approaches, which often rely on complex optical systems or impose stringent requirements on the material properties and electrical conductivity of the test object, piezoelectric ultrasonic transducers remain dominant in industrial nondestructive testing owing to their compact structure, mature driving and signal-processing electronics, relatively low cost, and wide applicability.[36–39]

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The core component of a piezoelectric transducer is the piezoelectric material itself, which enables the interconversion between electrical and acoustic energy via the direct and inverse piezoelectric effects.[40,41] Key material parameters, including the electromechanical coupling coefficient, piezoelectric coefficients, mechanical quality factor, Curie temperature, and depolarization behavior, strongly influence the transducer's bandwidth, sensitivity, and thermal stability. However, the reliable operating-temperature limit of a complete transducer also depends on interfaces, packaging, exposure duration, and thermal cycling.[42–46]

In addition to the intrinsic degradation of piezoelectric materials, multiple factors under high-temperature conditions can lead to transducer failure. First, electrodes and lead wires are prone to diffusion, melting, or delamination from the piezoelectric element when exposed to elevated temperatures and oxidizing atmospheres. Meanwhile, organic bonding layers as well as polymer-based matching layers and backing materials that are widely used at room temperature tend to soften or decompose at high temperatures, resulting in interfacial debonding or acoustic-impedance mismatch.[47,48] Second, the pronounced differences in CTE among the metal housing, backing, matching layer, and piezoelectric element can, under repeated thermal cycling, readily induce the accumulation of residual interfacial stresses, leading to cracking and even catastrophic packaging failure.[49,50] In addition, liquid or paste-type coupling agents commonly used at room temperature to acoustically couple the transducer to the test object often volatilize or carbonize rapidly at high temperatures, causing interruption of the acoustic transmission path.[51,52]

Therefore, for high-temperature applications, coordinated optimization is required in both material systems and transducer structures. On the one hand, it is necessary to develop piezoelectric materials that simultaneously exhibit excellent performance, excellent thermal stability, and resistance to radiation. On the other hand, high-temperature transducer structures must be designed to simultaneously ensure acoustic performance and thermo-mechanical reliability, including the selection of heat-resistant matching layers and backing materials, the adoption of appropriate high-temperature bonding or joining processes, and the development of novel transducer architectures with enhanced resistance to thermal shock.[53]

Figure 1 shows a comprehensive overview of a high-temperature piezoelectric ultrasonic transducer. The publication statistics shown in Fig. 2 were obtained from the Web of Science database. Two independent searches were performed using the Topic field with the queries TS = ("high temperature nondestructive testing") and TS = ("high temperature piezoelectric ultrasonic transducer"), respectively, covering publications from 1990 to 2025. Research articles and review articles were included, whereas conference proceedings, book chapters, editorials, letters, and other document types were excluded. The searches were conducted on May 9, 2026. The resulting annual publication counts were used to illustrate the overall research trend rather than as an exhaustive bibliometric census, because the absolute numbers are sensitive to the selected search terms and indexing criteria. It can be seen that the number of relevant publications has increased rapidly and continuously since 1997, indicating that ultrasonic inspection under high-temperature conditions has attracted sustained and widespread attention from the research community. Several relatively systematic review articles have been published in the field of HTPUTs and related technologies.[54–60] However, most of these reviews are confined to a single perspective, focusing either on "materials" or on "structural" design, with insufficient emphasis on the systematic relationships among piezoelectric material parameters, transducer configurations, and specific engineering applications. Moreover, since 2020, approximately 300 related papers have been published annually, including many emerging high-temperature application scenarios that have not been adequately covered in previous reviews, which further highlights the

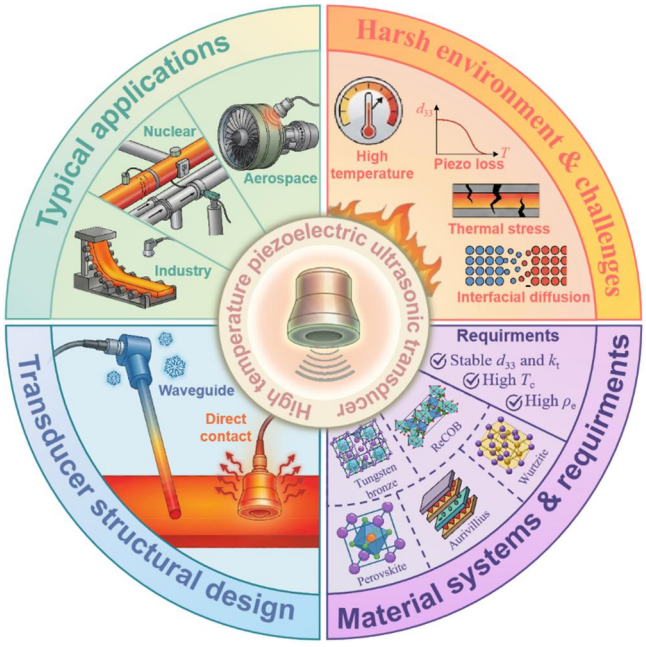


FIG. 1. A comprehensive overview of high-temperature ultrasonic piezoelectric transducers.

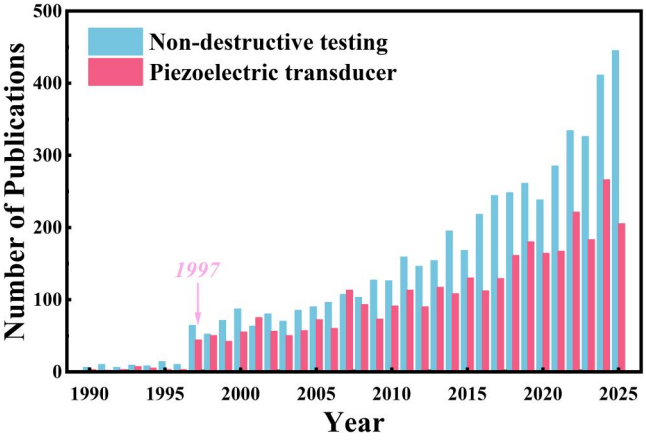


FIG. 2. Annual publication counts retrieved from the Web of Science database from 1990 to 2025 using the Topic queries "high temperature nondestructive testing" and "high temperature piezoelectric ultrasonic transducer." Research articles and review articles were included. Detailed search criteria are provided in the main text.

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need for an updated and more comprehensive review. Against this background, the present work starts from the theoretical models of piezoelectric ultrasonic transducers and systematically reviews recent domestic and international progress in high-temperature piezoelectric ultrasonic transducers from the perspectives of materials, structures, and applications. The aim is to establish clearer linkages among "materials–structures–applications" based on previous studies, thereby providing more targeted references for the design and engineering implementation of high-temperature ultrasonic testing systems.

As shown in Fig. 3, this review is roughly divided into the following parts. Section II briefly reviews the fundamental design principles of PUTs and several high-temperature effects that affect performance indicators. Section III systematically summarizes representative high-temperature piezoelectric materials, their key parameters, intrinsic temperature boundaries, and transducer-level validation status. Section IV reviews structural design strategies for high-temperature ultrasonic transducers and key considerations in material selection. Section V introduces several representative applications of high-temperature piezoelectric ultrasonic transducers and analyzes their performance advantages and limitations through typical case studies. Finally, Sec. VI summarizes the contributions of this review and provides an outlook on future development directions of HTPUTs.

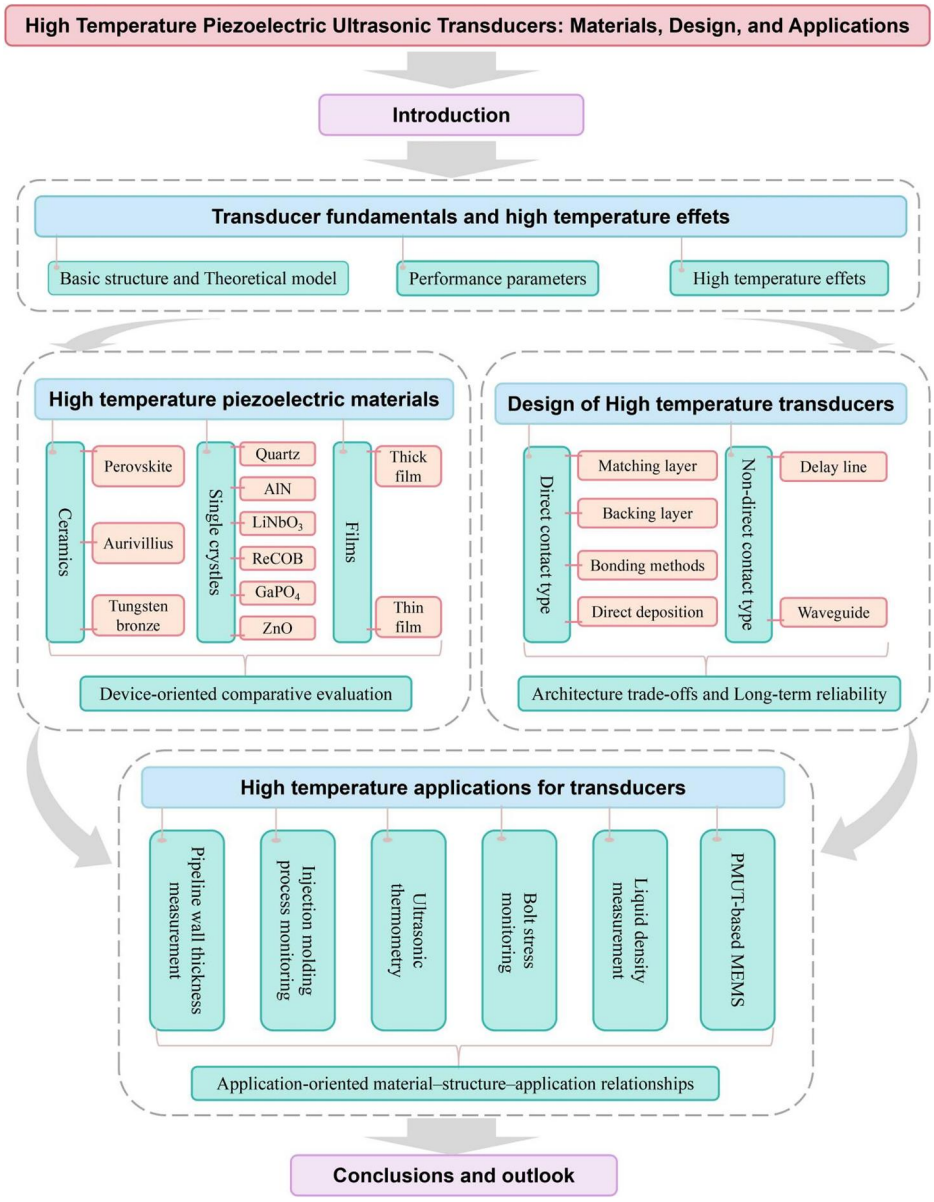


FIG. 3. Diagram of the outline in this review.

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II. TRANSDUCER FUNDAMENTALS AND HIGH-TEMPERATURE EFFECTS

A. Basic structure and theoretical model

A piezoelectric ultrasonic transducer (PUT) is an electromechanical energy-conversion device that generates and receives ultrasonic waves through the inverse and direct piezoelectric effects. In a typical thickness-mode PUT, a piezoelectric plate or disk with electrodes on both surfaces serves as the active element. When an alternating electric field is applied, the piezoelectric element vibrates and radiates ultrasonic waves; conversely, incident ultrasonic waves induce mechanical stress in the element and generate electrical signals at the electrodes.[61–64]

In practical applications, PUTs appear in a wide variety of structural forms.[65–68] Among these configurations, the piston-type thickness-mode transducer is one of the most widely used. As shown in Fig. 4(a), a representative PUT generally consists of a piezoelectric element, electrodes, a matching layer, a backing layer, electrical leads, packing materials, connectors, and a housing. The piezoelectric element determines the basic electromechanical conversion behavior, whereas the matching layer, backing layer, bonding interfaces, and housing strongly influence acoustic transmission, damping, bandwidth, mechanical support, and environmental protection.[56] For HTPUTs, these passive components are particularly important because their thermal stability, oxidation resistance, acoustic properties, and thermal-expansion compatibility directly affect the long-term reliability of the device.

As schematically shown in Fig. 4(b), a thickness-mode PUT can be understood as a layered acoustic structure composed of a front matching layer, a piezoelectric element, and a rear backing layer. The piezoelectric layer is usually polarized along the thickness direction, and its thickness directly determines the fundamental resonance frequency of the transducer,

$$t = n\frac{c_p}{2f}. \tag{1}$$

The fundamental thickness mode corresponds to n = 1, where n is an odd number and c_p is the speed of sound wave propagation in the piezoelectric material.

For ultrasonic imaging applications, transducers with broad bandwidth and high sensitivity are preferred. Acoustic matching at two mechanical ports—comprising both front matching and back support—is an effective strategy for improving transducer performance. For normal incidence, the pressure-amplitude transmission coefficient is given by

$$T = \frac{2Z_l}{Z_p + Z_l}, \tag{2}$$

where Z_l is the acoustic impedance of the load medium and Z_p is the acoustic impedance of the piezoelectric element. Because the acoustic impedance of piezoelectric materials often differs substantially from that of the target medium, a matching layer may be inserted between them. According to transmission-line theory, in the ideal one-dimensional, lossless, single-frequency case, a quarter-wavelength matching layer can achieve optimal transmission when its acoustic impedance satisfies:[69]

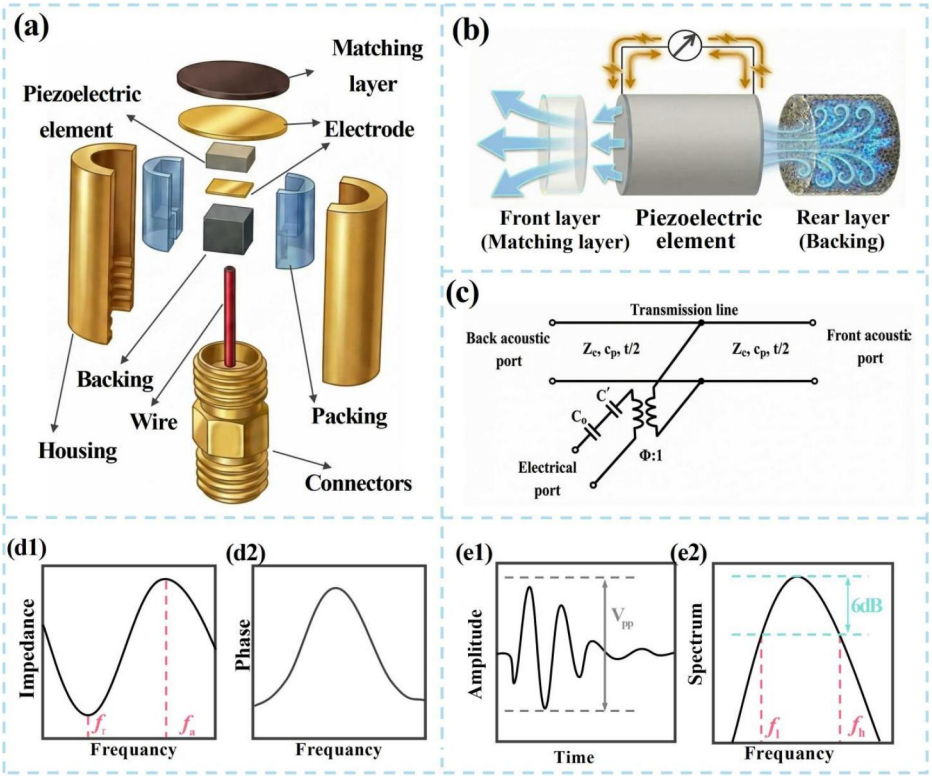


FIG. 4. Schematic illustrations of a piezoelectric ultrasonic transducer: (a) exploded view of the transducer structure; (b) three-port network representation illustrating energy transmission; (c) KLM equivalent electrical circuit model. (d1) and (d2) Typical impedance–frequency and phase–frequency spectra obtained from electrical characterization, where f_r and f_a denote the resonance and anti-resonance frequencies, respectively. (e1) and (e2) Typical pulse–echo time-domain waveform and frequency-domain spectrum used for acoustic characterization, where V_{pp} represents the peak-to-peak amplitude and f_l and f_h denote the lower and upper −6 dB frequencies, respectively.

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$$Z_m = (Z_p Z_l)^{1/2}. \tag{3}$$

For broadband operation, the acoustic impedance of the single matching layer should be modified to be

$$Z_m = (Z_p Z_l^2)^{1/3}. \tag{4}$$

Sometimes, two matching layers may be used to further improve the front acoustic matching of the transducer.[70] The acoustic impedances of the two layers should be

$$\begin{aligned} Z_{m1} &= (Z_p^4 Z_l^3)^{1/7} \\ Z_{m2} &= (Z_p Z_l^6)^{1/7}. \end{aligned} \tag{5}$$

To further increase the transducer performance, a double $\lambda/8$ matching scheme has been proposed. Theoretically, two $\lambda_m/8$ matching layers are equivalent to a single quarter matching layer and work as a simplified graded matching.[71]

Similarly, on the rear surface of the piezoelectric element, a small portion of the acoustic energy is transmitted into air, while most is reflected. For normal incidence, the pressure-amplitude reflection coefficient is given by

$$R = \frac{Z_a - Z_p}{Z_a + Z_p}, \tag{6}$$

where Z_a is the acoustic impedance of air, which is close to zero. It is obvious that a majority of the energy will be reflected back to the front face.

In practice, front matching alone cannot eliminate rear-side reflection, because the acoustic impedance mismatch between air and the piezoelectric element is severe. The reflected waves cause multiple reverberations and prolong the pulse decay, resulting in ringing. A backing layer is therefore introduced to absorb backward-propagating acoustic energy and shorten the pulse response. However, stronger backing damping usually improves bandwidth at the cost of sensitivity. Thus, backing design requires a trade-off between bandwidth, sensitivity, mechanical support, and thermal stability. High-temperature backing materials will be discussed in Sec. IV.

In addition to acoustic matching at the front and rear mechanical ports, electrical impedance matching is also required at the electrical port. Since a piezoelectric transducer usually exhibits complex impedance near resonance, the reactive component may cause source–load mismatch and reduce effective voltage or power transfer. Therefore, electrical matching is commonly used to compensate for the reactive component and match the transducer input impedance to the source/load impedance, typically 50 Ω, thereby improving energy transfer and signal coupling efficiency.

To describe the electromechanical–acoustic coupling behavior of ultrasonic transducers, several one-dimensional equivalent circuit models have been developed, including the Mason, Redwood, and KLM models.[72–74] Among them, the KLM model is widely used because it provides an intuitive representation of the piezoelectric layer as an electromechanical transformer connected to front and rear acoustic transmission lines, as shown in Fig. 4(c). Matching layers, backing materials, and acoustic loads can be incorporated as cascaded transmission-line elements, allowing the effects of material parameters, multilayer structures, and acoustic/electrical loading conditions to be analyzed within a unified framework. The main constants used in the KLM model are listed as follows:

$$C_0 = \frac{\varepsilon_{33}^S A}{t}, \tag{7}$$

$$k_t^2 = \frac{e_{33}^2}{c_{33}^D \varepsilon_{33}^S}, \tag{8}$$

$$\omega_0 = \frac{\pi c_p}{t}, \tag{9}$$

$$\phi = k_t \sqrt{\frac{\pi}{\omega_0 C_0 Z_c}} \sin c\left(\frac{\omega}{2\omega_0}\right), \tag{10}$$

$$C' = -\frac{C_0}{k_t^2}\left(\sin c\left(\frac{\omega}{\omega_0}\right)\right)^{-1}, \tag{11}$$

where A is the transducer area; t is the thickness of the piezoelectric material; ρ is the density of the piezoelectric material; c_p is the speed of sound in the piezoelectric material; Z_c is the acoustic impedance of the piezoelectric layer ($=\rho c_p$); e_{33}^S is the clamped dielectric permittivity; e_{33} is the piezoelectric constant; c_{33}^D is the elastic constant of piezoelectric layer; ω_0 is the resonant angular frequency. With the KLM model, the electric impedance of the transducer is given as

$$Z_{in} = \frac{1}{j\omega C_0} + \frac{1}{j\omega C'} + \frac{Z_1 Z_2}{(\phi^2 (Z_1 + Z_2))}, \tag{12}$$

where Z_1 and Z_2 denote the input acoustic impedances of the transmission lines toward the front and back acoustic ports, respectively. From a design perspective, the KLM model enables an intuitive approach to transducer optimization: the two acoustic ports can be exploited to engineer the front matching layer and the backing structure, whereas the electrical port can be used to achieve electrical impedance matching. Simulation tools based on this model, such as PiezoCAD, have been widely employed in ultrasonic transducer design.

B. Performance parameters

1. Transducer performance parameters and characterization methods

The performance of an ultrasonic transducer is primarily determined by its electromechanical conversion capability and acoustic response. For quantitative evaluation, measurable parameters such as effective electromechanical coupling coefficient, center frequency, bandwidth, sensitivity, signal-to-noise ratio, and spatial resolution are commonly used. These parameters can be extracted from electrical impedance–phase measurements and pulse–echo experiments and provide a practical basis for comparing transducers with different materials, structures, and operating conditions.[75,76]

a. Electrical properties. Electrical impedance–phase measurements are commonly used to characterize the electrical response of ultrasonic transducers. As shown in Figs. 4(d1) and 4(d2), the resonance frequency (f_r) corresponds to the minimum impedance, whereas the anti-resonance frequency (f_a) corresponds to the maximum impedance. These two frequencies are key indicators of the electromechanical coupling behavior of the transducer. In addition, the impedance magnitude and phase near resonance determine the

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matching condition between the transducer and the external driving/receiving circuitry and are therefore important for practical device design.[77] Based on the measured resonance and anti-resonance frequencies, the effective electromechanical coupling coefficient (k_{eff}) can be further calculated,

$$k_{eff} = \sqrt{\frac{f_a^2 - f_r^2}{f_a^2}}, \tag{13}$$

where f_r is the resonance frequency and f_a is the anti-resonance frequency of the transducer.

b. Acoustic properties. Pulse–echo testing is widely used to evaluate the acoustic response of ultrasonic transducers. In the pulse–echo mode, the same transducer usually serves as both transmitter and receiver. As shown in Fig. 4(e1), the transducer emits a short ultrasonic pulse and then receives echoes reflected from interfaces such as the back wall of a calibration block, a planar reflector, or a defect. The received echo contains information on the entire transmit–propagate–reflect–receive process, and therefore reflects key acoustic performance parameters, including sensitivity, damping behavior, ringing duration, frequency response, and bandwidth. After applying a fast Fourier transform (FFT) to the time-gated echo signal, the corresponding frequency-domain spectrum can be obtained, as shown in Fig. 4(e2).

The center frequency characterizes the dominant spectral component of a transducer's pulse response and serves as a key parameter for determining its operating frequency and applicable scenarios. It also governs the fundamental trade-off between penetration capability and spatial resolution in ultrasonic testing.[78] The center frequency, f_c, is usually defined by the following formula:

$$f_c = \frac{f_h + f_l}{2}, \tag{14}$$

where f_h and f_l, respectively, represent the frequency points corresponding to when the amplitude is reduced to half in the spectrum.

The bandwidth reflects the transducer's capability to accommodate a range of frequency components and is directly associated with the pulse duration, ringing behavior, and axial resolution. The most commonly used metric is the −6 dB bandwidth relative to the peak amplitude. The absolute bandwidth (BW) and fractional bandwidth (FBW) are defined as follows:

$$BW = f_h - f_l \qquad FBW = \frac{f_h - f_l}{f_c} \times 100\%. \tag{15}$$

Sensitivity must be defined according to the measurement boundary. Intrinsic transducer sensitivity describes the transmitting or receiving capability of the transducer itself under specified acoustic and electrical loading conditions. In contrast, pulse–echo measurements include the complete transmit–propagate–reflect–receive path and are therefore also affected by coupling loss, propagation attenuation, reflector characteristics, and acquisition conditions. Accordingly, the voltage ratio commonly obtained from pulse–echo testing is better regarded as a pulse–echo response or pulse–echo sensitivity rather than an intrinsic transducer sensitivity. It can be expressed as

$$S_{PE} = 20\log_{10}\left(\frac{V_o}{V_i}\right), \tag{16}$$

where V_o is the peak-to-peak amplitude of the selected echo, and V_i is the peak-to-peak excitation voltage. Because SPE is defined from a voltage-amplitude ratio, the factor 20 is used in the logarithmic conversion. The measured S_{PE} is also influenced by the propagation path, coupling condition, reflector, excitation waveform, and receiving electronics. Therefore, meaningful comparison requires these conditions to be specified or kept consistent. Insertion loss similarly represents the combined losses along a defined transmit–receive path, whereas system-level sensitivity further includes the effects of cables, amplification, filtering, and acquisition electronics.

Signal-to-noise ratio (SNR) can be defined using either an amplitude ratio or a power ratio, provided that the corresponding logarithmic convention is stated explicitly. In this review, SNR is defined on an RMS-voltage basis as

$$SNR = 20\log_{10}\left(\frac{V_s}{V_n}\right), \tag{17}$$

where V_s is the root mean square (RMS) value of the echo signal within the selected signal gate, and V_n is the RMS value of the noise measured within a noise-only gate (e.g., a time window without echoes) under the same acquisition settings. If signal and noise powers are used instead, the corresponding definition is $\mathrm{SNR} = 10\log10(P_s/P_n)$.

Spatial resolution is an important parameter that reflects the performance of an ultrasonic transducer. It consists of lateral and axial components. The lateral resolution refers to the ability of the transducer to distinguish two targets separated in a direction perpendicular to wave propagation, whereas the axial resolution denotes its ability to resolve two targets separated along the propagation direction. For a single A-line pulse–echo waveform, the axial resolution can be estimated as

$$R_{axial} = \frac{c}{2BW}. \tag{18}$$

The lateral resolution can be calculated based on the beam width. The narrower the beam width, the higher the lateral resolution. Its definition is as follows:

$$R_{lateral} = \frac{c}{f_c} f_{\#}, \tag{19}$$

where $f_{\#}$ is defined as the ratio of the focal length to the effective aperture of an ultrasonic transducer. Axial resolution is primarily governed by the pulse duration and effective bandwidth, whereas lateral resolution depends mainly on wavelength, effective aperture, and focusing conditions. Increasing the center frequency can improve both resolutions, but it also increases acoustic attenuation and reduces penetration depth. Consequently, in practical applications, the achievable spatial resolution and penetration depth represent a trade-off among these competing factors.[79]

For meaningful comparison of HTPUT performance across different studies, the measurement conditions associated with bandwidth, pulse–echo sensitivity, and SNR should be clearly reported. At minimum, the test description should include the excitation waveform, frequency, voltage, and number of cycles; receiver gain, filtering,

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and acquisition settings; coupling method and contact condition; propagation medium and distance; reflector type and geometry; signal averaging and time-gating procedures; and the test temperature, dwell time, and thermal-cycling history. The definitions used for bandwidth, sensitivity, and SNR should also be explicitly stated. When these conditions differ substantially, absolute performance values should not be directly compared; temperature-dependent retention or normalized changes obtained under the same experimental configuration are generally more meaningful for evaluating high-temperature stability.

2. Specific material parameters for transducer design

Piezoelectric materials form the core functional component of ultrasonic transducers. In materials science, a wide range of parameters are used to characterize the diverse properties of piezoelectric materials. Among these, the dielectric, piezoelectric, and acoustic properties are the three most critical categories for transducer design. The relevant parameters are summarized as follows:

$$C_{33}^{S} = \frac{\varepsilon_{33}^{S} A}{t}, \tag{20}$$

$$k_t^2 = \frac{\text{Stored mechanical energy}}{\text{Input electrical energy}} = \frac{e_{33}^2}{c_{33}^{D}\varepsilon_{33}^{s}}, \tag{21}$$

$$f_0 = \frac{\omega_0}{2\pi} \quad \omega_0 = \frac{\pi c_p}{t}, \tag{22}$$

$$Z_p = \rho c_p, \tag{23}$$

where C_{33}^{S} is the clamped capacitance; ε_{33}^{s} is the clamped dielectric permittivity; A is the area; t is the thickness; k_t is the thickness-mode electromechanical coupling coefficient; e_{33} is the piezoelectric stress constant; c_{33}^{D} is the elastic constant; f_0 is the resonant frequency; ω_0 is the resonant angular frequency; c_p is the speed of sound inside the piezoelectric material; Z_p is the acoustic impedance; and ρ is the density.

Because piezoelectric materials can be electrically modeled as capacitive structures, the input of a piezoelectric ultrasonic transducer also exhibits capacitive behavior. According to Eq. (20), this capacitance is jointly determined by the dielectric permittivity, effective area, and thickness, and therefore affects the electrical impedance matching of the transducer. The electromechanical coupling coefficient k quantifies the efficiency with which a piezoelectric material converts electrical energy into mechanical energy and vice versa. A higher k generally enables more efficient energy conversion and a broader bandwidth. The piezoelectric coefficient d is closely associated with the transducer's sensitivity. Meanwhile, as shown in Eqs. (22) and (23), the resonance frequency and acoustic impedance are governed by intrinsic material properties such as acoustic velocity and density, thereby influencing acoustic matching and energy transmission. For high-temperature applications, several temperature-related quantities should be distinguished. The Curie temperature (T_c) represents the intrinsic ferroelectric phase-transition boundary, whereas the depolarization temperature (T_d) reflects the temperature at which significant polarization loss occurs and may therefore be lower than Tc. Neither parameter should be directly interpreted as the operating-temperature limit of a complete transducer. The maximum demonstrated temperature refers to the highest temperature experimentally validated for a specific material or device, whereas reliable continuous operation additionally depends on exposure duration, thermal cycling, property retention, and the stability of interfaces and packaging.

C. High-temperature effects on performance parameters of PUTs

In Sec. II B, the key performance parameters of PUTs have been systematically summarized, and these parameters are known to undergo pronounced variations with increasing temperature, which are collectively referred to here as high-temperature effects. It should be emphasized that the influence of high-temperature effects on transducer performance is highly complex and multifaceted. Although an increase in temperature inevitably induces changes in performance, such changes do not necessarily follow a monotonic trend of either improvement or degradation. Instead, the direction and magnitude of the performance variation strongly depend on multiple factors, including the intrinsic properties of the material, the transducer configuration, and the operating mode. As a result, it is difficult to establish a unified and deterministic relationship between temperature and performance. Nevertheless, the primary pathways of high-temperature effects can still be examined to clarify their influence on individual performance metrics. This helps explain the diversity and uncertainty of transducer performance evolution under high-temperature conditions. The equivalent-circuit framework introduced in Sec. II A remains applicable at elevated temperatures as long as the transducer operates within the same vibration mode and the assumptions of the linear model remain valid. Temperature does not require a different circuit topology; instead, it enters the model through the temperature dependence of the dielectric, piezoelectric, elastic, and acoustic parameters. Consequently, the values of the equivalent-circuit parameters and acoustic loads vary with temperature, leading to corresponding changes in the electrical impedance spectrum and acoustic response.

An important distinction in evaluating high-temperature transducer behavior is whether the observed performance change is reversible or irreversible. Reversible temperature drift mainly originates from temperature-dependent variations in dielectric, elastic, piezoelectric, and acoustic properties and may largely recover after cooling if no permanent structural change occurs. In contrast, irreversible degradation involves permanent changes in the material or device, including depolarization, thermal aging, oxidation, diffusion, interfacial debonding, microcracking, and thermal-fatigue damage. Therefore, an instantaneous performance change at elevated temperature should not necessarily be interpreted as permanent degradation; heating–cooling recovery, dwell-time dependence, and thermal-cycling behavior are required to distinguish these two cases. This reversibility criterion provides one dimension for interpreting high-temperature behavior, while the physical origin of the performance change provides another.

From the latter perspective, the pathways through which high-temperature environments affect PUT performance can be categorized into three main classes. First, the intrinsic properties of the piezoelectric material exhibit pronounced temperature dependence, including the piezoelectric coefficients, electromechanical coupling factors, dielectric permittivity, elastic constants, density, and acoustic velocity. These parameters enter the KLM model described in Eqs.

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(7)–(12), and their temperature-induced variations are reflected in the electrical impedance response, including shifts in fr and fa and the resulting change in k_{eff}. Second, the multilayer structure and associated interfaces are also temperature-sensitive. Thermal-expansion mismatch can generate interfacial stresses, bonding layers may soften or degrade, and the acoustic properties of matching and backing layers may drift, thereby altering acoustic transmission and damping. Third, elevated temperature can modify the external propagation medium and front-end electronics, further affecting the amplitude, spectral characteristics, and SNR of the pulse–echo response. Because these temperature-dependent effects vary strongly with material system, composition, frequency, thermal history, device structure, coupling condition, propagation medium, and circuitry, a universal quantitative mapping from temperature to experimentally observable transducer performance cannot be established. Table I therefore summarizes these relationships using a "mechanism–response–performance parameter" framework while separately indicating whether the associated changes are reversible or irreversible.

III. HIGH-TEMPERATURE PIEZOELECTRIC MATERIALS

Sections II B and II C have identified the key piezoelectric material parameters that require particular attention in the design of PUTs, with special emphasis on the properties required for operation under high-temperature conditions. Based on these criteria, a targeted screening has been conducted among material systems that are currently widely used. In this section, the selected high-temperature piezoelectric material systems are systematically reviewed and compared along three technological routes—ceramics, single crystals, and thin/thick films—to provide a foundation for the subsequent design and performance optimization of high-temperature ultrasonic transducers.

A. High-temperature piezoelectric ceramics

1. Perovskite structure piezoelectric ceramics

Perovskite-type piezoelectric materials have attracted much attention in recent years due to their unique crystal structure and

TABLE I. Relationships among mechanisms, temperature responses, and performance parameters for the three major pathways of high-temperature effects. (The table only summarizes typical response trends. The specific direction and magnitude of changes depend on the material system, packaging structure, temperature range, holding time, and thermal cycling conditions, etc.).

Category	Mechanistic/factor	Possible temperature response	Reversibility/degradation type	Key controlling conditions	Potentially affected performance parameters
Active properties of the piezoelectric material	Piezoelectric coefficient	May increase, decrease, or show nonmonotonic variation depending on phase stability, domain-wall mobility, and depolarization	Partly reversible at moderate temperature; irreversible after depolarization or thermal aging	Temperature range, dwell time, thermal cycling, distance from T_d/T_c	• Sensitivity • SNR
	Electromechanical coupling coefficient	Fluctuations with increasing temperature; a reduction in retention after prolonged high-temperature exposure	Irreversible after depolarization, cracking, or phase transition	Temperature range, vibration mode, poling stability, thermal history	• k_{eff} • Bandwidth • Ring-down duration
	Dielectric constant	Commonly increases with temperature	Often reversible if no irreversible phase transition, domain degradation, or defect redistribution occurs	Temperature range, phase-transition behavior, defect state, electrical boundary conditions	• Robustness of impedance matching
	Dielectric loss	Often highly temperature-sensitive and may increase noticeably at elevated temperatures	Often reversible at low/moderate temperature; may become irreversible after defect migration, thermal aging	Electrical resistivity, atmosphere, dwell time	• k_{eff} • SNR • Sensitivity
	Electrical conduction/leakage	May exhibit thermally activated behavior; the effective conductivity	Partly reversible after cooling; may become irreversible after oxygen-	Material resistivity, oxygen-vacancy concentration, atmosphere	• Sensitivity retention • Drift and hysteresis behavior

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TABLE I. (*Continued.*)

Category	Mechanistic/factor	Possible temperature response	Reversibility/degradation type	Key controlling conditions	Potentially affected performance parameters
		typically increases with temperature	vacancy migration, reduction reactions		
Multilayer structure and interfacial state of the transducer	Temperature drift of elastic/acoustic parameters of the piezoelectric layer	Thickness-mode resonance condition is affected; together with thermal expansion, contributes to frequency drift	Generally reversible in the absence of permanent damage; becomes irreversible after cracking, depolarization, or phase transition	Temperature range, crystal orientation/material system, thermal history, structural integrity	• f_c • f_r/f_a
	Temperature drift of matching-layer acoustic properties (Z, c, α, etc.)	Deviation from optimal acoustic matching; pulse-echo spectral shape may be reshaped	May be reversible with temperature if the material remains structurally stable; becomes irreversible after oxidation, densification, cracking, or microstructural degradation	Material type, porosity, oxidation resistance, thermal cycling	• Bandwidth • Axial resolution • Lateral resolution • f_c
	Temperature drift/aging of backing damping characteristics	Damping may increase or decrease; pulse-echo spectral shape may be reshaped	May be reversible if only temperature-dependent property drift occurs; becomes irreversible after softening, decomposition, oxidation, or permanent pore-structure change	Backing material type, temperature range, oxidation, dwell time, thermal cycling	• Bandwidth • Ring-down duration • Sensitivity
	Softening/creep/aging of the bonding layer	Reduced stiffness and altered damping; effects are often more pronounced after long dwell thermal cycling	Aging, creep, remelting, decomposition, or interfacial cracking are often irreversible	Couplant/bonding type, pressure, atmosphere, heating rate, thermal cycling	• Bandwidth • Ring-down duration • Sensitivity • f_c
	Interfacial debonding/delamination	More sensitive to thermal cycling and thermal shock; may reduce effective coupling area or impede acoustic energy transmission	Generally irreversible once debonding, voids, or delamination occurs	CTE mismatch, interfacial bonding strength, thermal cycling severity, layer thickness	• Sensitivity • SNR • Degradation of the acoustic transmission
	Thermal-expansion mismatch	Steady heating may introduce reversible thermal stress; thermal cycling can accumulate thermal fatigue and trigger irreversible damage/failure	Thermal stress itself may be reversible during a single cycle; residual stress accumulation, microcracks, and fatigue damage are usually irreversible	CTE mismatch, heating/cooling rate, layer geometry, thermal cycling	• f_c • f_r/f_a • Stability

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TABLE I. (*Continued.*)

Category	Mechanistic/factor	Possible temperature response	Reversibility/degradation type	Key controlling conditions	Potentially affected performance parameters
External propagation medium and circuitry	Failure of packaging /sealing / protective layers	May introduce environmental coupling uncertainty and accelerate degradation; long-term effects are more evident	Minor thermal drift may be reversible; seal failure, coating cracking, and packaging degradation are generally irreversible	Packaging material, atmosphere, temperature range, dwell time, thermal cycling	• Long-term stability • Reliability
	Sound speed in the propagation medium	Medium sound speed (e.g., metal, water, oil) varies with temperature	Mostly reversible if the medium remains stable and does not undergo phase change or permanent damage	Medium type, temperature range, phase state, pressure	• Axial resolution • Lateral resolution
	Attenuation coefficient of the propagation medium	Attenuation changes with temperature; high-frequency components may be preferentially suppressed, reducing effective high-frequency content	Mostly reversible if the medium state is stable; may become irreversible or poorly repeatable after phase change, oxidation, or bubble formation	Medium state, frequency, temperature gradient, flow condition	• f_c • Sensitivity • Axial resolution • Lateral resolution
	Acoustic impedance/scattering conditions of the medium	Reduced echo stability and increased amplitude fluctuations (due to unstable scattering/reflection conditions)	Often reversible under stable and repeatable conditions; may become irreversible after oxidation, poor wetting, contamination, or persistent bubbles	Interface state, wetting behavior, oxidation, medium uniformity	• SNR • Sensitivity • Repeatability
	External circuitry (drive/matching/interconnect/front-end)	With temperature, the equivalent circuit parameters and system transfer characteristics drift; the dominant manifestation is increased measurement bias and uncertainty	Short-term drift may be calibrated; electrode diffusion, solder degradation, insulation failure, and interconnect damage are often irreversible	Electrode material, wire/interconnect stability, packaging, atmosphere, dwell time	• SNR • Sensitivity • Stability

designability. As an important physical property of these compounds, the piezoelectric effect endows them with a key role in energy conversion and acoustoelectric coupling.[80–82] A typical perovskite structure is shown in Fig. 5(a), in which A-site and B-site ions are prone to equivalent or nonequivalent substitution, thereby controlling domain structure, phase transition behavior, and defect chemistry on the lattice scale, thus realizing tunable properties such as macroscopic dielectric, piezoelectric, and thermal stability.[83]

Lead zirconate titanate [$Pb(Zr_xTi_{1-x})O_3$, PZT] is one of the most representative piezoelectric materials. It is a binary solid solution formed by the ferroelectric phase $PbTiO_3$ and the antiferroelectric phase $PbZrO_3$. In its ABO_3 perovskite structure, Pb^{2+} occupies the A-site, while Zr^{4+} and Ti^{4+} share the B-site. In the 1950s, Jaffe and

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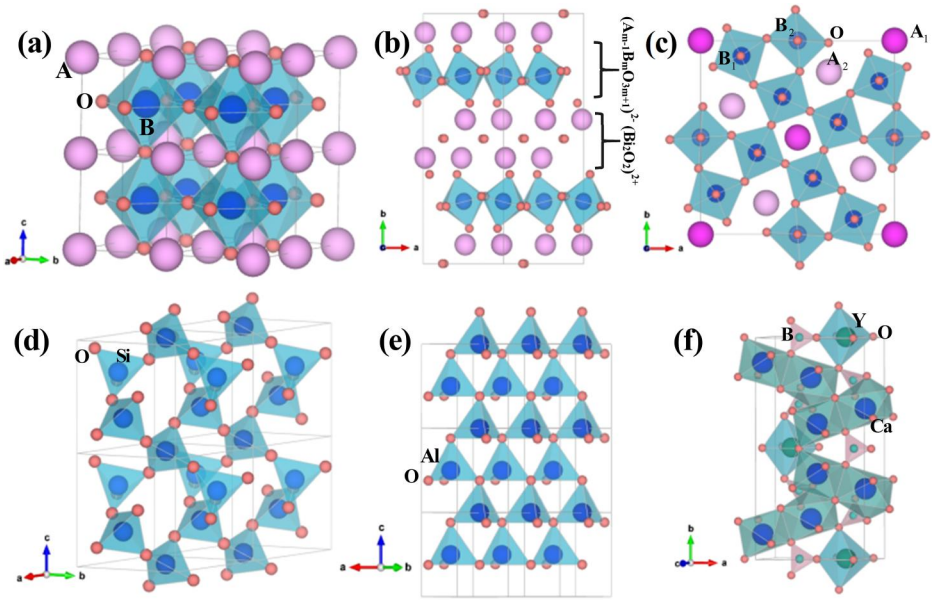


FIG. 5. Crystal structures of (a) perovskite, (b) bismuth layered (m = 1) (c) tungsten bronze, (d) quartz, (e) AlN, and (f) YCOB.

his collaborators systematically studied and popularized the PZT system. Owing to its excellent piezoelectric, dielectric, and ferroelectric properties, as well as its relatively low cost, PZT has become the most widely used piezoelectric material system.[84] However, conventional commercial compositions such as PZT-4 and PZT-8 have Curie temperatures of only about 328 and 300 °C, respectively, and even the high-Curie temperature composition PZT-5A reaches only about 368 °C. Moreover, when operated above half of their Curie temperatures, these materials suffer severe thermal depolarization, leading to the loss of piezoelectricity.[85,86] Therefore, improving the maximum operating temperature and enhancing the thermal stability of PZT has long been an important research focus. Guo *et al.* investigated Sm-doped PMN-PZ-PT ceramics and achieved excellent room-temperature performance ($d_{33} \approx 720$ pC/N, $k_{33} \approx 0.80$, $T_c \approx 293$ °C), as shown in Figs. 6(a) and 6(b). Both *in situ* heating and *ex situ* annealing tests of d_{33} showed less than 20% variation below 280 °C.[87] Huang *et al.* proposed a donor–acceptor co-doped PZT system, denoted as $(x[(1-y)BiYO_3$–$yFe_2O_3]$–$(1-x)PZT)$. This material exhibits a room temperature d_{33} of 467 pC/N and a high T_c of 380 °C. The *in situ* d_{33} sensitivity η remains below 10% within the temperature range of 30–300 °C, and d_{33} increases to approximately 790 pC/N at 331 °C before any signs of depolarization occur [Figs. 6(c) and 6(d)].[88] Liu *et al.* achieved a high T_c of 398 °C in 0.1PYN–PZT system. The material exhibits a room-temperature d_{33} of 450 pC/N, which remains nearly stable to 350 °C and peaks at ~900 pC/N near 380 °C before dropping, indicating that the depolarization temperature is above 380 °C [Figs. 6(e) and 6(f)].[89]

Although the T_c and T_d of modified PZT systems have been substantially increased, their high-temperature stability remains strongly composition-dependent. Recent compositions retain substantial piezoelectric activity to approximately 350–380 °C, whereas transducer-level operation is generally validated at lower temperatures. Therefore, applications at higher temperatures require materials with not only higher intrinsic thermal boundaries but also improved depolarization resistance and long-term stability. In 2001, Eitel *et al.* prepared $BiScO_3$-$PbTiO_3$ (BSPT) piezoelectric ceramics based on the correlation between tolerance factor t and T_c, with T_c as high as about 450 °C (nearly 100 °C higher than commercial PZT) and room temperature d_{33} of about 460 pC/N.[90,91] Since then, much research has been carried out on the properties and structure of BSPT, and remarkable progress has been made in piezoelectric characteristics, temperature stability and high T_c mechanism.[92–94] With these advantages, BSPT has become an important candidate for high-temperature piezoelectric applications above approximately 300 °C and has attracted extensive research attention, although its long-term transducer-level validation remains limited. Yang *et al.* introduced $CaTiO_3$ into the BSPT system, achieving a high-temperature piezoelectric response with d_{33} increasing from 475 pC/N at 20 °C to 853 pC/N at 360 °C, and stabilizing at 669 pC/N at 440 °C [Fig. 7(a)]. The k_p remained between 0.65 and 0.67 from 150 to 350 °C, and the k_t was about 0.5.[95] Wang *et al.* modified the BSPT system with $Bi(Zn_{1/2}Ti_{1/2})O_3$, achieving d_{33} of 430 pC/N and T_c of 445 °C at room temperature. The sample exhibited excellent thermal stability with $T_d \approx 418$ °C [Fig. 7(b)].[96] Zhao *et al.* constructed a hierarchical domain structure in the PIN–BS–PT system, achieving a room-temperature d_{33} of 410 pC/N, with a variation within ±10% from 50 to 350 °C [Fig. 7(c)].[97] Gong *et al.* introduced BMN into the BSPT system, achieving $d_{33} = 365$ pC/N, $\tan\delta \approx 0.57\%$, $T_c = 435$ °C, and maintaining d_{33} above 420 pC/N even at 400 °C [Fig. 7(d)], with minimal variation in k_t.[98]

Overall, modified PZT systems offer mature processing and high electromechanical performance, but their high-temperature use is still restricted by depolarization, dielectric loss, and long-term stability. BSPT-based ceramics provide a more promising route for higher-temperature bulk HTPUTs, but their processing reproducibility, cost, and device-level reliability still require further validation.

2. Bismuth layered structure piezoelectric ceramics

Since the first synthesis by Aurivillius in 1949,[99] bismuth layer-structured oxides (BLSFs) have attracted extensive attention owing to

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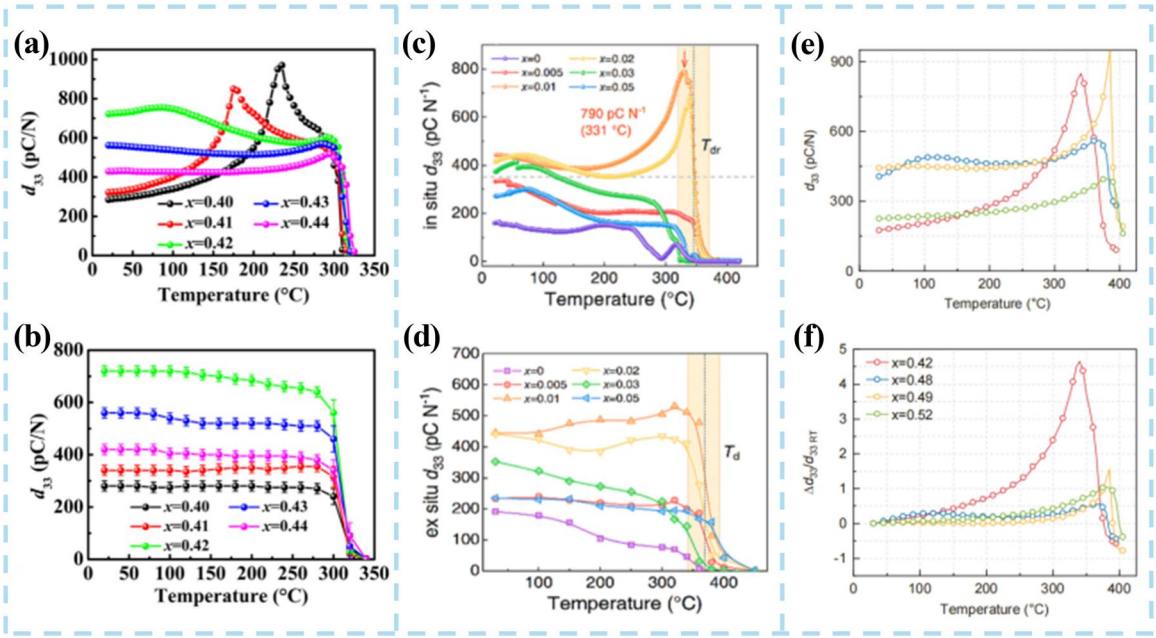


FIG. 6. The temperature dependence of (a) *in situ* d_{33} and (b) *ex situ* d_{33} for Sm-modified PMN-PZ-PT ceramics. Reproduced with permission from Guo *et al.*, J. Materiomics **7**, 683–692 (2021). Copyright 2021 Elsevier.[87] The temperature dependence of (c) *in situ* d_{33} and (d) *ex situ* d_{33} for *x*BYF-modified PZT ceramics. Reproduced with permission from Huang *et al.*, Chem. Eng. J. **477**, 147192 (2023). Copyright 2023 Elsevier.[88] Temperature stability of (e) d_{33} and (f) normalized d_{33} for PYN-PZ-*x*PT ceramics. Reproduced with permission from Liu *et al.*, J. Eur. Ceram. Soc. **44**, 873–881 (2024). Copyright 2024 Elsevier.[89]

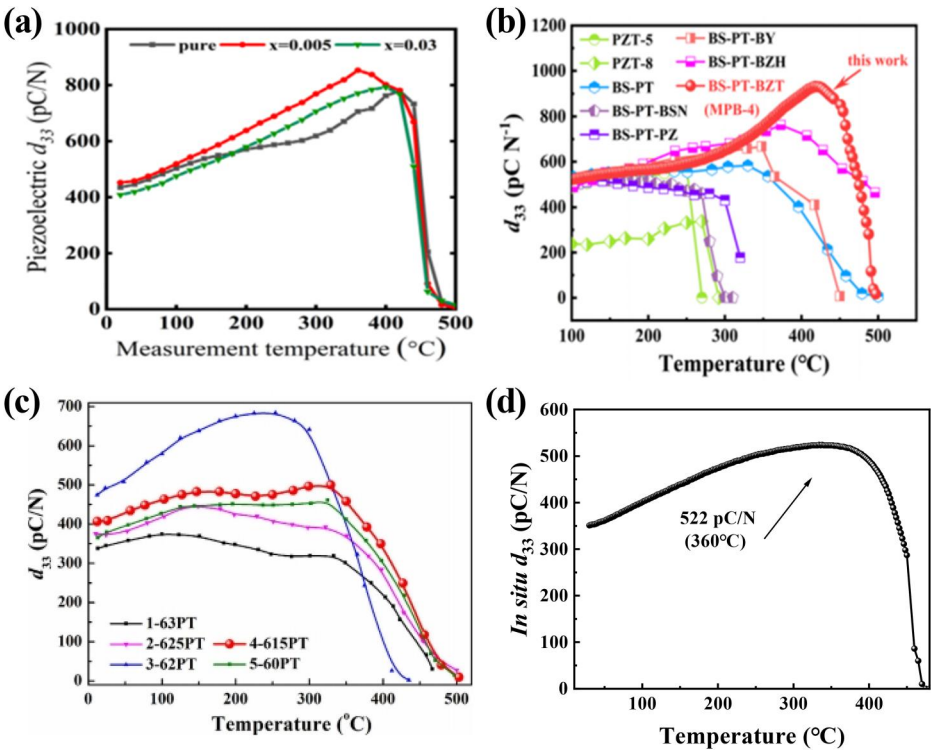


FIG. 7. (a) The *in situ* d_{33} of BS–PT–*x*CTO ($x = 0$, 0.005, and 0.03) at high temperature. Reproduced with permission from Yang *et al.*, ACS Appl. Mater. Interfaces **13**, 33253–33261 (2021). Copyright 2021 American Chemical Society.[95] (b) Comparisons of the temperature-dependent *in situ* d_{33} between the BS–PT–BZT piezoceramic and other representative lead-based perovskite piezoceramics. Reproduced with permission from Wang *et al.*, J. Mater. Chem. C **12**, 17595–17602 (2024). Copyright 2024 Royal Society of Chemistry.[96] (c) Variation of the *in situ* d_{33} for *x*PIN-*y*BS-PT at high temperature. Reproduced with permission from Zhao *et al.*, Acta Mater. **181**, 238–248 (2019). Copyright 2019 Elsevier.[97] (d) The high-temperature d_{33} of 0.02BMN-BS-0.635PT sample monitored by *in situ* techniques. Reproduced with permission from Gong *et al.*, ACS Appl. Mater. Interfaces **17**, 66790–66800 (2025). Copyright 2025 American Chemical Society.[98]

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TABLE II. Electrical parameters of bismuth layered structure ceramics.

m	Compound	T_c (°C)	d_{33} (pC/N)	ε_r (1 kHz)	tan δ (%)	Q_m	Ref.
2	Bi_3TiNbO_9	907	3	115	0.2	1350	103
2	$CaBi_2Nb_2O_9$	936	5	97	0.2	2600	104
2	$SrBi_2Nb_2O_9$	438	10	137	···	1600	105
2	$BaBi_2Nb_2O_9$	241	···	250	0.1	···	106
2	$PbBi_2Nb_2O_9$	590	15	260	···	···	107
2	$Na_{0.5}Bi_{2.5}Nb_2O_9$	776	10	155	0.5	1250	108
3	$Bi_4Ti_3O_{12}$	637	16	148	0.4	1022	109
4	$Bi_5FeTi_3O_{15}$	759	7	118	0.1	···	110
4	$CaBi_4Ti_4O_{15}$	790	7	169	0.2	5017	111
4	$SrBi_4Ti_4O_{15}$	521	15	218	0.5	···	112
4	$BaBi_4Ti_4O_{15}$	400	23	260	1.0	···	113
4	$PbBi_4Ti_4O_{15}$	560	11	147	···	···	114
4	$Na_{0.5}Bi_{4.5}Ti_4O_{15}$	657	16	145	0.5	3000	115
4	$K_{0.5}Bi_{4.5}Ti_4O_{15}$	555	21	212	0.6	1602	116

their unique layered crystal structure and high Curie temperatures. These compounds consist of alternating perovskite-like (ABO_3) layers and bismuth oxide $(Bi_2O_2)^{2+}$ layers, and their general formula can be expressed as $(Bi_2O_2)^{2+}(A_{m-1}B_mO_{3m+1})^{2-}$. The A-site can be occupied by +1, +2, or +3 valence cations suitable for 12-fold coordination—such as Na^+, K^+, Ca^{2+}, Sr^{2+}, Ba^{2+}, Pb^{2+}, Bi^{3+}, La^{3+}, or their composite ions—while the B-site is typically occupied by cations with octahedral coordination, including Fe^{3+}, Ti^{4+}, Nb^{5+}, and W^{6+}, or their combinations.[100,101] The parameter m is an integer representing the number of octahedral layers in the perovskite block $(A_{m-1}B_mO_{3m+1})^{2-}$ between two adjacent $(Bi_2O_2)^{2+}$ layers, commonly taking values from 1 to 5. Representative compounds include Bi_2WO_6 (m = 1), Bi_3TiNbO_9 (m = 2), $Bi_4Ti_3O_{12}$ (m = 3), and $SrBi_4Ti_4O_{15}$ (m = 4) [see Fig. 5(b) for the crystal structure (m = 1)]. Macroscopically, bismuth layer-structured oxides are characterized by high Curie temperatures, strong spontaneous polarization, pronounced anisotropy in piezoelectric and dielectric responses, low aging rates, excellent temporal and thermal stability of resonance frequency, and high mechanical quality factors. Nevertheless, as high-temperature piezoelectric materials, they still suffer from two intrinsic limitations: (1) relatively low piezoelectric activity, originating from the geometrical constraint of their layered (quasi-two-dimensional) lattice on polarization rotation and enhancement mechanisms; and (2) a high coercive field, which hinders complete polarization.[102]

There are many types of bismuth layer-structured piezoelectric ceramics, and to date, dozens of such systems have been reported. Their Curie temperatures span a wide range: for example, $BaBi_2Ta_2O_9$ exhibits a Curie temperature of only 110 °C, $SrBi_4Ti_4O_{15}$ shows a Curie temperature of 521 °C, while $CaBi_2Nb_2O_9$ reaches as high as 936 °C. The piezoelectric properties of these systems also vary significantly. Table II summarizes the key parameters of several prototype bismuth layer-structured oxide ceramics synthesized by the conventional solid-state method. It should be noted that for bismuth layer-structured oxides with Curie temperatures below 400 °C, their piezoelectric performance offers no advantage compared with perovskite-based ceramics such as BSPT and PZT. Consequently, research on bismuth layer-structured oxide ceramics has primarily focused on systems with Curie temperatures exceeding 500 °C, aiming to exploit their potential in high-temperature applications.

Researchers have long been devoted to improving the piezoelectric performance of BLSFs while maintaining their high Curie temperatures, particularly for compositions with T_c above 900 °C.[117] Wu *et al.* modified CBN ($CaBi_2Nb_2O_9$) with a high-entropy strategy, resulting in a d_{33} of 20.6–31.5 pC/N and T_c between 901 and 908 °C. *In situ* XRD confirmed a stable orthorhombic phase from 25 to 800 °C, with d_{33} retaining 90% after annealing at 800 °C [Fig. 8(a)].[118] Fang *et al.* enhanced both d_{33} (20 pC/N) and T_c (960 °C) in CBN-0.005Na ceramics, with d_{33} retaining more than 85% after annealing at 900 °C for 1 h and $\Delta d_{33} < 2\%$ after 25 thermal cycles between room temperature and 650/750 °C [Figs. 8(b)–8(d)].[119]

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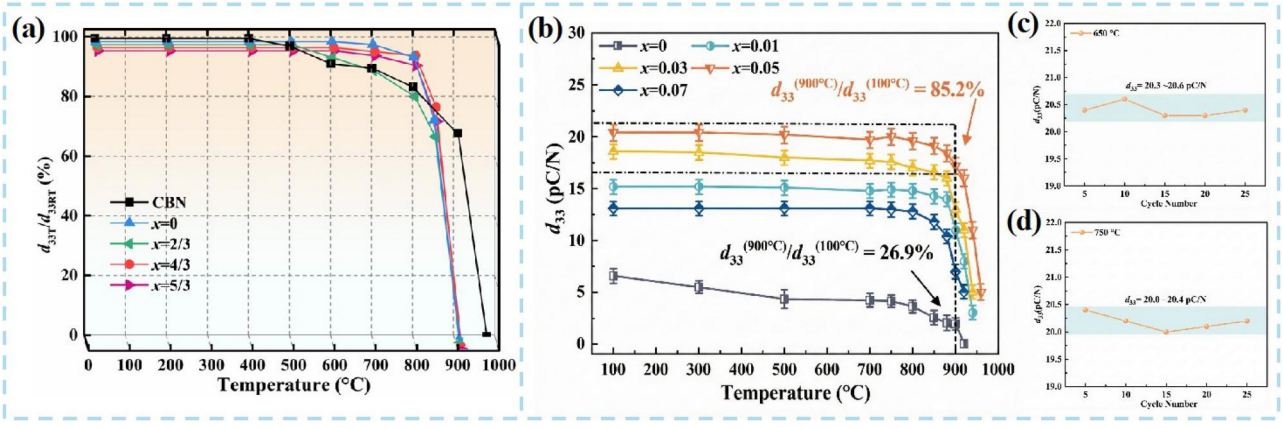


FIG. 8. (a) The thermal depolarization behavior for the $Ca_{1/3}(Na_{0.5}Bi_{0.5})_{1/3}Bi_{7/3}Ti_{1/3}Nb_{5/3-x}Ta_xO_9$ ($0 \le x \le 5/3$) ceramics. Reproduced with permission from Wu *et al.*, Adv. Mater. **37**, 2419134 (2025). Copyright 2025 John Wiley and Sons.[118] (b) The d_{33} of $Ca_{1-x}Na_xBi_2Nb_2O_9$ ceramics after 1 h of annealing at different temperatures and under different thermal cycling conditions: (c) 650 °C; (d) room temperature to 750 °C. Reproduced with permission from Fang *et al.*, Acta Mater. **293**, 121108 (2025). Copyright 2025 Elsevier.[119]

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Overall, BLSF ceramics are attractive for HTPUTs because of their high Curie temperature, low dielectric loss, and good thermal depolarization resistance. However, their relatively low piezoelectric activity and electromechanical coupling usually limit acoustic output and sensitivity. Therefore, BLSF-based materials are more suitable for applications where thermal stability and long-term signal retention are prioritized over high transmitting efficiency.

3. Tungsten bronze structure piezoelectric ceramics

Tungsten-bronze ferroelectrics are characterized by a framework of corner-sharing BO_6 octahedra with multiple crystallographic sites that can accommodate different cations, providing substantial compositional tunability.[120–125] Among these materials, tetragonal tungsten-bronze structures are the most widely investigated for piezoelectric applications. Their relatively high Curie temperatures and flexible compositional design make them potential candidates for high-temperature piezoelectric sensing.

Lead metaniobate ($PbNb_2O_6$) is one of the earliest discovered and most widely used tungsten bronze–type ferroelectrics. At room temperature, it exists in two phases: an orthorhombic ferroelectric phase (metastable) with a Curie temperature (T_c) of 570 °C, which transforms into a paraelectric phase above T_c, and a trigonal nonferroelectric phase (stable at room temperature) that converts to a tetragonal phase at 1200 °C. $PbNb_2O_6$ exhibits a high T_c, strong resistance to depolarization, and a large d_{33}/d_{31} ratio, thereby providing excellent hydrostatic piezoelectric performance.[126] Meanwhile, its mechanical quality factor (Q_m) is very low, which makes it suitable for applications such as ultrasonic nondestructive testing and medical ultrasound. However, $PbNb_2O_6$ also possesses inherent drawbacks, notably the difficulty of sintering bulk ceramics and relatively low piezoelectric coefficients and electromechanical coupling factors. To address these issues, compositional modification through doping is often adopted to enhance sintering activity and poling efficiency, thereby improving overall performance.[127,128]

Overall, tungsten bronze-type ferroelectrics are promising for high-temperature sensing because of their relatively high Curie temperature, compositional flexibility, and lead-free potential. However, their piezoelectric response and electromechanical coupling are generally lower than those of modified PZT or BSPT systems, and further device-level validation is still needed for HTPUT applications.

B. High-temperature piezoelectric single crystals

1. Quartz

Quartz (SiO_2) is one of the earliest piezoelectric materials employed in electronic devices such as oscillators, resonators, and filters.[129] A typical quartz structure is shown in Fig. 5(d). It possesses a high melting point of 1750 °C and a density of 2.65 g/cm^3, along with excellent chemical and physical stability, extremely low elastic vibration losses, a high mechanical quality factor, and minimal mechanical and dielectric dissipation. However, d_{33} is only 2 pC/N. Its phase transitions are closely related to its usable temperature range: α-quartz is stable below 573 °C, and β-quartz between 573 and 870 °C—both exhibiting piezoelectricity. However, above 870 °C, quartz transforms into tridymite, at which point the piezoelectric effect disappears. In addition, mechanical twinning tends to occur near 350 °C, leading to performance degradation; thus, the operational temperature of quartz-based devices is typically limited to below 350 °C.[130,131] Early piezoelectric devices utilized natural quartz, but constraints in resource availability and crystal size hindered large-scale applications. Consequently, hydrothermally grown synthetic quartz has become the dominant alternative, enabling stable supply and consistent material quality for modern electronic and acoustic devices.

2. AlN

Aluminum nitride (AlN) is a nonferroelectric material with a hexagonal wurtzite crystal structure [as shown in Fig. 5(e)], characterized by lattice parameters of approximately $a \approx 3.110$ Å and $c \approx 4.978$ Å. The lack of a center of inversion symmetry in its crystal structure endows AlN with piezoelectric properties.[132,133] It exhibits excellent thermal stability, with a melting point exceeding 2000 °C and no phase transition occurring below this temperature, thereby theoretically maintaining its piezoelectric effect up to the melting point. Beyond piezoelectricity, AlN also possesses a high thermal conductivity, high electrical resistivity, optical transparency across the ultraviolet to infrared spectrum, and a relatively low thermal expansion coefficient. Moreover, its excellent compatibility with complementary metal oxide semiconductor (CMOS) fabrication processes enables facile monolithic integration, allowing AlN-based piezoelectric micromachined ultrasonic transducers (PMUTs) to achieve significant miniaturization compared to conventional designs.[134,135] However, the piezoelectric response of AlN is relatively weak ($d_{33} \approx 5.6$ pC/N), and its fabrication remains challenging due to the difficulty in growing high-quality bulk crystals. Currently, the main synthesis methods include hydride vapor phase epitaxy (HVPE), molecular beam epitaxy (MBE), metal-organic chemical vapor deposition (MOCVD), and physical vapor transport (PVT). Among these, the PVT technique has emerged as an effective approach for producing large-sized AlN crystals owing to its high growth rate, superior crystallinity, and enhanced operational safety.[136,137] Kim *et al.* measured the properties of AlN single crystals from 23 to 1020 °C, finding that bulk resistivity remained high (8.8×10^{13} Ω cm at room temperature, 5×10^{10} Ω cm at 1000 °C). The electromechanical coupling coefficient k33 slightly increased from 0.391 to 0.395, and d33 increased from ∼6.5 to ∼13.5 pC/N at 1000 °C.[138] Barth *et al.* fabricated AlN and $Al_xSc_{1-x}N$ thin films, showing that d_{33} increased with Sc content, reaching a maximum at $x = 0.4$.[139]

3. $LiNbO_3$

In 1949, American researchers first developed ferroelectric single crystals such as $LiNbO_3$, $NaNbO_3$, and $KNbO_3$, which simultaneously exhibited high Curie temperatures and strong piezoelectric properties. Among these materials, lithium niobate ($LiNbO_3$) has become the most widely used.[140] At room temperature, $LiNbO_3$ crystallizes in a trigonal ferroelectric phase with an ABO_3-type perovskite-related structure. $LiNbO_3$ single crystals are commonly grown by the Czochralski method, while alternative growth and post-treatment approaches have been developed to improve compositional uniformity.[141–144] In terms of properties, $LiNbO_3$ has a Curie temperature of ∼1150 °C, and its electromechanical coupling coefficients vary significantly with the crystal cut orientation. The 36° rotated Y-cut (36° Y-cut) is most commonly used because it corresponds to a thickness-

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extensional vibration mode with high sensitivity and a large thickness electromechanical coupling coefficient ($k_t \approx 0.49$). The longitudinal piezoelectric coefficient of $LiNbO_3$ is approximately $d_{33} \approx 6$ pC/N.[145]

4. ReCOB

Rare-earth calcium oxyborate crystals, ReCOB [general formula $ReCa_4O(BO_3)_3$, where Re denotes rare-earth elements such as Y, Gd, or La], have attracted considerable attention in the fields of optics and high-temperature sensing since their synthesis, owing to their large effective nonlinear coefficients, high laser damage thresholds, excellent physicochemical stability, and nonhygroscopic nature.[146,147] A typical YCOB structure is shown in Fig. 5(f). These crystals exhibit melting points of approximately 1500 °C and undergo no phase transition below the melting point. Moreover, they maintain extremely high electrical resistivity even at elevated temperatures (10^6 Ω cm at 1000 °C), reflecting remarkable thermal and electrical stability. Their piezoelectric performance is also notable, with piezoelectric coefficients of approximately 3–16 pC/N and electromechanical coupling factors ranging from 6% to 31%.[148,149] The intrinsic piezoelectric response originates from the polarization of ReO_6 and CaO_6 octahedra within the crystal lattice. The piezoelectric properties of ReCOB crystals depend strongly on the rare-earth composition and the associated lattice distortion and defect structure.[150] ReCOB crystals can be grown by the Czochralski method, enabling relatively large bulk crystals.[151]

ReCOB crystals incorporating rare-earth elements with different ionic radii exhibit markedly distinct physical properties. Table III summarizes the key parameters of ReCOB single crystals with various Re^{3+} ionic radii.

5. $GaPO_4$

Gallium orthophosphate ($GaPO_4$) is an α-quartz isomorph whose crystal structure is identical to that of quartz. Both are nonferroelectric materials and therefore share many similar physical properties, including high electrical resistivity and a high mechanical quality factor, which can reach approximately 2000 at room temperature.[156,157] $GaPO_4$ exhibits no Curie temperature but undergoes an $\alpha \rightarrow \beta$ phase transition near 970 °C. Below this transition point, it maintains excellent piezoelectric performance and frequency stability, along with outstanding radiation resistance. The primary drawback of $GaPO_4$ lies in its relatively low electromechanical coupling coefficient, with k_t being only about 0.15.[158,159] In terms of crystal growth, early studies employed the hydrothermal method to synthesize $GaPO_4$ crystals; however, this approach often led to the retention of hydroxyl (OH-) groups within the crystal lattice, which degraded structural order and limited device performance. To address this issue, Li *et al.*[160] proposed a high-temperature flux growth method that effectively suppressed OH- incorporation, thereby improving crystal quality and enhancing the material's applicability in high-temperature piezoelectric devices.

6. ZnO

Zinc oxide (ZnO) is a nonferroelectric piezoelectric material with a hexagonal wurtzite structure, isostructural with AlN. Its lattice constants at room temperature are approximately $a \approx 3.25$ Å and $c \approx 5.21$ Å. The piezoelectricity of ZnO, like that of AlN, originates from the absence of a center of inversion symmetry in its crystal structure. ZnO exhibits excellent thermal stability, with a decomposition temperature of about 1975 °C and no structural phase transitions at high temperatures. Based on lattice-scale considerations, its piezoelectricity is expected to persist up to this temperature range.[161] However, its electrical conductivity increases significantly at elevated temperatures, which constitutes the primary limitation for engineering applications, and its practical operating temperature is generally restricted to below 500 °C. In terms of properties, ZnO exhibits moderate piezoelectric coefficients ($d_{33} \approx 10$–12 pC/N) and a relatively low thermal expansion coefficient, facilitating good thermal compatibility with a wide range of substrates.[162] Regarding fabrication, bulk single crystals are typically produced by hydrothermal or vapor transport methods, with the former enabling the growth of large, low-defect crystals at relatively low temperatures. Overall, ZnO combines good processability with advantageous multiphysical properties; however, the sharp increase in electrical conductivity at high temperatures defines its upper application limit, necessitating mitigation through optimized electrode design and defect/doping engineering.[163]

C. High-temperature piezoelectric films

With the continuous advancement of new piezoelectric materials and growth techniques, the advantages of bulk materials are rapidly being transferred and realized in film form.[164] The thickness of piezoelectric materials has a decisive influence on device performance, as

TABLE III. General properties of several ReCOB single crystals.

Material	r (Re^{3+})	d_{26} (pC/N)	$\varepsilon_{22}^T/\varepsilon_0$	k_{26} (%)	Electrical resistivity (Ω cm, 900 °C)	Ref.
TmCOB	0.880	7.7–9.6	11.2	19.0–19.4	6×10^7	152
ErCOB	0.890	7.6–9.5	11.8	18.5–19.4	1.3×10^7	150
YCOB	0.900	7.8–9.0	12.0	18.6–19.0	2×10^7	150
GdCOB	0.938	10.5–13.0	13.5	21.1–24.7	5.7×10^6	150
SmCOB	0.958	9.9–12.7	13.9	19.6–27.0	2.8×10^6	154
NdCOB	0.983	12.7–15.2	15.5	24.5–30.0	1.1×10^6	155
PrCOB	0.990	13.0–15.8	15.3	24.7–31.5	1×10^5	150
LaCOB	1.032	9.6–11.8	15.0	19.0–25.0	1.2×10^6	153

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illustrated in Fig. 9, which presents the forms of piezoelectric materials with different thicknesses. For films, the thickness of thin films typically ranges from several hundred nanometers to a few micrometers, whereas that of thick films lies within the range of several to tens of micrometers. According to material systems, piezoelectric films can generally be classified into crystal films and polymer films (such as PVDF). Given that polymer films generally lack sufficient thermal stability, they are usually excluded from consideration for high-temperature applications.[166] For inorganic piezoelectric films, commonly used material systems include PZT, $LiNbO_3$, AlN, and ZnO, among which AlN and ZnO adopt the wurtzite structure.[167,168] As discussed in Secs. III A and III B, the advanced properties of these materials have been described in detail. However, current film growth and deposition technologies remain a major bottleneck limiting further performance improvement.

D. Device-oriented comparative evaluation of high-temperature piezoelectric materials

Sections III A–III C have systematically reviewed the composition, properties, and research progress of several representative high-temperature piezoelectric materials. However, the discussion has generally focused on parameter summaries within individual material systems. To further highlight the differences and tradeoffs among different material systems, as well as their implications for transducer design, this section provides a cross-comparative summary from the perspective of device implementation.

The suitability of a high-temperature piezoelectric material cannot be determined from a single parameter such as d_{33} or T_c. Instead, material selection should be based on application-dependent combinations of electromechanical coupling, dielectric and mechanical losses, electrical resistivity, acoustic properties, and their stability with temperature. For broadband pulse–echo NDE, relatively high k_t, sufficient piezoelectric response, low dielectric loss, high resistivity, and appropriate acoustic impedance are generally desirable, while an excessively high Q_m may increase ringing and limit usable bandwidth. For narrowband or resonant sensing, higher Q_m, low dielectric loss, and small temperature coefficients of resonance-related material properties become more important. For high-output or long-propagation-distance ultrasonic excitation, strong piezoelectric response and electromechanical coupling must be balanced against dielectric and mechanical losses. In extreme-temperature or long-duration applications, the retention of these properties with temperature, together with depolarization resistance, electrical resistivity, and thermal stability, becomes more important than their room-temperature values alone. Therefore, the relevant figure of merit is inherently application dependent and should be regarded as a combination of temperature-dependent material parameters rather than a single universal scalar quantity.

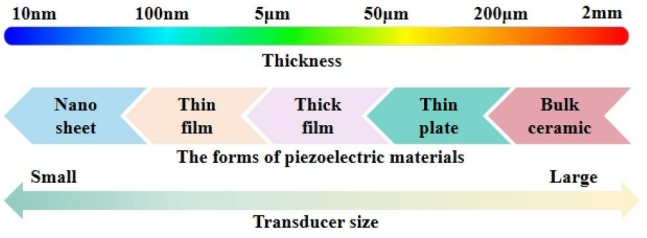


FIG. 9. Forms of piezoelectric materials with different thicknesses. The schematic was drawn by the authors based on the general classification reported in Ref. 165.

Table IV compares these material systems in terms of intrinsic temperature boundaries, major advantages, available device-relevant data, transducer-level validation, and remaining design-parameter gaps. To avoid conflating material stability with device capability, T_c, T_d, structural phase transitions, and other material-level thermal limitations are distinguished from the temperature ranges demonstrated at the transducer level. Based on this comparison, it can be seen that different material systems differ significantly in their maturity and applicability boundaries for high-temperature transducer design. PZT-based materials have the most mature device application basis and relatively complete conventional design parameters, but their practical high-temperature use is usually limited by depolarization.[84–89] BSPT-based ceramics show greater potential in terms of high Curie temperature and high piezoelectric activity, but their high cost must be taken into consideration.[90–98] Bismuth-layered[99–116] and tungsten-bronze-structured[120–128] ceramics possess higher theoretical temperature tolerance or lead-free advantages, but the electromechanical coupling, acoustic parameters, and high-temperature loss data required for transducer design remain relatively insufficient. In contrast, single crystals such as $LiNbO_3$,[140–145] quartz,[129–131] AlN,[132–139] ZnO,[161–163] $GaPO_4$,[156–160] and ReCOB[146–153] place greater emphasis on high-temperature stability, low loss, or depolarization-free operation. Among them, $LiNbO_3$ has been relatively well validated in high-temperature transducer applications, although the growth and processing of single crystals are generally more complex than those of ceramics. Therefore, the relative advantages of these material systems should be interpreted in the context of the target operating temperature, transducer architecture, and dominant application requirements.

Meanwhile, the role of Table IV is not merely to support preliminary material selection, but more importantly to provide guidance for future research on high-temperature piezoelectric materials. For HTPUT designers, Table IV enables separate assessment of intrinsic material temperature boundaries and experimentally demonstrated transducer capability, together with performance advantages, data availability, and device maturity, thereby guiding material selection. For researchers working on high-temperature piezoelectric materials, the final column highlights the current parameter gaps between existing material systems and the requirements of transducer design. In future materials research, in addition to improving conventional indicators such as piezoelectric coefficients, T_c, and T_d, greater emphasis should be placed on systematically supplementing high-temperature piezoelectric, dielectric, elastic, acoustic, and loss-related parameters, so that materials research can more effectively support the design of high-temperature transducers.

IV. DESIGN OF HIGH-TEMPERATURE PIEZOELECTRIC ULTRASONIC TRANSDUCERS

In practical high-temperature NDE applications, the piezoelectric material provides the fundamental electromechanical conversion capability, whereas the long-term stability of an HTPUT is governed by the coordinated performance of its multilayer structure, component materials, interfaces, and packaging. In particular, the interfaces among the piezoelectric element, electrodes, matching and backing

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TABLE IV. Device-oriented comparison of high-temperature piezoelectric material systems: temperature boundaries, available design data, transducer validation, and missing parameters.

Material system	Intrinsic temperature boundaries	Main reported advantages	Device-relevant data currently available	Transducer-level validation	Main missing parameters for transducer design
PZT-based perovskite ceramics[84–89]	Conventional PZT generally has T_c of 300–400 °C. Modified systems can retain piezoelectricity to approximately 350–380 °C, although T_d depends strongly on composition and thermal history.	High piezoelectric response, relatively strong electromechanical coupling, mature fabrication and poling processes, and relatively low cost. This is currently the ceramic system with the most mature application basis.	Parameters such as d_{33}, k_p/k_t, T_c/T_d, dielectric permittivity, dielectric loss, Q_m, density, sound velocity, and acoustic impedance are relatively complete.	HTPUTs based on PZT are commonly used at approximately 150–300 °C, while modified high-T_c PZT systems can extend the upper operating limit toward ∼350 °C. The device basis is relatively mature, although long-term reliability near the upper limit remains insufficient.	Complete high-temperature piezoelectric/dielectric/elastic parameter matrices, temperature coefficients, long-term thermal-aging data, depolarization behavior, and high-temperature resistivity data for different modified PZT systems still need to be unified.
BSPT-based perovskite ceramics[90–98]	BSPT-based ceramics generally have T_c of 400–500 °C. Selected compositions show T_d above 400 °C and good piezoelectric retention at 300–400 °C.	Compared with conventional PZT, these materials exhibit a higher Curie temperature and stronger high-temperature piezoelectric response, and can maintain relatively good piezoelectric activity at higher temperatures.	d_{33}, k_p/k_t, T_c/T_d, dielectric permittivity, dielectric loss, and related parameters have been widely reported, and some systems have high-temperature performance data.	BSPT-based HTPUTs are mainly considered for approximately 250–400 °C operation. Transducer-level validation has increased in this range, but long-duration and engineering-scale demonstrations remain less mature than those of PZT.	Complete high-temperature elastic/piezoelectric/dielectric constant matrices, k_t/k_{33} retention, dielectric-loss and resistivity evolution, and temperature dependence of sound velocity/acoustic impedance still require systematic supplementation.
Bismuth-layered piezoceramics[99–116]	Representative high-temperature BLSFs generally have T_c of 500–900 °C or higher. Some compositions retain strong piezoelectricity after annealing at 800–900 °C.	High Curie temperature, good thermal stability, low aging rate, and lead-free or low-lead advantages in some systems.	T_c, d_{33}, Q_m, dielectric permittivity, dielectric loss, and partial high-temperature stability data have been reported.	Transducer-level validation remains relatively limited; most studies still focus on material preparation and intrinsic property optimization.	k_t/k_{33}, elastic constants, sound velocity, acoustic impedance, high-temperature dielectric loss, high-temperature resistivity, poling stability, and temperature coefficients for transducer design remain insufficient.
Tungsten-bronze piezoceramics[120–128]	T_c varies strongly with composition and is generally about 300–650 °C. Systematic T_d and high-temperature retention data remain limited.	Strong compositional tunability and potential as lead-free high-temperature piezoelectric materials.	T_c, d_{33}, dielectric properties, and phase-structure regulation data are relatively abundant, but most data remain at the material level.	Complete high-temperature ultrasonic transducer demonstrations are still rare, and the device application basis is relatively weak.	High-temperature k_t/k_{33}, complete elastic constants, sound velocity, acoustic impedance, Q_m, dielectric loss, resistivity, thermal-stability parameters, and reproducible processing data are still clearly insufficient.

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TABLE IV. (*Continued.*)

Material system	Intrinsic temperature boundaries	Main reported advantages	Device-relevant data currently available	Transducer-level validation	Main missing parameters for transducer design
Quartz single crystal[129–131]	Quartz undergoes the α–β transition at 573 °C and transforms to tridymite near 870 °C. Mechanical twinning and property degradation may occur near 350 °C.	Good thermal stability, low dielectric loss, high resistivity, good frequency stability, and a mature material-constant system.	Elastic, dielectric, and piezoelectric constants and temperature coefficients are relatively complete, and cut-dependent basic data are abundant.	It has a solid application basis in high-temperature resonators, delay lines, and frequency-stable devices, but is less suitable as an active transducer material requiring high acoustic output.	Effective electromechanical coupling, acoustic loss, temperature dependence of sound velocity/acoustic impedance for different cuts, and parameter selection for specific transducer operating modes still require further organization.
AlN single crystal[132–139]	AlN has no Curie or depolarization transition and no phase transition below its melting point above 2000 °C. Stable piezoelectric properties have been reported to approximately 1000 °C.	Good thermal and chemical stability, high resistivity, low dielectric loss, and suitability for high-temperature low-loss electromechanical-acoustic conversion.	Basic thermal, elastic, and partial piezoelectric parameters are available.	Bulk AlN HTPUT validation remains limited. AlN-based thin-film and MEMS ultrasonic devices have mainly been demonstrated in the several-hundred-degree range, typically around 200–600 °C.	High-temperature piezoelectric constants, electromechanical coupling coefficients, dielectric loss, resistivity, sound velocity, acoustic impedance, acoustic attenuation, and cut/orientation-dependent parameters remain incomplete.
$LiNbO_3$ single crystal[140–145]	$LiNbO_3$ has a high T_c of approximately 1150 °C. Its high-temperature stability depends strongly on crystal cut and thermal history.	Good electromechanical coupling, flexible cut design, and significantly better temperature capability than conventional PZT. It is one of the most widely validated single-crystal materials for high-temperature ultrasonic devices.	For different cuts, kt, impedance response, temperature-dependent properties, dielectric parameters, and elastic parameters have been widely reported.	$LiNbO_3$ is one of the most extensively validated materials for HTPUTs, with device demonstrations commonly covering approximately 500–800 °C. Clear pulse–echo signals have been experimentally demonstrated up to 800 °C.	Cut-dependent complete piezoelectric/elastic/dielectric constant matrices, sound velocity, acoustic attenuation, dielectric loss, temperature-dependent resistivity, and long-term thermal-cycling data at high temperature still need to be unified.
ReCOB-type rare-earth calcium oxyborate single crystals[146–155]	ReCOB crystals have no Curie transition or phase transition below their melting point of approximately 1500 °C. Their high-temperature limitation is mainly related to electrical	No ferroelectric Curie depolarization problem, relatively high resistivity at high temperature, good thermal stability, and suitability as an extreme-temperature	Some piezoelectric constants, elastic constants, dielectric constants, resistivity data, and high-temperature stability data are available.	Some high-temperature sensing or transducer-related studies have been reported, but large-scale device validation and engineering application basis remain limited.	Complete material-constant matrices, sound velocity, acoustic impedance, acoustic attenuation, high-temperature dielectric loss, temperature coefficients, and batch-to-batch

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TABLE IV. (*Continued.*)

Material system	Intrinsic temperature boundaries	Main reported advantages	Device-relevant data currently available	Transducer-level validation	Main missing parameters for transducer design
	and chemical stability.	piezoelectric material platform.			reproducibility data for different crystals and cuts still need to be supplemented.
$GaPO_4$ single crystal[156–160]	$GaPO_4$ has no Curie temperature and undergoes an α–β phase transition near 970 °C. It maintains good piezoelectric and frequency stability below this transition.	Quartz-like structure with better high-temperature stability, as well as relatively good resistivity and frequency stability.	Basic piezoelectric, elastic, and dielectric parameters have been reported.	Mainly used in high-temperature resonant and frequency-stable sensors; high-temperature ultrasonic transducer-level validation is relatively limited.	High-temperature electromechanical coupling, sound velocity, acoustic impedance, acoustic attenuation, dielectric loss, cut-dependent parameters, and long-term temperature-stability data remain insufficiently systematic.
ZnO single crystal[161–163]	ZnO has no Curie transition and no structural phase transition below its decomposition temperature of approximately 1975 °C. However, the rapid increase in electrical conductivity generally limits its practical operating temperature to below approximately 500 °C.	Intrinsic piezoelectricity from the wurtzite structure, absence of depolarization, relatively high sound velocity, and good thermal stability.	Basic piezoelectric, elastic, and dielectric parameters have been reported.	Direct high-temperature HTPUT validation is limited; most work remains at the level of material parameters and basic acoustic-device studies.	Complete high-temperature material-constant matrices, sound velocity, acoustic impedance, acoustic attenuation, resistivity, dielectric loss, temperature coefficients, and comparability of data under different defect states and growth conditions remain insufficient.

layers, bonding layers, housing, and electrical interconnections often become critical reliability bottlenecks because they must simultaneously maintain acoustic transmission, electrical continuity, mechanical integrity, and chemical stability at elevated temperatures. Consequently, a high intrinsic temperature limit of an individual material does not necessarily guarantee reliable device operation if softening, diffusion, oxidation, creep, cracking, delamination, or loss of electrical contact occurs elsewhere in the transducer. These challenges are further intensified by thermal-expansion mismatch, which can generate residual stresses and increase the risk of interfacial damage during prolonged exposure and thermal cycling. Therefore, the design of HTPUTs should be considered at the complete device level, with particular attention to acoustic matching, thermal stability, interfacial compatibility, and packaging reliability. Based on whether the active transducer is directly exposed to the high-temperature environment, HTPUTs can be broadly classified into direct-contact and nondirect-contact configurations, as shown in Fig. 10. This section reviews the representative architectures, component selections, and fabrication strategies of these two categories, together with their applicable temperature ranges and major performance–reliability tradeoffs.

A. Direct contact type

Direct contact high-temperature ultrasonic transducers refer to devices in which the piezoelectric element, through the matching layer and coupling interface, is in direct contact with the hot surface of the test object to transmit acoustic energy and perform both transmission and reception. Such configurations typically offer high acoustic coupling efficiency, short signal paths, and compact system architecture. However, their long-term stability strongly depends on the coordinated design of materials and fabrication processes for the critical components, particularly the high-temperature reliability of the matching layer, backing layer, and bonding interfaces.

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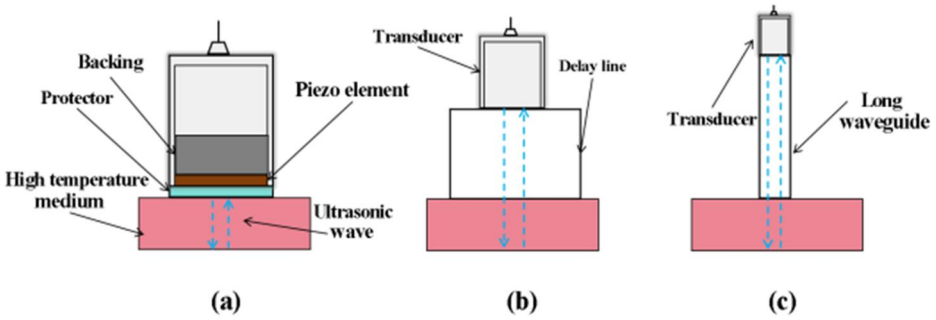


FIG. 10. Ultrasonic transducers for extreme conditions: (a) with a thin protective layer; (b) with a delay line; (c) with a waveguide.

1. Matching layer

In practical engineering, piezoelectric elements are typically not placed in direct contact with the measured object. Instead, an acoustic matching layer is introduced between the two. This matching layer serves two key functions: first, it protects the fragile piezoelectric element from wear, corrosion, and mechanical impact; second, it provides acoustic matching to mitigate the significant acoustic impedance mismatch between the piezoelectric material and the high-temperature medium. The requirements for acoustic matching have been elaborated in detail in Sec. II A. It is worth noting that the difference in acoustic matching between high-temperature and room temperature arises because temperature changes alter the speed of sound in the medium, leading to a shift in acoustic impedance. In principle, the aforementioned "ideal matching" strictly holds only near a specific temperature.

In addition, as a protective layer, the material should possess sufficient mechanical strength, corrosion resistance, and hardness, while maintaining stability under high-temperature service conditions and exhibiting a CTE closely matched to that of the piezoelectric element. For high-temperature sensors, thermal expansion compatibility is particularly critical; otherwise, thermal cycling and elevated thermal stresses may lead to interfacial delamination or even spalling between the protective layer and the piezoelectric component.[169,170] In working environments involving molten metals, it is also essential to ensure good wettability of the matching layer with the liquid metal. Insufficient wettability may lead to interfacial microbubbles. These bubbles can significantly attenuate the transmitted and received ultrasonic energy and may ultimately cause measurement failure.[171,172] Given that a single material can hardly satisfy all these requirements simultaneously, the design of matching layers for HTPUTs should be regarded as a multi-objective optimization problem rather than a simple acoustic-impedance matching problem. The relative priority of thermal stability, wettability, acoustic impedance matching, mechanical reliability, oxidation resistance, and interfacial compatibility depends strongly on the specific application scenario. For long-term monitoring of high-temperature solid structures such as steel blocks or pipelines, thermal stability, CTE compatibility, oxidation resistance, and mechanical robustness are often more critical, because interfacial cracking or delamination may directly interrupt the acoustic transmission path. In contrast, for applications involving molten metals or high-temperature liquids, wettability and chemical compatibility may become dominant factors, since poor wetting can introduce interfacial bubbles and unstable reflection conditions. For moderate-temperature or short-duration measurements, acoustic impedance matching may still play a more important role in improving signal amplitude and bandwidth. Therefore, the design priority of the matching layer should be determined according to the operating temperature, contact medium, service duration, thermal cycling condition, and required measurement accuracy.

As a practical example, in the NDT of steel blocks or pipelines, the inspected medium is steel, with an acoustic impedance of approximately $Z_3 = 45$ MRayl. Most piezoelectric materials exhibit acoustic impedance of around $Z_2 = 30$ MRayl. According to Eq. (3), the optimal impedance of the matching layer in this case should be $Z_1 = 37$ MRayl. However, the most commonly used protective material, Inconel 625, has an acoustic impedance of about 45 MRayl, which deviates significantly from the theoretically required value. This indicates that achieving ideal acoustic matching through a quarter-wavelength ($\lambda/4$) metallic protective layer is impractical in many cases. Consequently, a protective layer with a half-wavelength ($\lambda/2$) thickness has been proposed. Although it does not provide acoustic matching, its increased thickness offers superior mechanical robustness. This type of transducer was first proposed by Mrasek *et al.*,[173] in which the piezoelectric element was $LiNbO_3$ and the protective layer was Inconel 600 with a thickness of 0.6 mm, corresponding to a half-wavelength ($\lambda/2$) at an operating frequency of 5 MHz. Another example is Kažys *et al.*'s[174] ultrasonic transducer made of bismuth titanate as the piezoelectric material. The matching layer is made of AISI 321 stainless steel with a thickness of $\lambda/2$. The operating temperature of the transducer can reach 500 °C.

According to the study by Amini *et al.*,[175] black alumina can be used as a matching layer in ultrasonic transducers owing to its excellent stability and oxidation resistance at temperatures up to 800 °C. With an acoustic impedance of ~36 MRayl, it can partially alleviate the impedance mismatch between the transducer and the test medium. Based on this property, alumina was selected as the matching layer in a transducer employing a 36° Y-cut $LiNbO_3$ piezoelectric element with a center frequency of 3 MHz. The thermal expansion coefficient of alumina, about $8.5 \times 10^{-6}\,°C^{-1}$, is close to that of $LiNbO_3$ (10.3×10^{-6}–$15.5 \times 10^{-6}\,°C^{-1}$), which helps maintain interfacial bonding and structural integrity over a wide temperature range, thereby ensuring reliable operation. Experimental results demonstrated that the transducer could still produce a clear echo signal at 800 °C and remained fully functional after cooling to room temperature. For comparison, zirconia was also tested as a potential matching layer. Although zirconia has a favorable acoustic impedance and high thermal stability, it failed to achieve satisfactory acoustic coupling with the piezoelectric element under the same brazing conditions,

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indicating that interfacial compatibility and joining reliability can be as important as intrinsic acoustic parameters.

When high-temperature ultrasonic transducers are operated in molten metal environments, the wettability and chemical compatibility of the protective or matching layer become particularly important. For example, a nickel matching layer exhibits excellent wettability with certain molten metals. However, its high acoustic impedance may still cause significant impedance mismatch and reduce the received echo amplitude, even when a quarter-wavelength design is used. This example illustrates that, in molten media, improving wetting and maintaining a stable solid–liquid acoustic interface may sometimes take precedence over ideal acoustic impedance matching.[176,177]

These examples show that matching-layer design in HTPUTs should not follow a fixed universal priority order. Instead, the design should be adapted to the specific application requirements. In many long-term high-temperature applications, ideal acoustic impedance matching may need to be partially sacrificed to ensure thermal stability, mechanical reliability, oxidation resistance, wettability, or interfacial durability. Therefore, practical matching-layer design should balance acoustic transmission efficiency with long-term reliability under the target service conditions. Overall, matching-layer design for HTPUTs requires more than acoustic-impedance matching. Thermal stability, acoustic attenuation, oxidation resistance, interfacial reliability, and fabrication feasibility must also be considered to ensure stable transmission under high-temperature operation.

2. Backing layer

The backing layer, attached to the rear side of the piezoelectric element, primarily serves to provide mechanical support and to suppress the ringing effect by absorbing backward-propagating acoustic energy, thereby broadening the operational bandwidth. The corresponding trade-off is a reduction in transducer efficiency, as part of the acoustic energy radiated by the piezoelectric element is absorbed by the backing. The two key parameters in backing design are acoustic impedance and acoustic attenuation coefficient. The acoustic impedance must be carefully balanced to optimize the trade-off between bandwidth enhancement and transduction efficiency, while the attenuation coefficient should be sufficiently high to effectively absorb the transmitted acoustic energy and prevent the occurrence of "ghost" echoes during signal reception.[178,179] For high-temperature applications, backing materials must also satisfy engineering constraints such as thermal stability, radiation resistance, and thermal expansion compatibility. Here, thermal stability refers not only to mechanical and chemical stability but also to maintaining relatively constant acoustic impedance and attenuation coefficient across the entire operating temperature range.

In conventional ultrasonic transducers designed for room-temperature applications, the backing layer is typically composed of a polymer matrix filled with heavy metal or glass powders and can be directly bonded to the piezoelectric element. The filler phase, such as heavy metal or glass powders, increases the effective density and helps control multiple scattering. The polymer matrix serves as both the absorbing and binding phase, ensuring reliable interfacial connection with the piezoelectric element. By selecting the type, particle size, and volume fraction of the powders, as well as the composition and ratio of the polymer matrix, the acoustic impedance and attenuation coefficient of the backing can be precisely tuned. A typical formulation employs a tungsten–epoxy composite system.[180,181] It should be noted, however, that such polymer-based backings tend to undergo thermal softening and degradation at elevated temperatures, with an operational limit typically around 150–200 °C, rendering them unsuitable for high-temperature environments.[182]

One of the simplest backing materials that can meet high-temperature requirements is stainless steel (AISI 316). However, dense AISI 316 exhibits very low ultrasonic attenuation (approximately 0.13 dB/cm at 2 MHz), which causes multiple internal reflections within the backing and deteriorates the echo quality. To address this issue, Kazys *et al.* designed the rear surface of the backing layer as a concave structure, such as a conical shape with a diffusion angle of ∼120° facing the piezoelectric element, to scatter and dissipate internal reflections [as shown in Fig. 11(a)].[183] Another solution involves using a porous version of stainless steel AISI 316. Feuillard *et al.* systematically investigated the acoustic properties of porous AISI 316 with different porosities up to 400 °C. They determined that a porosity of 25% yielded optimal performance, with a sound velocity of 4402 m/s, an acoustic impedance of 26.7 MRayl, and an attenuation coefficient of 4.6 dB/cm at room temperature. These parameters remained nearly unchanged up to 400 °C, indicating that the scattering-dominated loss mechanism is largely insensitive to temperature.[184] Both approaches effectively overcome the drawback of low acoustic attenuation in AISI 316.

Boubenia *et al.* proposed a high-temperature backing material based on metal matrix composites with tunable acoustic impedance and high attenuation.[185] In this design, one metal serves as the matrix while another acts as the inclusion phase. The first reported system was a tin–tungsten composite with varying tungsten volume fractions. By adjusting the W content, the effective acoustic impedance of the composite could be tuned within the range of 25–31 MRayl, while the attenuation coefficient could be adjusted from 20 dB/cm to as high as 191 dB/cm. However, due to the low melting point of tin, the operating temperature of this composite was limited to ∼230 °C. To increase the temperature tolerance, the authors subsequently developed an aluminum–tungsten composite (5% W), exhibiting an acoustic impedance of 8–10 MRayl and a remarkably high attenuation coefficient of 156 dB/cm. After annealing at 550 °C for 12 h, no defects were observed, indicating that the upper temperature limit is primarily constrained by the melting point of the metal matrix and can be effectively extended by selecting a matrix material with higher melting point.

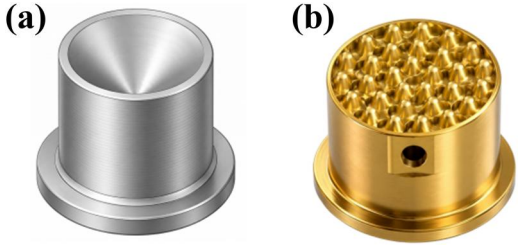


FIG. 11. High-temperature ultrasonic transducer with (a) an integral concave backing and (b) an irregular backside. The schematics were newly drawn by the authors based on the structural concepts reported in Refs. 174 and 183.

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Amini *et al.* proposed using 3 mol. % yttria-stabilized porous zirconia (YSZ) as a backing material.[186] The fabrication method involves introducing polyethylene microspheres into ceramic powder; during sintering, the microspheres are burned out, leaving behind pores that form a tunable porous structure with adjustable acoustic properties. The pore size is determined by the radius of the microspheres, enabling the attenuation coefficient to be adjusted from approximately 1.5 to 5.5 dB/cm. The porosity governs the acoustic impedance: as the porosity decreases from 30% to 10%, the effective acoustic impedance increases from about 20 MRayl to 34 MRayl. Moreover, the coefficient of thermal expansion of the porous medium is approximately equal to that of the dense matrix ($11 \times 10^{-6}\,^{\circ}C^{-1}$), which is close to that of $LiNbO_3$ ($15.4 \times 10^{-6}\,^{\circ}C^{-1}$), facilitating interfacial stability during thermal cycling. Experimental results showed that the YSZ backing maintained stable acoustic and structural properties at 800 °C, demonstrating strong potential for high-temperature applications.

Kazys *et al.* tested a backing layer made of graphite bronze.[174] Graphite alloys work at temperatures up to 800 °C, and its acoustic impedance is 13.7 MRayl. Although there is no literature indicating its specific sound attenuation coefficient, experiments show that under the design of an irregular backside [as shown in Fig. 11(b)], the graphite alloy can achieve backside sound wave absorption at a thickness of 1 cm.[174]

Overall, backing-layer design for HTPUTs involves a trade-off between bandwidth enhancement, sensitivity retention, mechanical support, and thermal reliability. Although high acoustic attenuation is beneficial for suppressing ringing and broadening bandwidth, the backing material must also maintain stable acoustic, mechanical, and interfacial properties under high-temperature exposure and thermal cycling.

3. Bonding methods

The piezoelectric element must maintain reliable acoustic and mechanical contact with the matching and backing layers throughout the operating temperature range, while the transducer itself must also remain effectively coupled to the medium under inspection to ensure stable transmission and reception of ultrasonic energy. Under high-temperature conditions, these interfaces become particularly critical because they must simultaneously satisfy acoustic, mechanical, chemical, and, where applicable, electrical requirements. Stable acoustic transmission requires intimate contact with limited void formation and interfacial discontinuity, whereas mechanical reliability requires sufficient bonding strength together with the ability to accommodate differential thermal expansion. Chemical compatibility and oxidation resistance are also essential during prolonged thermal exposure, while interfaces that additionally serve as electrodes or conductive paths must maintain electrical continuity. These requirements are often coupled and may compete with one another. For example, a rigid ceramic bonding layer may provide excellent intrinsic thermal resistance but limited stress-relaxation capability, whereas a ductile metallic interlayer can accommodate thermal-expansion mismatch but may introduce additional constraints associated with diffusion, oxidation, wetting, or remelting. Therefore, the selection of a high-temperature coupling or bonding method should be based on the overall interfacial performance rather than solely on the nominal temperature rating of the coupling material.

Thermal-expansion mismatch should also be considered as a multilayer thermomechanical problem rather than solely through the CTE values of two adjacent materials. The magnitude and evolution of interfacial stress depend not only on the CTE difference and temperature excursion, but also on the elastic properties and thicknesses of the bonded layers, device geometry, interfacial constraint, heating and cooling rates, and time-dependent stress relaxation or creep. During a single heating–cooling excursion, part of the thermally induced stress may be reversible; however, repeated thermal cycling can progressively accumulate residual stress and interfacial damage, eventually leading to microcracking, delamination, loss of acoustic contact, or electrical failure. In practical HTPUT design, these risks can be mitigated by selecting thermally compatible materials, introducing compliant or ductile interlayers, optimizing layer thickness and stiffness transitions, and controlling bonding and thermal-cycling conditions. These strategies must nevertheless be balanced against acoustic transmission, bonding strength, electrical conductivity, and chemical stability.

At present, the commonly used high-temperature coupling approaches can be broadly classified into liquid coupling, dry coupling, and solid coupling. Although dry coupling is sometimes grouped together with solid-state coupling, it is treated separately in this review because it relies on direct pressure-assisted solid–solid contact without an intermediate coupling material and therefore involves distinct interfacial requirements and failure mechanisms.

a. Liquid coupling. Liquid coupling offers several significant advantages for connecting transducer components. First, as a fluid, it can readily fill microscopic interfacial gaps, thereby preventing air entrapment, minimizing reflection and scattering, and enhancing ultrasonic transmission efficiency. This characteristic also relaxes the stringent requirements on interface flatness. In addition, when two components exhibit markedly different coefficients of thermal expansion, liquid coupling can help accommodate differential thermal expansion through its fluidity, thereby reducing stress transfer across the interface. Finally, the process of liquid coupling is relatively straightforward, requiring only the injection of the liquid and appropriate sealing.

Liquid couplants can be classified into two types: those that remain liquid throughout the entire temperature range (e.g., silicone oil) and those that are solid at low temperatures but liquid only at high temperatures (e.g., glass solders). Silicone oil has been extensively studied as a liquid couplant because it can provide effective acoustic transmission up to approximately 250 °C. However, beyond this temperature, silicone oil gradually evaporates, volatilizes, and may even decompose or carbonize, ultimately resulting in a decline or loss of coupling performance.[187] In principle, glass solders can be applied at higher temperatures. Considering factors such as medium properties, chemical reactivity and compatibility, diffusion, evaporation, thermal expansion, and mechanical behavior upon solidification, the glass solder NaPoLi—composed of sodium oxide, phosphorus oxide, and lithium oxide—has been identified as a potential liquid coupling agent.[175,188] Nevertheless, at elevated temperatures, NaPoLi undergoes chemical reactions with most transducer components, except when platinum or gold protective coatings are applied to the interface. NaPoLi has demonstrated good acoustic coupling performance at temperatures up to 500 °C. Compared with NaPoLi, nonalkaline glass

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solders (LG-K1 and LG-K2) exhibit much lower chemical reactivity and remain thermally stable up to about 600 °C.[173] Nevertheless, the problem of easy leakage of liquid couplant interferes with their long-term stability, and their poor radiation resistance also limits their application in scenarios such as nuclear power plants.

b. Dry coupling. Dry coupling refers to the process of directly pressing together the components to be bonded under high pressure at elevated temperatures, without the use of couplant, to achieve intimate contact and thereby establish a stable channel for acoustic energy transmission.[189,190] Dry coupling eliminates degradation of an intermediate couplant, although oxidation, contact-pressure evolution, surface degradation, and long-term interfacial stability still require consideration.[191]

Dry coupling imposes stringent process requirements. For reliable coupling, the contacting interfaces should be pretreated to achieve a flatness variation of less than 1 μm and subsequently subjected to optical-grade polishing so that the surface roughness is reduced to below 0.01 μm. During the coupling process, both high-temperature and high pressure are typically required. Elevated temperature helps establish a stable contact, but its upper limit must remain below the critical thresholds tolerated by other components of the device.[192] Meanwhile, the required contact pressure can reach several hundred megapascals,[193] making brittle piezoelectric elements prone to cracking or even catastrophic fracture during loading. Finally, the CTE of the two interfaces must be considered, as any mismatch may induce thermal stresses at high temperatures, potentially leading to device failure.[194]

As shown in Fig. 12(a), to address the pressurization problem during coupling, a structural design employing a disk spring has been proposed to maintain a constant pressure between sensor elements over a wide temperature range.[195] When the screw is tightened, the disk spring is compressed, providing a stable elastic force. As the temperature increases, thermal expansion of the structure occurs, and the disk spring correspondingly deforms to automatically compensate for the displacement, thereby maintaining an approximately constant clamping force. This approach effectively avoids the unexpected excessive pressure caused by thermal expansion in conventional structures, which could otherwise damage the device.

In addition, a cyclic loading method has been proposed.[195] Instead of applying the full target pressure in a single step, the load is applied in three successive cycles, which significantly reduces the maximum pressure experienced by the piezoelectric crystal. During the first loading cycle, the rough interface is flattened and undergoes plastic deformation. By the second and third cycles, the interface has already adhered, and only a smaller additional pressure is required to achieve effective coupling. Experimental results demonstrate that, without cyclic loading, a pressure of 229 MPa was required, whereas under three-cycle pressurization, the maximum pressure for the first cycle was reduced to 76 MPa, and only 12.7 MPa was needed in the third cycle [as shown in Fig. 12(b)].

At present, thin ductile metal foils are commonly inserted at the bonding interface to mitigate thermal stresses arising from mismatched CTEs. The three most frequently considered foils are gold, silver, and aluminum. Among them, aluminum foil is generally excluded due to its susceptibility to oxidation at elevated temperatures.[174] Silver foil, by contrast, combines favorable ductility with high

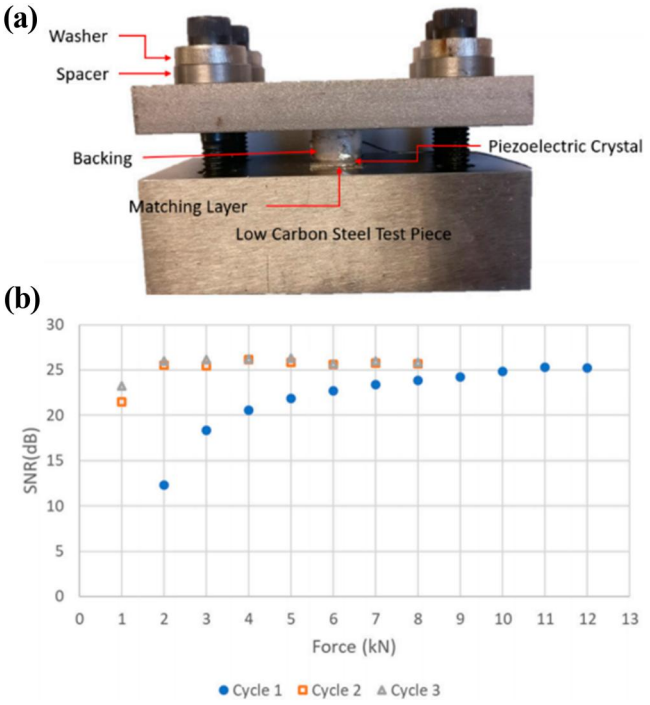


FIG. 12. (a) Photograph of the disk spring structural design. (b) SNR as a function of applied pressure under different loading cycles. Reproduced with permission from Bhadwal *et al.*, Sensors **19**, 5383 (2019). Copyright 2019 Authors, licensed under a Creative Commons Attribution (CC BY 4.0) license.[195]

electrical conductivity, enabling it to function as an electrode when positioned at the interface of piezoelectric crystals. In addition, its acoustic impedance (~37.8 MRayl) closely matches that of many piezoelectric materials, thereby facilitating efficient acoustic energy transmission.[189] Gold foil offers superior ductility and higher thermal stability than silver, allowing more effective compensation for CTE mismatch; however, its use is often constrained by the significantly higher cost of the transducer.[195,196] Furthermore, studies have demonstrated that annealed foils exhibit greater flexibility than rolled foils, thereby reducing the coupling pressure required to maintain effective bonding.

A novel approach known as ultrasonically assisted diffusion bonding has recently attracted increasing attention.[183] Similar to traditional dry coupling, the bonded interfaces must undergo rigorous pretreatment, including polishing to optical flatness and subsequent deposition of nickel, silver, or gold layers. In this method, a Langevin-type ultrasonic transducer is used in conjunction with a horn. Under these conditions, the temperature is gradually increased to soften the interlayer gold foil. When the temperature reaches approximately 100 °C, a 17 kHz ultrasonic excitation is applied, and heating is continued up to about 200 °C.[174] The imposed ultrasonic field generates tangential displacement at the contact interface, thereby enhancing atomic diffusion and promoting bonding. Depending on the surface preparation quality, a relatively low applied pressure of 40–60 MPa is sufficient to achieve stable joining. Owing to the introduction of ultrasonic vibration, both the pressure and temperature required for

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bonding are significantly reduced, reaching levels compatible with the thermal and mechanical limitations of piezoelectric elements.

c. Solid coupling. Solid coupling refers to the connection of various components of a transducer through solid couplant. It has its own unique advantages. Compared with liquid coupling, this method provides a more stable connection, is suitable for higher-temperature applications, and can maintain long-term effectiveness. Relative to dry coupling, it imposes less stringent requirements on interface processing, as the surface does not need to be polished to the same degree of precision. The proper choice of solid couplant can enhance the coupling interface's tolerance to mismatches in CTEs. This technology encompasses multiple implementation forms, including adhesive bonding, brazing/soldering, and the use of novel solid couplants.

(1) Adhesive bonding

Bond *et al.* reported the use of a commercial conductive adhesive, Duralco-124, for bonding electrodes to wires. This material is composed of an epoxy resin matrix loaded with silver powder, where the metallic filler provides high electrical conductivity and the resin contributes to mechanical adhesion and limited thermal resistance. Despite the manufacturer's claim that the adhesive can operate reliably up to 340 °C, experimental results demonstrated signal loss when the temperature exceeded ∼140 °C. The failure was attributed to the degradation of the conductive adhesive under repeated thermal cycling, which disrupted electrical continuity.[176] These observations suggest that although Duralco-124 offers ease of processing and favorable conductivity, its actual temperature tolerance in long-term high-temperature applications is significantly lower than the rated specification.

Fei *et al.* used a commercial high-temperature conductive adhesive (YH-A018), rated for continuous operation up to 500 °C, to bond the back electrode to the alumina backing material. The adhesive was cured in two sequential steps, first at room temperature for 12 h and then at 120 °C for 2 h, to ensure reliable conductivity and adhesion. However, although the adhesive itself is capable of withstanding higher temperatures, the entire device was evaluated only up to 200 °C, a limit imposed by the thermal tolerance of other encapsulating components, particularly the isolation epoxy resin, which is rated at 210 °C.[197]

In addition, an alumina-based high-temperature ceramic adhesive (Cotronics 989), with a nominal temperature resistance of ∼1648 °C, is therefore widely used for bonding components of high-temperature ultrasonic transducers.[198] Giurgiutiu *et al.* applied it to bond a $GaPO_4$ piezoelectric plate to titanium, and the results showed that clear ultrasonic signals were still obtained at 426 °C, but the signals disappeared at higher temperatures, which the authors attributed to the bonding layer. Parks *et al.* used the same adhesive to bond AlN and SiC, where it remained effective at 1000 °C for 8 h, and the subsequent performance degradation was attributed to the oxidation of the AlN.[199] These results indicate that Cotronics 989 demonstrates excellent high-temperature stability and strong potential for practical applications. However, many organic adhesive systems exhibit limited radiation resistance and much lower acoustic impedance than piezoelectric elements, metallic substrates, and protective layers.

(2) Soldering/Brazing

Soldering/Brazing refers to the reliable connection of piezoelectric elements to other components, such as matching layers and backing layers, through solder. There are two types of solders: soft solders with melting temperatures below 450 °C (soldering) and hard solders with melting temperatures above 450 °C (brazing). The process design must meet the temperature window: the solder's melting temperature should exceed the sensor's maximum working temperature to prevent joint remelting during service and remain below the failure critical temperature of each component to avoid material degradation or interface damage. In addition, the solder should be flexible enough to relieve stress due to the coefficient of thermal expansion (CTE) mismatch between the piezoelectric layer, matching layer, and backing layer, and chemically compatible with the thin electrode coating on the piezoelectric element surface to avoid dissolution or corrosion, thereby ensuring long-term stability and reliability of the joint.

The melting temperature of solder is determined by its chemical composition. One of the common solders is Pb/Sn/Ag alloy, with a melting point of about 310 °C, suitable for moderate temperature joints.[200,201] Another type of commercial solder widely used in high-temperature joining of dissimilar metals is 88Au-12Ge (mass fraction 88% Au/12% Ge).[202] Its melting point is 361 °C and the recommended welding temperature is 380 °C. The coupling layer has good ductility and a coefficient of thermal expansion of approximately 13.4×10^{-6}/°C, making it particularly suitable for metal-to-metal connections with CTE mismatches. To obtain stable wetting and interface reaction, a nickel layer of 1.5–2.5 μm should be plated on the metal surface before soldering, and the soldering should be completed in a vacuum environment.[203,204] Soldering is simpler than brazing, but the soldering temperature is lower and cannot accommodate higher temperature applications.

Brazing enables reliable joining at elevated temperatures for metal–metal, metal–nonmetal, and dissimilar metal/porous components. Amini *et al.* applied this process to the assembly of high-temperature ultrasonic transducers [Figs. 13(a) and 13(b)] and systematically compared several solders and their failure mechanisms.[175] The silver-based Active-braze® No. 10 caused cracking of the piezoelectric plate under multiple heating/cooling-rate schedules due to thermal-mismatch stresses between $LiNbO_3$ and porous ZrO_2; TiBraze® 800 exhibited insufficient wettability, leading to poor interfacial bonding and coupling failure. In contrast, the Al–Mg active brazing foil TiBraze® Al-665 produced a stable, crack-free bond among $LiNbO_3$, the porous ZrO_2 backing, and the Al_2O_3 quarter-wave matching layer. For the prototype transducer successfully

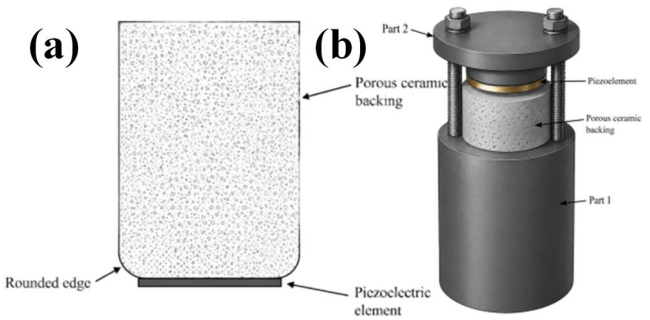


FIG. 13. Schematic illustrations of the brazed transducer assembly: (a) bonding configuration between the porous ceramic backing and the piezoelectric element, and (b) assembly jig used for the bonding operation. The schematics were newly drawn by the authors based on the assembly concept reported in Ref. 175.

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bonded with TiBraze® Al-665, steel-block contact tests over 27–800 °C yielded clear back-wall echoes across the entire temperature range. As temperature increased to ∼600 °C, the echo amplitude initially decreased but recovered at 800 °C, a trend attributed primarily to the temperature-dependent coupling with the test block rather than to intrinsic material degradation. Spectrally, the center frequency decreased slightly from 3.2 to 2.7 MHz, while the 6 dB bandwidth showed little temperature dependence, remaining around

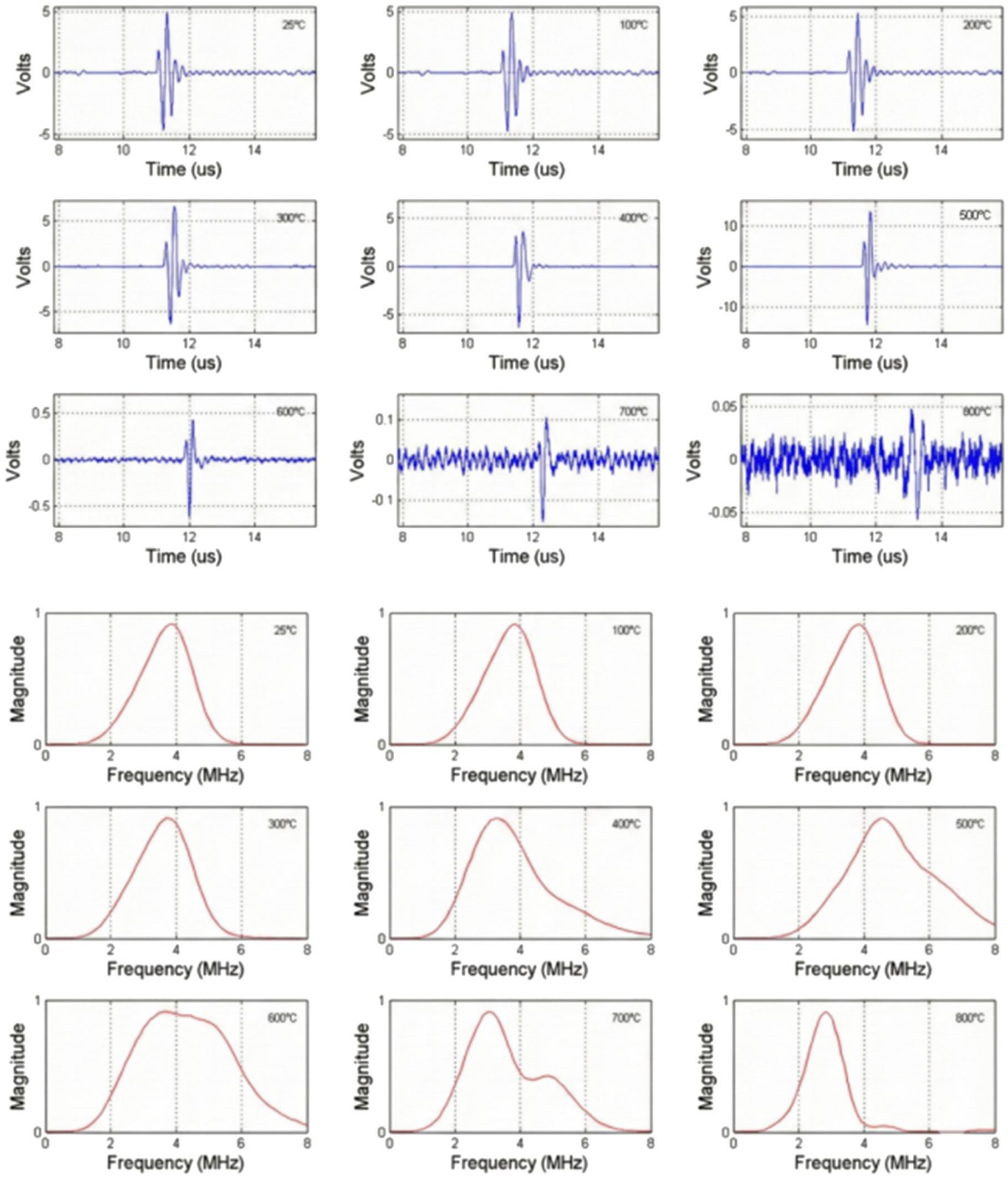


FIG. 14. Ultrasound performance at elevated temperatures of $LiNbO_3$ and porous zirconia transducers brazed with Al-665. Corresponding time and normalized frequency plots are included from room temperature to 800 °C. Reproduced with permission from Bosyj *et al.*, Instruments **3**, 2 (2018). Copyright 2018 Authors, licensed under a Creative Commons Attribution (CC BY 4.0) license.[205]

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70%–75% at 800 °C. Upon cooling to room temperature, the device retained functionality.[175]

Bosyj *et al.* further compared the joining performance of TiBraze® Al-665 with other solder (Morgan Gold-ABA®, 72Ag-28Cu), conducting brazing tests under two environments: vacuum and a reactive atmosphere.[205] The results showed that satisfactory acoustic and mechanical coupling was achieved only under two conditions: (i) TiBraze® Al-665 brazing in vacuum (Fig. 14); and (ii) 72Ag-28Cu

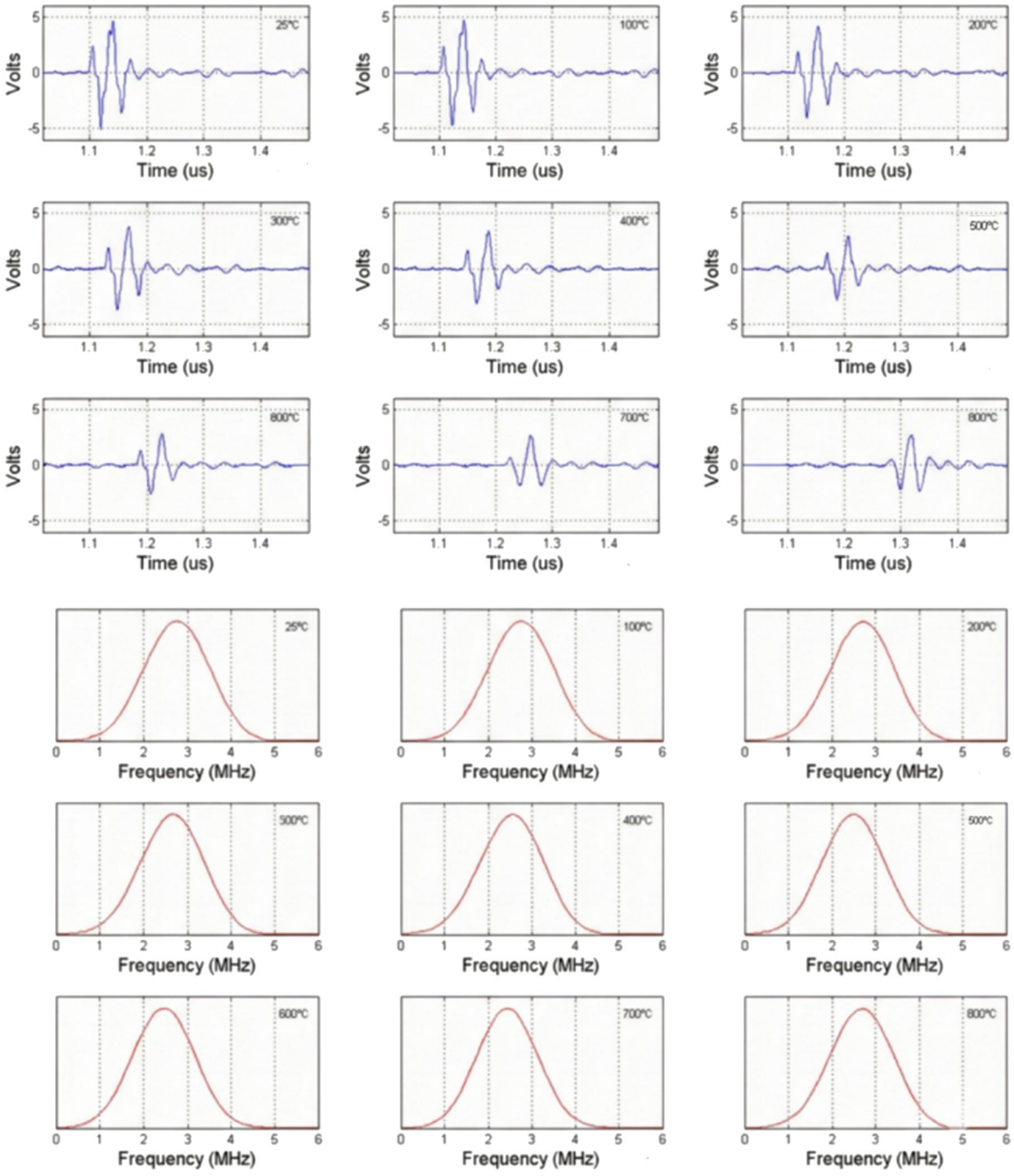


FIG. 15. Ultrasound performance at elevated temperatures of $LiNbO_3$ and porous zirconia transducers brazed with AgCu. Corresponding time and normalized frequency plots are included from room temperature to 800 °C. Reproduced with permission from Bosyj *et al.*, Instruments **3**, 2 (2018). Copyright 2018 Authors, licensed under a Creative Commons Attribution (CC BY 4.0) license.[205]

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brazing in a reactive atmosphere (Fig. 15). However, the two exhibited markedly different transducer performance and temperature stability. For Al-665, the prototype transducer performed excellently in the mid-temperature range (400–500 °C), with SNR maintained at 20–25 dB and echo amplitude approximately three times that at room temperature; when the temperature exceeded 500 °C, the amplitude decayed rapidly, and although the back-wall echo remained clearly discernible at 800 °C, the SNR dropped to ∼5 dB. The bandwidth, which was 40%–50% (∼2 MHz) at room temperature, fluctuated between 30% and 100% with temperature. The authors attributed this pronounced temperature dependence to annealing/softening of Al-665 above ∼345 °C, leading to interfacial stress relief and altered coupling conditions. By contrast, the 72Ag-28Cu prototype transducer achieved high-amplitude back-wall echoes with ∼60% bandwidth already at room temperature, and within the RT–800 °C range it exhibited weak temperature dependence in center frequency and bandwidth along with a stable SNR on the order of ∼20 dB, showing small overall variation with temperature and good repeatability. Meanwhile, the absence of a vacuum brazing requirement also reduced process complexity.[205]

(3) Novel nano-Ag couplant

Low-temperature silver sintering has been widely adopted in power-chip packaging in recent years, and both the formulation and processing performance of nano-Ag pastes have advanced rapidly.[206–208] The key mechanism is the high surface energy of silver nanoparticles, which enables them to undergo grain growth and densification at relatively low temperatures of approximately 250–300 °C, ultimately forming a dense metallic joint with high thermal (and electrical) conductivity. The resulting bonds can withstand temperatures exceeding 800 °C. These characteristics offer two major advantages: (i) the low processing temperature significantly reduces thermal shock and thermomechanical stress during assembly; and (ii) the improved high-temperature performance enhances the thermal robustness and service lifetime of the device.

In view of this, Gu *et al.* developed a high-temperature ultrasonic transducer by introducing a pressureless, low-temperature–sintered nano-Ag coupling layer between a $LiNbO_3$ wafer and a 35-steel substrate.[209,210] In the previous work, they compared the spreading properties of silver pastes of different viscosities and evaluated the effects of spreading properties and sintering temperature on the performance of the coupled layer after sintering. Based on these results, a paste containing 92 wt. % Ag with D50 ≈ 146 nm and viscosity ≈ 52.5 Pa s was selected. The coupling layer was prepared by SMT template screen printing, heated to 350 °C at a heating rate of 10 °C/min and kept for 50 min, and then cooled naturally to form a joint. For electrical connections, graphite conductive adhesive was used for the leads, and mica/glass-fiber–insulated high-temperature wires were employed to ensure oxidation resistance and stable electrical contact. After fabrication, time-domain echoes, V_{pp}, SNR, and impedance/phase were evaluated over 20–350 °C (as shown in Fig. 16). Notably, V_{pp} and SNR increased counterintuitively from 20 to 75 °C, which the authors attributed to a "temperature-activated coupling effect"—mild heating enhances the conductivity of the nano-Ag layer, relaxes residual sintering stresses, and improves Ag/$LiNbO_3$ interfacial conformity. At higher temperatures, performance gradually degraded, consistent with grain growth and increased porosity within the Ag network. The Q_m first decreased and then increased markedly between 100 and 350 °C, while the k_t increased initially and then decreased across the full range, peaking near 200 °C. Finally, during 350 °C durability tests

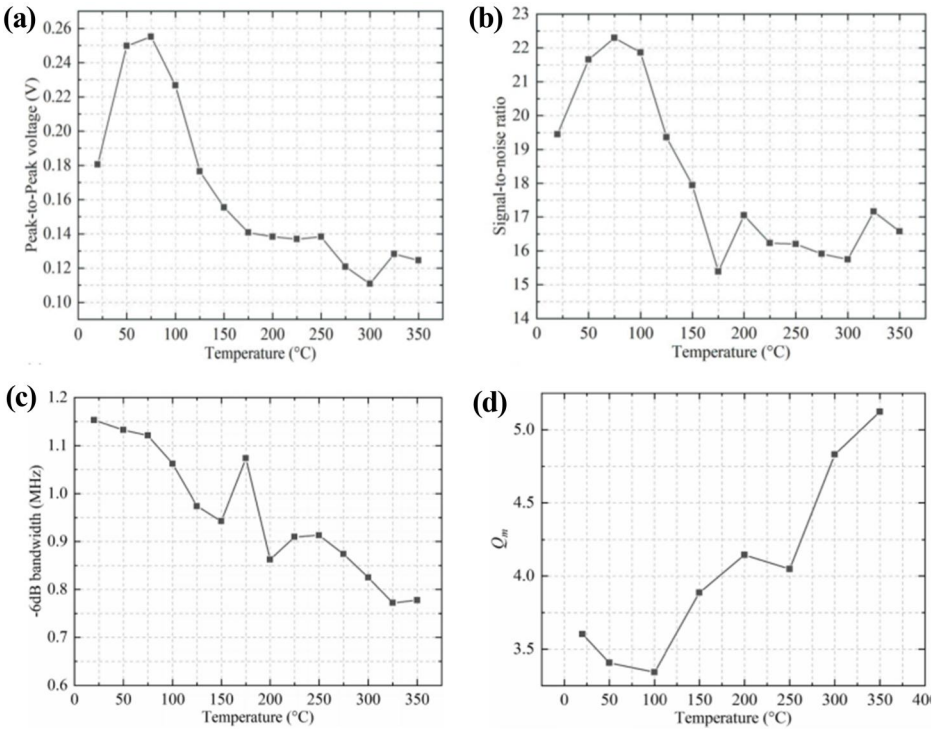


FIG. 16. Variations of transducer performance metrics with operating temperature: (a) V_{pp}, (b) SNR, (c) −6 dB bandwidth, and (d) Q_m. Reproduced with permission from Gu *et al.*, Ultrasonics **155**, 107736 (2025). Copyright 2025 Elsevier.[211]

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of 30 and 66 h, the transducer maintained clear primary/secondary echoes, with minimal changes in V_{PP} and SNR, indicating functional stability at 350 °C and promising high-temperature serviceability.[211]

In summary, preliminary applications of the low-temperature–sintered silver nanoparticles in ultrasonic transducers have demonstrated reliable acoustic and mechanical performance. However, research on this emerging technology remains limited, and a substantial gap persists between the demonstrated stable operating temperature of 350 °C and the theoretical target of 800 °C.

Overall, acoustic coupling is one of the main reliability bottlenecks in HTPUTs. The coupling layer must maintain stable transmission while resisting thermal degradation, oxidation, interfacial debonding, and thermal-expansion mismatch. Thus, the choice of liquid coupling, dry coupling, adhesive bonding, or brazing should depend on the operating temperature, contact condition, service duration, and repeatability requirement.

4. Direct deposition

Compared with bulk transducers, film ultrasonic transducers directly deposited onto the surface of the target structure offer a series of unique advantages. Through film growth processes, the piezoelectric layer can be tightly bonded to the metallic substrate in the solid state, enabling stable high-temperature acoustic coupling without the need for couplants or adhesive layers. This fundamentally eliminates issues such as volatilization, aging, or failure of coupling layers at elevated temperatures.[212] Direct deposition removes several conventional couplant- and adhesive-related temperature limitations, but the upper operating temperature remains jointly governed by the piezoelectric film, electrodes, substrate, and film–substrate interface. Since the transducer is monolithically fabricated with the structure under test, it no longer requires external fixtures or additional packaging and can operate *in situ* for long durations under harsh conditions such as thermal cycling and mechanical vibration. This architecture is therefore particularly attractive for continuous online monitoring and structural health monitoring of high-temperature components, provided that the film–substrate interface, electrodes, and electrical interconnections remain stable during prolonged thermal exposure and cycling.[213,214]

The film deposition process also exhibits excellent adaptability to substrate geometries, allowing conformal coverage over regions that are difficult for conventional rigid probes to contact—such as bolt heads, pipe walls, and complex curved surfaces—thus facilitating multipoint distributed or even arrayed deployment at critical locations. Moreover, both the film thickness and pattern can be precisely controlled, enabling the resonance frequency and the excited or received wave modes to be designed according to application needs, providing greater flexibility for specific frequency-band thickness measurements, guided-wave inspections, or localized focusing. Finally, such deposition processes are highly compatible with existing surface engineering and micro/nanofabrication technologies, allowing for monolithic integration of electrodes, interconnects, and backend circuitry, which offers strong potential for high-density integration and batch fabrication. However, extending these processes from small, well-controlled substrates to large or complex engineering surfaces remains challenging.[215,216]

Kobayashi *et al.* employed the sol-gel method to *in situ* deposit $Bi_4Ti_3O_{12}$/PZT composite thick films on various metallic substrates, including steel, titanium, nickel, and copper, with planar, cylindrical, and spherical geometries. After corona poling at 200–400 °C and the application of silver-paste top electrodes, thick-film ultrasonic transducers were fabricated without the need for couplants. The results are shown in Fig. 17. On a planar steel substrate, the transducer exhibited a center frequency of ∼3.6 MHz and a bandwidth of 44% at room temperature; when heated to 440 °C, the main echo amplitude decreased by only about 2 dB, and both the echo waveform and bandwidth remained nearly unchanged, indicating that the electrode/ceramic/steel interface remained stable even in the absence of couplants. On curved metallic substrates, the cylindrical surface yielded a center frequency of 30 MHz, a bandwidth of ∼47%, and an SNR of about 26 dB, while the spherical surface achieved a center frequency of 9.5 MHz, a bandwidth of 42%, and an SNR of about 40 dB; in both cases, clear pulse echoes were obtained at 440 °C.[217] These results demonstrate the feasibility of forming effective acoustic apertures on curved surfaces and maintaining clear ultrasonic responses under the tested high-temperature conditions, highlighting the high-temperature potential and conformability of directly deposited thick-film transducers.

Despite these advantages, directly deposited thin- and thick-film transducers still face several important fabrication and reliability challenges. Film porosity, crystallographic texture, residual stress, and film–substrate adhesion can strongly affect piezoelectric response, acoustic performance, and device consistency, while prolonged high-temperature exposure may also lead to electrode degradation, substrate oxidation, or interfacial reactions. For ferroelectric films, uniform poling can become difficult on thick, large-area, or curved structures, whereas AlN- and ZnO-based films are more sensitive to deposition-induced texture and defect control. In addition, monolithic integration reduces repairability compared with detachable bulk transducers and scaling the deposition process to large or complex engineering surfaces remains challenging because uniform film thickness, composition, stress, and electrode quality must be maintained. Therefore, the advantages of eliminating conventional coupling and bonding layers must be balanced against fabrication repeatability, interfacial reliability, maintainability, and manufacturing scalability.

Overall, directly deposited piezoelectric films provide a distinctive route for high-temperature ultrasonic sensing by eliminating conventional couplants and bonding layers and enabling conformal, miniaturized, and potentially highly integrated transducer configurations. However, their practical performance and long-term reliability depend strongly on the quality and stability of the film–substrate–electrode system, and the advantages of monolithic integration must be balanced against fabrication repeatability, thermomechanical reliability, maintainability, and manufacturing scalability. Therefore, directly deposited transducers are particularly attractive for applications requiring compact integration, complex-surface conformity, or permanent sensing, provided that the deposition process and interfacial reliability can be adequately controlled.

B. Nondirect contact type

Another type of ultrasonic transducer does not directly contact the test medium but is instead isolated from the extreme environment by a thick solid protective element. In addition to serving as a mechanical barrier, this element can significantly reduce the operating temperature of the piezoelectric component. Such protective elements

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are generally classified into two categories: delay lines and waveguides. The primary distinction between them lies in the relationship between their lateral characteristic dimension and the acoustic wavelength. A delay line has a lateral dimension much larger than the wavelength, approximating bulk-wave propagation within a large cross section, whereas a waveguide has a lateral dimension comparable to the wavelength, where guided-wave or modal effects must be considered. A delay line or waveguide can be employed as an isolation layer such that, with appropriate material selection and structural design, it simultaneously provides thermal insulation and preserves efficient signal transmission. In this way, conventional transducers designed for room-temperature operation can be extended for use in high-temperature environments.

1. Material selection

From the perspective of material selection, delay lines and waveguides are not fundamentally different; the specific choice is primarily determined by the intended application scenario and the required operating temperature range. Table V summarizes commonly used buffer rod materials along with their key acoustic and thermal parameters. It should be noted that these parameters are strongly temperature-dependent; the values listed in the table represent nominal room-temperature data, and appropriate temperature-corrected values should be used in practical design.

Under operating conditions not exceeding ~300 °C, high-temperature polymers such as polybenzimidazole (PBI) and polyimide (PI) can be employed as delay line materials. Both exhibit excellent thermal insulation properties and can effectively reduce the temperature of the piezoelectric element even without active cooling. However, their acoustic properties are relatively sensitive to temperature. For in-line measurement of polymer melts, experimental and modeling studies by Abu-Zahra and Rekuvienė *et al.* using solid PI delay lines consistently demonstrated that, as the melt temperature increases, variations in the acoustic impedance and reflection coefficient at the melt–PI interface occur, accompanied by an increase in

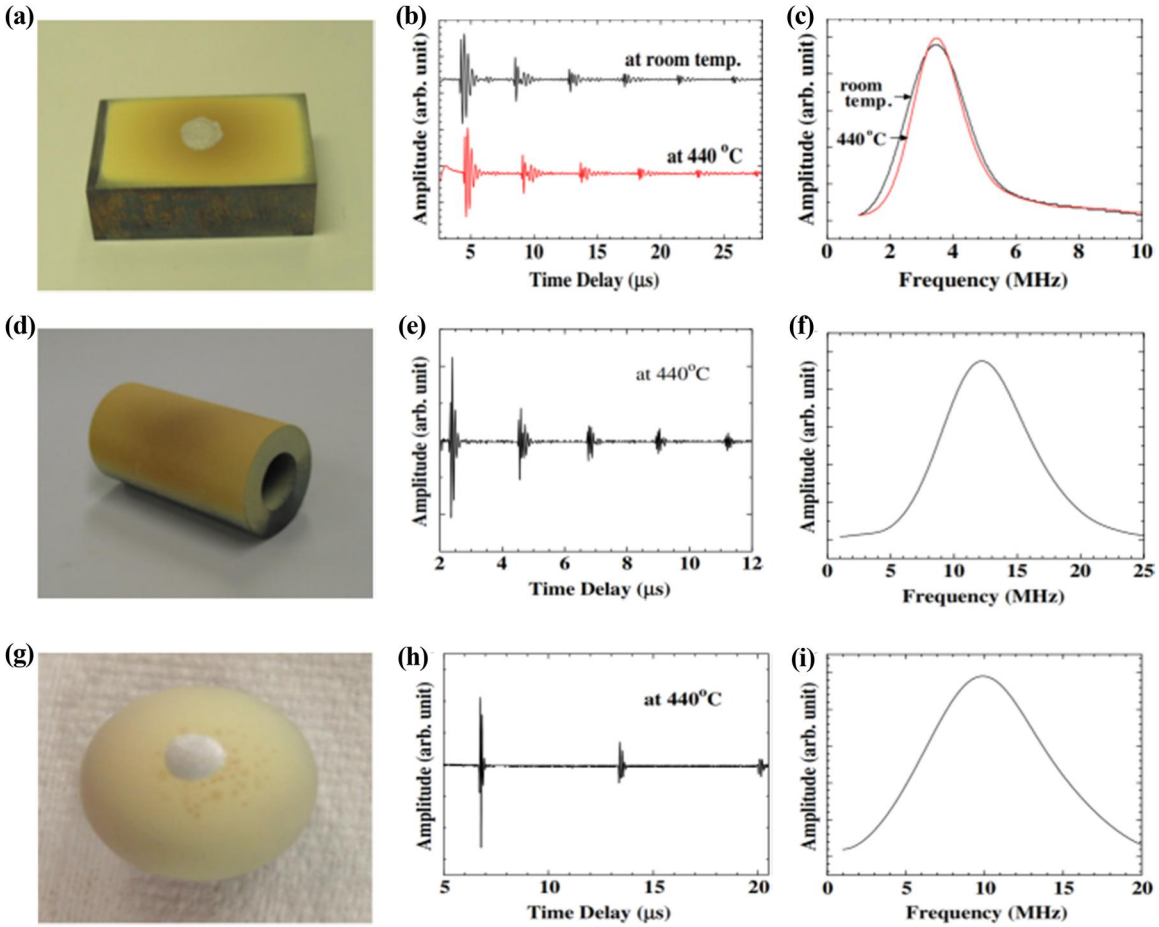


FIG. 17. (a) A 200 μm thick film deposited onto a planar steel substrate and its ultrasonic performance (b) in the time and (c) in the frequency domain at room temperature and 440 °C. (d) A 40 μm thick BIT/PZT film deposited onto a cylindrical steel shell of 25.4 mm outer diameter and its ultrasonic performance (e) in the time and (f) in the frequency domain at 440 °C. (g) A 90 μm thick BIT/PZT film deposited onto a spherical ball and its ultrasonic performance (h) in the time and (i) in the frequency domain at 440 °C. Reproduced with permission from Kobayashi and Jen, Smart Mater. Struct. **13**, 951–956 (2004). Copyright 2004 IOP Publishing Ltd.[217]

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TABLE V. Representative buffer-rod materials and their properties.[54]

Material	Maximum application temperature (°C)	Thermal conductivity ($W\,°C^{-1}\,m^{-1}$)	Longitudinal-wave velocity (m/s)	Acoustic impedance (MRayl)
Polyimide (PI)	300	0.22	2465.2	3.29
Polybenzimidazole (PBI)	343	0.40	2985.3	3.85
Glass ceramic ZERODUR	850	1.63	...	...
Aluminum (Al)	500	205	6320	16.9
Stainless steel	1200	18–20	6100	44–48
Titanium	1200	15–27	5740	27–29

effective attenuation within the delay line. These effects result in systematic shifts in echo amplitude and time-of-flight, manifested as a decrease in SNR, a narrowing of the effective bandwidth, and intensified trailing echoes.[218,219] In comparison, PBI is more commonly used as a high-temperature wedge or short buffer element. Santos *et al.* reported that it maintained stable operation during in-service pipeline inspection at around 200 °C, although additional attenuation and interfacial degradation were still observed at elevated temperatures.[220,221] Overall, PBI/PI-based designs offer excellent thermal management capabilities but are limited by their temperature sensitivity; therefore, calibration and compensation strategies should be implemented according to the target temperature range and interfacial conditions.

Glass–ceramic materials can operate at significantly higher temperatures; among them, ZERODUR K20, manufactured by Schott Advanced Optics (Germany), can be used at temperatures up to ~850 °C. As an example, a buffer rod made of this material with a length of 130 mm was tested by heating one end to 350 °C while the entire rod was placed in an ambient-temperature environment. The measured temperature distribution along the rod, as shown in Fig. 18(a), indicates that even with such a short length, the glass–ceramic buffer rod can reduce the temperature at the bonded interface with the ultrasonic transducer to about 34 °C, thereby enabling the use of conventional transducers for both excitation and reception.[54] It should be noted that ZERODUR K20 exhibits an extremely low coefficient of thermal expansion (approximately $2.4 \times 10^{-6}\,°C^{-1}$), which may cause thermal expansion mismatch with piezoelectric crystals or metallic structures. Therefore, interfacial stresses should be carefully evaluated during design, and appropriate buffering or transition layers should be incorporated to mitigate these effects. In addition, buffer rods can also be made of metals such as steel and titanium. The most commonly used material is steel, owing to its excellent mechanical strength and high operating temperature capability of up to 1200 °C. For moderately high-temperature environments not exceeding 500 °C, aluminum rods can also be employed. Transducers equipped with such rods have been successfully applied in process monitoring applications, including those in the food and polymer industries.[222,223]

2. Structural design

Delay lines and waveguides have significant differences in structural design. The delay line is positioned between the transducer and the high-temperature medium, serving primarily for thermal isolation to reduce the temperature of the piezoelectric element, and in some cases, to introduce a controlled echo delay.[224,225] Its lateral dimension is significantly larger than the operating wavelength, and it is typically fabricated as a cylindrical or rectangular buffer rod in which longitudinal wave propagation dominates. In long buffer rods, pulse echoes are prone to tailing due to mode conversion, cavity resonance, multiple reflections, and diffraction spreading, which can cause overlap with the primary echo from the target interface and significantly reduce SNR. Previous studies have shown that the intensity and time delay of trailing waves are highly sensitive to the cross-sectional shape of the buffer rod and the geometry of the transducer. Ono *et al.* systematically compared combinations of circular and rectangular buffer rods with circular and square piezoelectric transducers under 10 MHz excitation [Figs. 18(b) and 18(c)], and, based on waveguide theory, derived quantitative relationships between the trailing waves behavior and the cross-sectional dimensions. By appropriately selecting the cross-sectional size to prevent overlap between the trailing and target

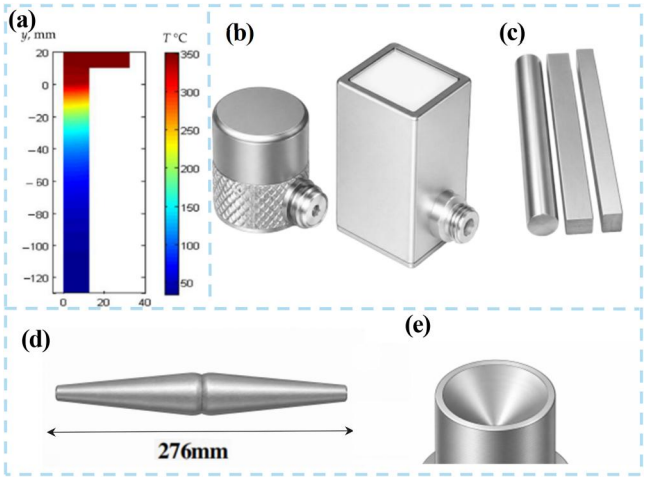


FIG. 18. (a) Temperature distribution along the buffer rod made of glass ceramic ZERODUR. Reproduced with permission from Kazys and Vaskeliene, Sensors **21**, 3200 (2021). Copyright 2021 Authors, licensed under a Creative Commons Attribution (CC BY) license.[54] (b) 10 MHz longitudinal wave ultrasonic transducers with circular and square shapes of active areas. (c) Al rods having circular, square, and rectangular cross section shapes without cladding and with SS cladding.[226] (d) External view of the double-conical coated buffer rod and (e) the spherical acoustic lens fabricated at the rod tip.[227] Panels (b)–(e) were newly drawn by the authors based on the structural concepts reported in Refs. 226 and 227.

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echoes, they achieved a marked improvement in SNR. Experimental results demonstrated that a circular buffer rod excited by a rectangular-aperture transducer yielded the best performance, achieving an SNR of ∼36 dB.[226] The team also proposed a coated buffer rod design consisting of a low-carbon steel core covered with a stainless-steel cladding applied by thermal spraying. The rod body was further modified from a conventional cylindrical shape to a double-conical form [Fig. 18(d)] to suppress trailing waves induced by multimode propagation and end-face reflections. In ultrasonic measurement and imaging experiments conducted in molten aluminum, a 276 mm-long buffer rod was employed. To satisfy the focusing requirements for imaging, the immersed end was machined into a hemispherical tip to generate a convergent acoustic beam [Fig. 18(e)]. Under testing conditions at 780 °C, high-quality imaging of engraved characters on a steel plate was successfully achieved.[227] These findings provide a direct guideline for engineering design: by optimizing the cross-sectional shape, dimension, and geometry of the buffer rod and transducer aperture geometry at a given operating frequency, trailing echo interference can be effectively suppressed and the signal-to-noise ratio enhanced.

When the temperature of the test object is significantly higher than the ambient temperature, a delay line alone may be insufficient to achieve adequate temperature reduction; in such cases, a long waveguide can be employed to provide effective thermal isolation. The underlying mechanism relies on heat conduction along the waveguide combined with convective and radiative heat dissipation, which together establish a pronounced temperature gradient along its length. The steady-state temperature gradient depends mainly on the thermal conductivity, waveguide geometry, and convective and radiative boundary conditions, whereas the specific heat primarily influences the transient thermal response. The commonly used waveguide configurations can be broadly categorized into two types: strip waveguides with rectangular cross sections and rod-type waveguides with axisymmetric circular cross sections. Cegla *et al.* investigated the temperature distribution in SS304 waveguides with various cross-sectional geometries and dimensions, and their results indicate that both a thin circular wire with a radius of 0.5 mm and a rectangular steel strip with a cross section of $1 \times 15\ \text{mm}^2$ can provide highly effective thermal isolation. In both cases, when one end of the waveguide was heated to 600 °C, the temperature measured along the waveguide had already decreased to ∼26 °C at a distance of about 0.5 m from the hot end. The corresponding temperature profiles were reported by Cegla *et al.*[228] It should be noted, however, that the small lateral dimensions of the waveguide can easily excite multiple guided-wave modes during propagation, some of which exhibit strong dispersion, potentially leading to waveform distortion and uncertainty in echo interpretation. The findings of Cegla *et al.* demonstrate that a nondispersive shear horizontal (SH) wave can be excited in rectangular waveguides with a large width-to-thickness ratio.[229] The key advantage of this mode is that ultrasonic pulses propagating within such waveguides experience negligible dispersion, thereby avoiding waveform distortion. The dispersion characteristics and pulse-propagation behavior of rectangular waveguides have been extensively analyzed in the relevant literature.[230]

Overall, delay lines and waveguides provide an effective thermal-isolation strategy for HTPUTs by separating the piezoelectric element from the hottest region. However, their design must balance heat insulation, acoustic attenuation, mode conversion, signal distortion, oxidation resistance, and mechanical robustness. Therefore, the selection of waveguide material and geometry should be determined by the target temperature, propagation distance, signal amplitude requirement, and allowable waveform distortion.

C. Architecture selection and performance tradeoffs

The selection among direct-contact, delay-line, and waveguide architectures involves coupled tradeoffs in operating-temperature capability, acoustic transmission, usable bandwidth, coupling stability, geometric flexibility, and suitability for continuous monitoring. Direct-contact configurations provide the shortest acoustic path and therefore generally minimize additional propagation loss and preserve pulse characteristics more effectively. Their compact structure is advantageous for localized and permanently installed monitoring, but the achievable operating temperature and long-term stability are directly constrained by the thermal durability of the piezoelectric element, electrodes, matching and backing layers, bonding interfaces, and packaging. Consequently, degradation of coupling, oxidation, CTE-induced cracking or delamination, and electrical interconnection failure can become dominant limitations during prolonged high-temperature exposure.

Delay-line and waveguide configurations reduce the thermal load on the active transducer by spatially separating it from the hot region, but this thermal isolation is achieved at the expense of a longer and more complex acoustic propagation path. Delay lines, whose transverse dimensions are typically much larger than the acoustic wavelength, predominantly support bulk-wave propagation and therefore provide relatively straightforward pulse identification and signal interpretation. However, attenuation, diffraction, multiple reflections, and trailing-wave interference may reduce signal amplitude and modify the practically usable bandwidth, while their relatively large cross section also increases heat conduction and limits geometric flexibility. Waveguides, in contrast, use smaller transverse dimensions and can provide stronger thermal isolation together with greater flexibility for long-distance, bent, or spatially distributed sensing, making them particularly attractive for remote and continuous monitoring. Their guided-wave nature, however, introduces dispersion, multimode propagation, mode conversion, waveform distortion, and additional calibration requirements. Thus, direct-contact architectures are generally favored when compactness and acoustic fidelity dominate; delay lines provide an intermediate solution when thermal isolation is required while retaining predominantly bulk-wave propagation; and waveguides become preferable when stronger thermal isolation, remote access, or geometric flexibility outweigh the increased complexity of guided-wave propagation.

D. Long-term reliability under thermal exposure and cycling

Long-term reliability of HTPUTs cannot be evaluated solely according to the highest demonstrated operating temperature. A more meaningful assessment should jointly consider the operating temperature, exposure duration, thermal-cycling history, performance retention, and the dominant degradation or failure mode. Short-term operation at an extreme temperature does not necessarily imply reliable continuous operation, whereas prolonged thermal exposure or

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repeated heating–cooling cycles may progressively induce oxidation, diffusion, creep, interfacial degradation, residual-stress accumulation, and packaging failure. Therefore, the reliability of an HTPUT should be evaluated within a temperature–time–cycle framework rather than using a single maximum-temperature value.

However, the available device-level reliability evidence remains highly heterogeneous. Many studies primarily report the highest temperature at which an ultrasonic response can still be obtained, while exposure duration, thermal-cycle count, post-cooling recovery, and quantitative performance retention are not always systematically documented. Moreover, the reported reliability indicators differ considerably among studies, including echo amplitude, SNR, center frequency, bandwidth, and qualitative functional survival. Consequently, the available results should not be interpreted as directly comparable lifetime values. Table VI summarizes representative HTPUTs and key transducer components for which explicit information on thermal exposure, cycling, performance retention, or degradation is available.

As summarized in Table VI, the available studies provide different levels of reliability evidence. Several transducer systems have maintained detectable ultrasonic responses at temperatures as high as 800–1000 °C, but the corresponding dwell time or number of thermal cycles is not always reported. In contrast, long-duration tests at lower temperatures provide more direct evidence of continuous operating stability. For example, the nano-Ag-coupled $LiNbO_3$ transducer maintained clear echoes with only minor changes in Vpp and SNR during durability tests of up to 66 h at 350 °C. In addition, changes in center frequency, bandwidth, or echo amplitude observed during heating may be at least partly reversible when the transducer retains its functionality after cooling. These results further demonstrate that the maximum demonstrated temperature and reliable continuous operating temperature represent different aspects of high-temperature capability.

More importantly, the available evidence indicates that long-term device reliability is often limited by interfaces and auxiliary components before the intrinsic thermal limit of the piezoelectric element is reached. Bonding-layer degradation, oxidation, CTE mismatch, interfacial stress evolution, electrode or interconnect instability, and packaging failure can progressively disrupt acoustic or electrical transmission even when the piezoelectric element itself remains functional. Prolonged thermal exposure and repeated thermal cycling can also promote different degradation pathways: extended dwell tends to enhance thermally activated aging, diffusion, oxidation, and creep, whereas repeated cycling more strongly promotes stress accumulation, interfacial fatigue, cracking, and delamination. Future HTPUT reliability studies should therefore report not only the maximum test temperature, but also exposure duration, heating and cooling conditions, number and amplitude of thermal cycles, retention of key

TABLE VI. Representative reliability evidence for HTPUTs and key transducer components under thermal exposure and cycling.

System/architecture	Temperature condition	Exposure duration/ thermal cycles	Performance retention/ drift	Dominant degradation/limiting factor
AlN/SiC bonded with Cotronics 989[199]	1000 °C	8 h; cycles not reported	Bond remained effective during high-temperature exposure; subsequent performance degradation occurred	Degradation mainly attributed to oxidation of AlN rather than adhesive failure
$LiNbO_3$ transducer brazed with TiBraze Al-665[175]	27–800 °C	Thermal excursion; dwell time and cycle count not reported	f_c decreased from ∼3.2 to 2.7 MHz; −6 dB BW remained ∼70%–75% at 800 °C; device remained functional after cooling	Temperature-dependent coupling and interfacial behavior; no permanent failure reported
$LiNbO_3$ transducer brazed with 72Ag–28Cu[205]	RT–800 °C	Thermal excursion; dwell time and cycle count not reported	Center frequency and BW showed weak temperature dependence; SNR remained ∼20 dB with good repeatability	No major failure reported; long-duration reliability not established
$LiNbO_3$/nano-Ag/steel transducer[211]	350 °C	30 and 66 h; cycles not reported	Clear primary/secondary echoes retained with minimal changes in V_{pp} and SNR	Grain growth and increased porosity of the Ag network identified as potential degradation mechanisms
Directly deposited $Bi_4Ti_3O_{12}$/PZT thick-film transducer[217]	440 °C	Duration and cycles not reported	Main echo amplitude decreased by only ∼2 dB; waveform and BW remained nearly unchanged	Long-term film/substrate adhesion, electrode stability, and cycling reliability not established

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electrical and acoustic performance parameters, post-cooling recovery, and the identified failure mode.

V. HIGH-TEMPERATURE APPLICATIONS FOR PIEZOELECTRIC ULTRASONIC TRANSDUCERS

When evaluating the reported applications, it is also important to distinguish their level of technological validation. In this review, the reported studies are broadly discussed as (i) laboratory proof-of-concept, where the sensing principle or device feasibility is demonstrated under controlled conditions; (ii) high-temperature laboratory validation, where the device is experimentally operated under representative elevated-temperature conditions; (iii) application-oriented prototype validation, where an integrated transducer or monitoring system is tested on representative equipment or processes; and (iv) long-term industrial deployment, which requires sustained operation under actual service conditions with demonstrated reliability, calibration stability, and maintainability. These levels should not be regarded as formal TRL assignments, but as a qualitative framework for distinguishing the maturity of the reported HTPUT applications.

A. Pipeline wall thickness measurement

Many industrial pipelines and pressure vessels experience corrosion or erosion during long-term operation, resulting in wall thinning, localized loss, and even cracks or leaks that threaten structural integrity.[231,232] Conventional NDT methods (e.g., eddy current and flux leakage) are usually performed only during periodic shutdowns, so they cannot enable real-time monitoring.[233] Although some intrusive long-term techniques place sensors directly in the corrosive environment to estimate average material loss, they cannot capture short-term corrosion-rate fluctuations driven by process variations. Therefore, corrosion should be treated as a continuous process requiring online monitoring. In this context, ultrasonic thickness measurement offers a fast, accurate, and nonintrusive approach to estimate wall thinning rates in real time for continuous monitoring. Ultrasonic thickness measurement is typically performed by measuring the time-of-flight between consecutive echoes in the time domain. Using the known ultrasonic wave velocity and the echo time, the thickness of the specimen can be calculated [as shown in Fig. 19(a)].[235] For long-term corrosion monitoring, wall-thickness changes of interest may be on the order of tens of micrometers, requiring measurement uncertainty at the micrometer-to-tens-of-micrometers level.

Conventional buffer-rod systems [with schematic and photograph shown in Figs. 19(b) and 19(c)] have been widely used for high-temperature thickness measurements. An alternative approach is to replace the buffer rod with an ultrasonic waveguide. Kim *et al.* designed and fabricated a pair of shear-horizontal (SH) transducers together with long strip-type waveguides. The SH vibration mode was selected because SH waves exhibit minimal dispersion, enabling the acquisition of ultrasonic signals with well-preserved waveform fidelity.[229] The SH transducer was mounted on one end of the waveguide strip, while a precision clamping device was designed and machined on the opposite side of the strip to ensure reliable gripping of the specimen [illustrated in Figs. 19(d) and 19(e)]. An ultrasonic waveguide system was constructed using two parallel waveguide strips, each connected to one transducer—one serving as the transmitter and the other as the receiver. The two strips were spaced ∼1 mm apart to suppress noise originating from waves propagating along the specimen surface and reaching the receiver. Compared with traditional pulse–echo techniques, this configuration receives only the signal that has propagated through the test pipe, effectively eliminating unwanted reflections from the waveguide ends. As a result, the measurement SNR is substantially improved.[236]

As shown in Fig. 19(f), the peak positions of the first and second back-wall echoes are automatically identified, and the corresponding arrival times are denoted as t_1 and t_2, from which the pipe-wall thickness is calculated. Because ultrasonic velocity varies with temperature, high-temperature monitoring requires prior calibration of the shear-wave velocity–temperature relationship; the measured dependence of shear-mode wave speed on temperature for carbon-steel pipes is presented in Fig. 19(g). After this calibration, the waveguide system was applied to long-term online monitoring of wall thinning under FAC conditions, with the fluid temperature maintained at 150 °C. As shown in Fig. 19(h), the blue trace represents real-time thickness data acquired at 1 h intervals, while the slope of the red segments indicates the FAC rate in each stage. As expected, the thinning rate increased as the flow velocity rose from 7 to 12 m/s. Figure 19(i) presents a further comparison with a commercial buffer-rod system: the blue trace shows the thinning results obtained using the buffer-rod device, which produced an anomalous increase in thinning rate after the flow velocity decreased from 10 to 7 m/s; in contrast, the red trace shows the measurements from the waveguide system, where the wall-thinning rate increases monotonically with increasing flow velocity, yielding a more physically consistent trend.[237] Overall, the waveguide system demonstrated stable long-term operation at 150 °C, with a thickness-measurement uncertainty of approximately ±20 μm.

Overall, high-temperature pipeline wall-thickness monitoring places particular emphasis on stable time-of-flight extraction, sound-velocity temperature compensation, and long-term acoustic coupling reliability. For moderate-temperature pipelines, conventional PZT or BSPT-based transducers can still be used when properly isolated or cooled, whereas waveguide and buffer-rod structures are more suitable for long-term monitoring under higher-temperature or thermally cycling conditions. The main remaining challenges are coupling degradation, interface instability, and measurement uncertainty caused by temperature-dependent sound velocity.

B. Injection molding process monitoring

Injection molding is a precise and cost-effective method for the mass production of plastic components, and it is estimated that more than one-third of all plastic products worldwide are manufactured using this process. In fields such as medical devices, where extremely tight dimensional tolerances are required, accurate real-time control of the injection-molding cycle is particularly critical.[238,239] Commercial injection-molding systems typically employ temperature and pressure sensors installed at various locations of the molding machine; however, these sensors provide only limited process information and are often insufficient to effectively suppress common defects such as warpage and shrinkage. Moreover, they rely on intrusive measurements that require drilling into the mold, thereby compromising mold integrity.[240,241] In contrast, ultrasonic transducers offer a nonintrusive and nondestructive means of monitoring, capable of capturing rich information related to the mold structure, cavity conditions, and the state and morphology of the polymer melt—factors that directly determine key quality attributes such as shrinkage

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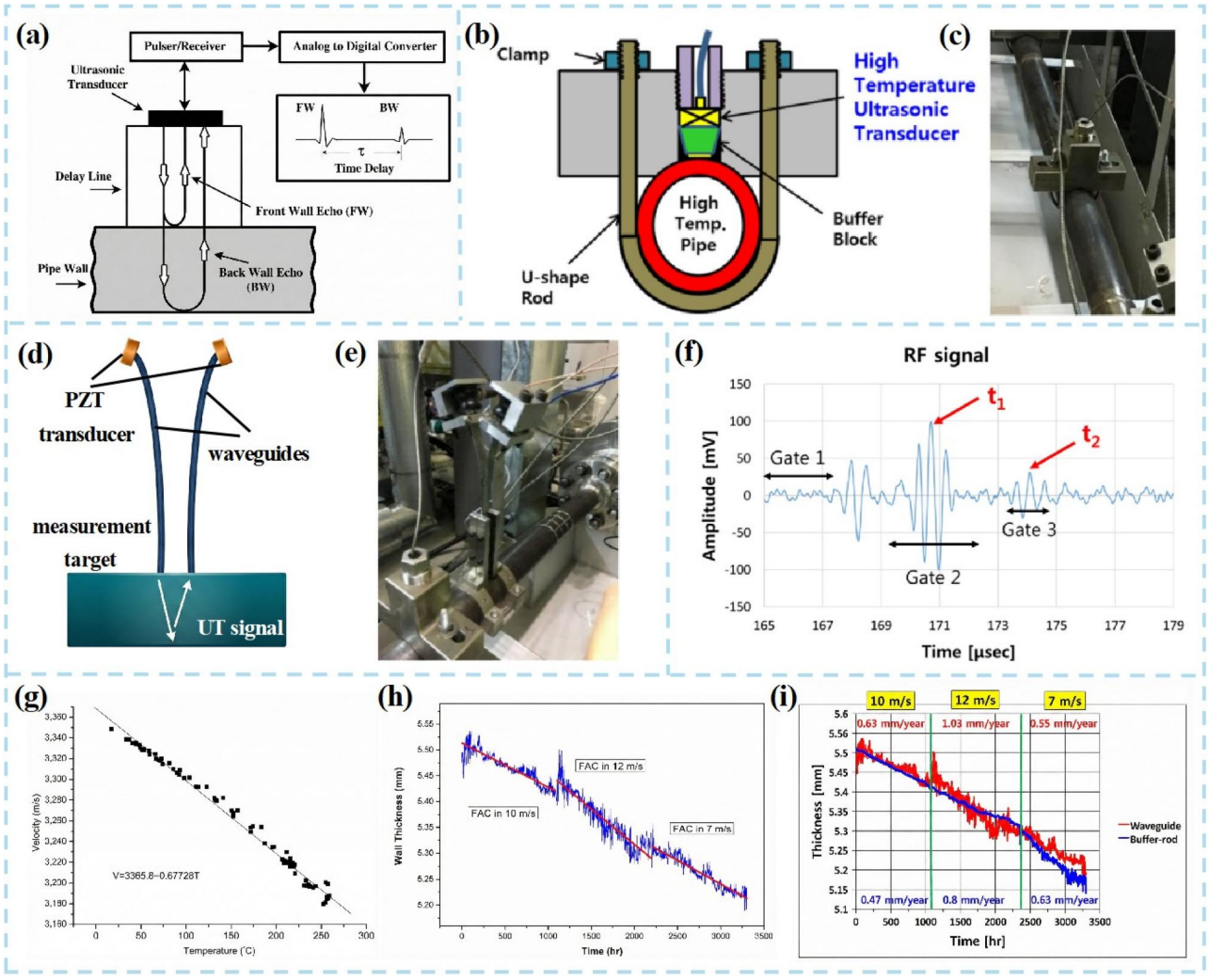


FIG. 19. (a) Working principle of a typical ultrasonic thickness gauge. Reproduced with permission from Honarvar *et al.*, Ultrasonics **53**, 1251–1258 (2013). Copyright 2013 Elsevier.[235] (b) An assembly drawing diagram of a high-temperature pipe using a buffer-rod type measurement system; (c) an installed buffer-rod type system for thickness monitoring on a pipe. (d) Conceptual diagram of a pair of waveguide strips for high-temperature thickness monitoring; (e) installed waveguide system for thickness monitoring on a pipe. (f) Typical ultrasonic rf signals from a developed waveguide with the pitch/catch method at 150 °C. (g) Calibration of the shear horizontal wave velocity with varying temperatures for a carbon steel pipe SA 106. (h) Monitoring of carbon steel pipe wall thickness under different flow velocities. (i) The pipe thickness measurement data using the buffer-rod and waveguide type systems. (b)–(i) are reproduced with permission from Oh *et al.*, Sensors **19**, 1762 (2019). Copyright 2019 Authors, licensed under a Creative Commons Attribution (CC BY) license.[237]

and warpage. Figure 20(a) summarizes typical configurations in which ultrasonic probes are integrated into different parts of an injection-molding machine (e.g., mold inserts, barrel, nozzle, and tie bars) and illustrates their corresponding application scenarios.[242]

Compared with conventional ultrasonic probes, film transducers offer notable advantages such as smaller form factor, superior high-temperature tolerance, and adaptability to curved surfaces such as pipes, enabling installation in regions that are inaccessible to traditional probes.[243,244] Kobayashi *et al.* were the first to employ film-based ultrasonic transducers for real-time monitoring of the injection-molding process. They fabricated seven ultrasonic transducers on the barrel; six sensors were located on the feeding and heating zones, and they also placed a sensor between them [Fig. 20(b)]. Two sensors were also fabricated on the mold insert of the machine, as shown in Fig. 20(c). They successfully measured the ultrasonic signals at the barrel and the velocity of the polymer melt in the cavity during solidification for the polyethylene molded part.[245]

Ono *et al.* arranged seven sol-gel–derived transducers on the barrel and two additional transducers on the moving-mold insert [Fig. 20(d)]. The nth multiple reflection from the cavity surface is denoted as L_{nm}, whereas echoes that propagate through the polymer melt and are reflected by the fixed mold insert are denoted as L_{2m}.

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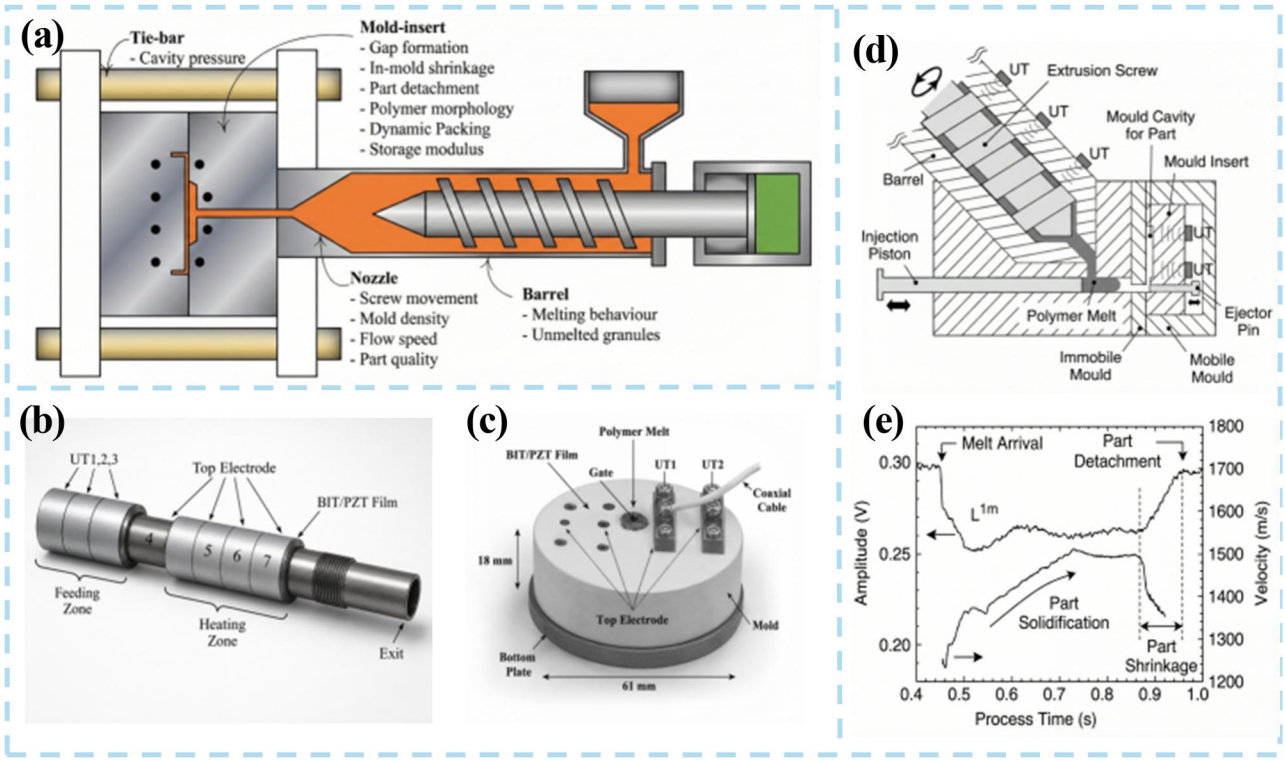


FIG. 20. (a) Monitored parameters using ultrasonic sensors at different locations in an injection-molding system. Reproduced with permission from Kariminejad *et al.*, Sensors **21**, 5193 (2021). Copyright 2021 Authors, licensed under a Creative Commons Attribution (CC BY) license.[242] Sol–gel ultrasonic sensors used for micromolding: (b) barrel-mounted sensor and (c) mold-inset sensor. Panels (b) and (c) were newly drawn by the authors based on the structural concepts reported by Kobayashi *et al.*, in IEEE Symposium on Ultrasonics (IEEE, 2003), Vol. 2, pp. 693–698. Copyright 2003 IEEE.[245] (d) Sol–gel ultrasonic transducers at the nozzle and mold inset; (e) ultrasonic velocity and echo amplitude as functions of process time. Panels (d) and (e) are reproduced with permission from Ono *et al.*, Trans. Inst. Meas. Control **29**, 383–401 (2007). Copyright 2007 SAGE Publications.[241]

Figure 20(e) shows the evolution of sound speed and the amplitude of the first reflected echo (L_{1M}) obtained from the first transducer on the insert during processing: within less than half a second, the polymer fills the cavity, allowing part of the ultrasonic energy to pass through the melt and reach the fixed mold; during cooling, the sound speed rises to ∼1500 m/s, indicating progressive solidification of the material; after solidification, the amplitude of L_1 gradually increases, signifying that the molded part begins to detach from the cavity wall due to shrinkage; subsequently, the decrease in sound speed suggests the formation of an air gap between the cavity surface and the molded part.[241]

Wu *et al.* modified the conventional extended nozzle of an injection-molding machine into a T-shaped nozzle to provide space for integrating a film-type ultrasonic sensor. Figure 21(a) shows the extended nozzle equipped with the ultrasonic transducer, which is capable of withstanding temperatures up to 350 °C and pressures up to 300 bar.[246] Thermocouples were simultaneously mounted on the nozzle surface for comparison measurements. The results indicate that thermocouples lack sufficient sensitivity to effectively monitor the detailed evolution of the injection-molding process. In contrast, the ultrasonic sensor mounted on the nozzle can clearly distinguish the various molding stages [Fig. 21(b)], including mold closing, injection, packing, solidification and ejection, screw movement, and mold opening. These stages can be identified from the characteristic variations in ultrasonic wave speed and echo amplitude as a function of processing time.[247]

Cheng *et al.* further conducted a systematic comparison of how different injection speeds affect PP (polypropylene) and ABS (acrylonitrile butadiene styrene), based on the aforementioned T-shaped extended nozzle. The experimental setup is shown in Fig. 21(c). Figures 21(d) and 21(e) present the evolution of ultrasonic wave velocity as a function of processing time under two conditions: high injection speed (#1) and low injection speed (#2). The results indicate that the ultrasonic velocity in ABS is overall higher than that in PP; moreover, the magnitude of velocity variation in PP is larger than that in ABS under different injection speeds. Tensile testing further reveals that for the ductile material PP, specimens molded at higher injection speeds no longer fracture as they do under low-speed conditions. In contrast, for the brittle material ABS, specimens molded at higher injection speeds exhibit a greater deflection distance [Fig. 21(f)]. These findings demonstrate that the ultrasonic velocity variations observed under different injection speeds can serve as a reliable indicator of the tensile performance of the molded parts.[248]

Therefore, ultrasonic monitoring of injection molding requires transducers that are compact, fast in response, and compatible with

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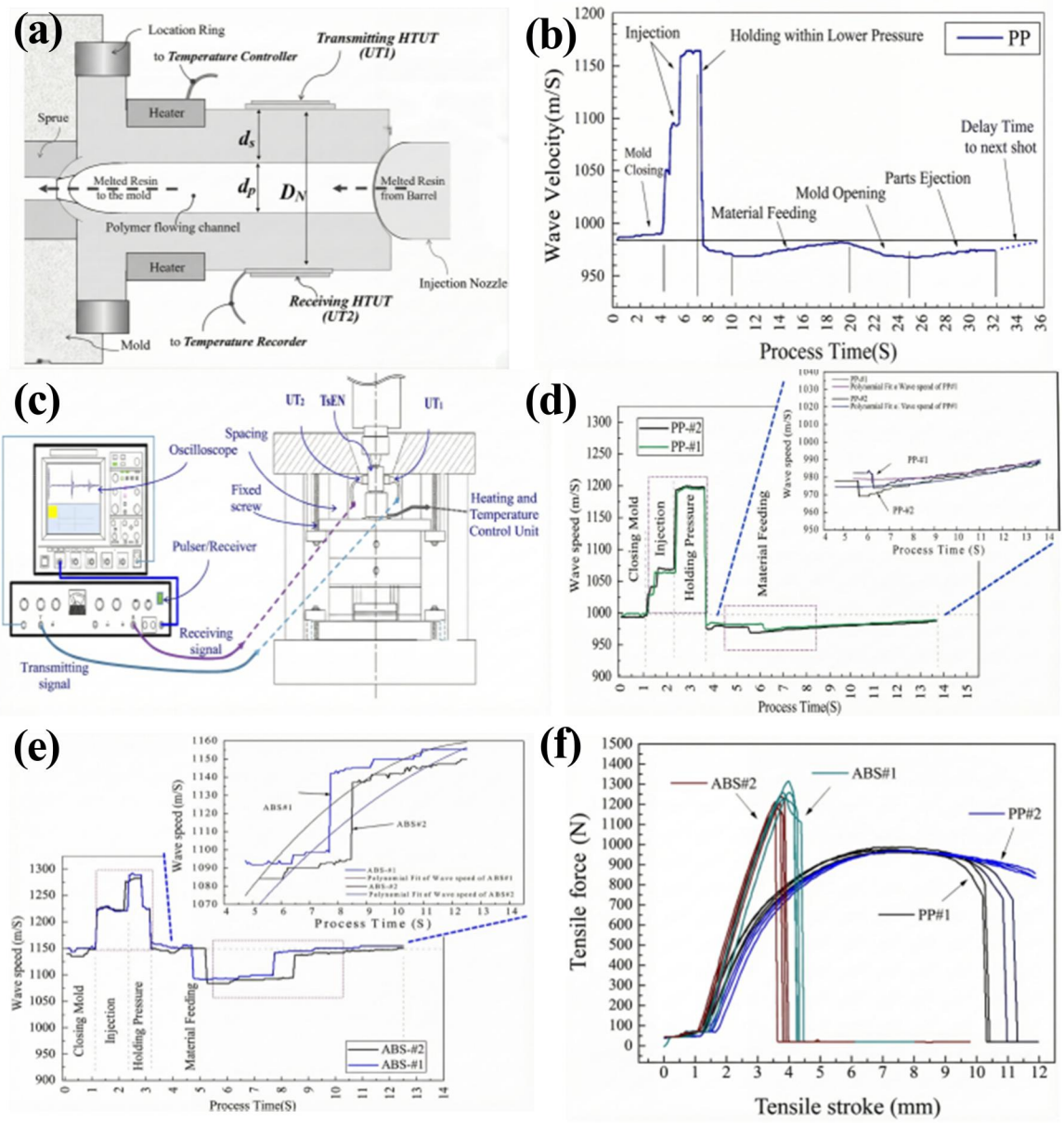


FIG. 21. Fabricated sol-gel ultrasound sensor at the nozzle: (a) cross section of the nozzle with the probes, (b) ultrasonic velocity at the nozzle during injection molding process. Reproduced with permission from Wu *et al.*, Sens. Actuators A **263**, 430–438 (2017). Copyright 2017 Elsevier.[247] Fabricated sol-gel ultrasound sensor at the extension nozzle: (c) the T-shaped extension nozzle and data acquisition system for ultrasound sensor, (d) ultrasonic velocity vs process time for PP #1 and PP #2 with different injection speeds, (e) ultrasonic velocity vs process time for ABS#1 and ABS#2 with different injection speeds, and (f) tensile test for PP and ABS molded parts with different injection speeds. Reproduced with permission from Cheng and Wu, J. Mater. Process. Technol. **282**, 116650 (2020). Copyright 2020 Elsevier.[248]

curved or confined mold/nozzle surfaces. Directly deposited thick-film transducers based on sol–gel PZT/$Bi_4Ti_3O_{12}$, ZnO, or AlN are particularly attractive for this application, while conventional bulk probes may still be used in lower-temperature or externally mounted configurations. The key challenges remain film/substrate adhesion, residual stress, thermal cycling reliability, and calibration under continuously changing melt states.

C. Ultrasonic thermometry

With the continued advancement of aerospace engineering, nuclear energy, and advanced manufacturing, the demand for accurate and stable temperature sensing in extreme environments—such as high-temperature, high pressure, strong oxidation, corrosion, and radiation—has been steadily increasing.[249–251] At present, the most mature technological solutions are thermocouples and optical

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temperature sensors. Thermocouples are favored for their ease of fabrication, cost effectiveness, and low thermal inertia; however, their measurement accuracy is susceptible to thermal drift and material degradation during long-term operation.[252,253] Optical temperature sensors offer fast response, high spatial resolution, and a wide operational temperature range, yet their measurements are easily influenced by background radiation, which limits further application in complex environments.[254,255]

The basic principle of ultrasonic thermometry is illustrated in Fig. 22(a): an ultrasonic transducer is coupled to a high-melting-point acoustic waveguide, and a groove is machined near the distal end of the waveguide. As the ultrasonic pulse propagates along the waveguide, it encounters the groove and the waveguide end face, where local acoustic-impedance mismatches generate two reflected echoes. Because the distance between the groove and the end face is fixed, the difference in arrival times of these two echoes can be used to determine the sound velocity in the waveguide at a given temperature. By exploiting the temperature dependence of sound velocity, the temperature of the environment surrounding the waveguide can be inferred. Since the sound-velocity–temperature relationship is difficult to obtain accurately from theory, practical implementations typically require a pre-calibration procedure using standard temperature sensors to establish a calibrated reference curve describing the variation of sound velocity with temperature.[256]

Traditionally, refractory metals such as tungsten have been used as waveguide materials for ultrasonic thermometry; however, they are highly susceptible to oxidation in oxidative atmospheres, leading to drift in material properties over time and, consequently, measurement

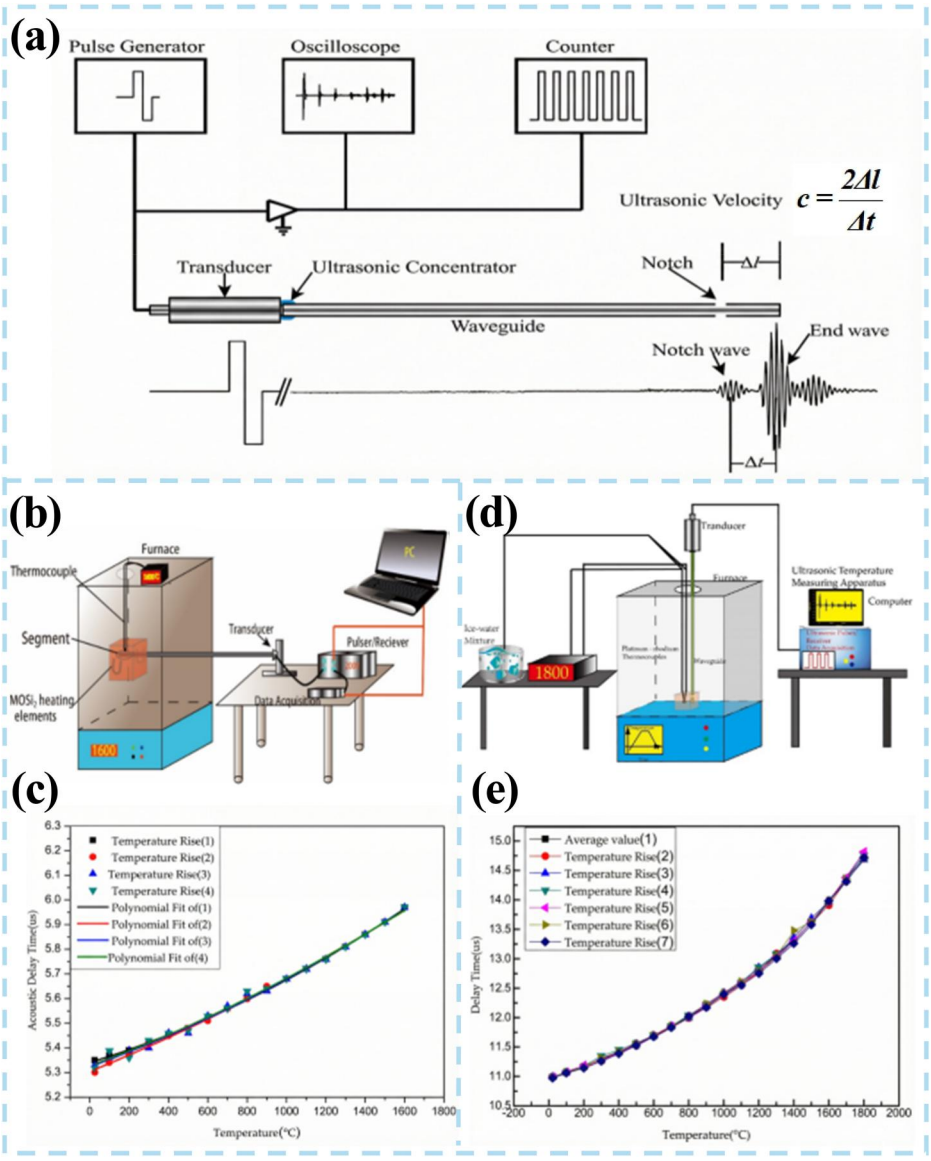


FIG. 22. (a) Schematic illustration of the principle of ultrasonic thermometry. Reproduced with permission from Wei *et al.*, Ultrasonics **113**, 106361 (2021). Copyright 2021 Elsevier.[256] Al_2O_3 ultrasonic temperature measurement sensor: (b) schematic diagram of the experimental setup and (c) measurement results. Reproduced with permission from Wei *et al.*, Sensors **16**, 1905 (2016). Copyright 2016 Authors, licensed under a Creative Commons Attribution (CC BY) license.[257] Iridium–rhodium alloy ultrasonic temperature measurement sensor: (d) schematic diagram of the experimental setup and (e) measurement results. Reproduced with permission from Wei *et al.*, Ultrasonics **113**, 106361 (2021). Copyright 2021 Elsevier.[256]

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failure. To address this issue, Wei *et al.* employed an Al_2O_3 waveguide with a diameter of 1 mm and a length of 600 mm, together with a 2.5 MHz transducer, to construct the pulsed ultrasonic thermometry system shown in Fig. 22(b). Based on this setup, a time-delay–temperature calibration curve was established [as illustrated in Fig. 22(c)]. The results indicate that within the temperature range of 26–1600 °C, the time delay increases almost linearly with temperature; moreover, above 1000 °C, the measurements exhibit high repeatability with a sensitivity of approximately 0.2–0.3 ns/(°C cm).[257] Subsequently, the same group proposed using an Ir/Rh alloy as the waveguide material and assembled a similar ultrasonic thermometry system [as shown in Fig. 22(d)]. Figure 22(e) shows that a temperature calibration curve spanning from room temperature to 1800 °C was established, exhibiting excellent repeatability over the entire range, with a sensitivity of ~0.0035 μs °C^{-1} and a temperature resolution of about 3 °C above 1000 °C. This approach effectively extended the upper temperature limit of ultrasonic thermometry to 1800 °C under oxidizing conditions.[256]

Conventional straight-waveguide thermometry essentially provides an integrated average of the temperature over the propagation path between the notch and the end face. Consequently, when a pronounced vertical temperature gradient exists inside the furnace, it becomes difficult to obtain the true local temperature. To overcome this limitation, Periyannan *et al.* proposed a bent waveguide configuration [as shown in Fig. 23(a)]: the short horizontal segment at the bend can be regarded as residing within an approximately uniform temperature zone. Figure 23(b) shows the echo signals for various bend lengths, in which reflections originating from the bend and from the end face can be clearly distinguished. Figure 23(c) schematically illustrates the arrangement of bent waveguides with different bend lengths inside a furnace with a uniform temperature distribution. Using a K-type thermocouple, the time-delay–temperature relationship was calibrated [Fig. 23(d)], and the results show that the sensitivity increases from 2.1 ns/°C for a bend length of 29 mm to 9 ns/°C for a bend length of 130 mm. Furthermore, Fig. 23(e) shows the deployment of multiple bent waveguides at different heights inside a furnace with a temperature gradient to enable stratified temperature measurements. The corresponding calibration curves in Fig. 23(f) exhibit an overall measurement error of only 2%–3%. The resulting temperature resolutions for bend lengths of 29, 57, and 130 mm are approximately 4.7, 2.1, and 1 °C, respectively.[258]

In ultrasonic thermometry, the upper operating temperature is often determined less by the piezoelectric element itself than by the thermal stability, oxidation resistance, and acoustic repeatability of the waveguide. Since the transducer can usually be placed at the relatively low-temperature end, conventional piezoelectric elements may be used together with high-temperature waveguides such as Al_2O_3 or refractory alloys. The main unresolved issues include oxidation, signal attenuation over long propagation paths, temperature-gradient-induced errors, and long-term calibration drift.

D. Bolt preload monitoring

As early as the 1970s, researchers proposed the use of ultrasonic techniques to measure bolt prestress. The theoretical basis lies in the strong dependence of ultrasonic wave velocity in solids on the material's stress state.[259,260] By monitoring the propagation characteristics of ultrasonic waves in the fastened component, this method enables the estimation of bolt prestress without disassembling the bolt or applying

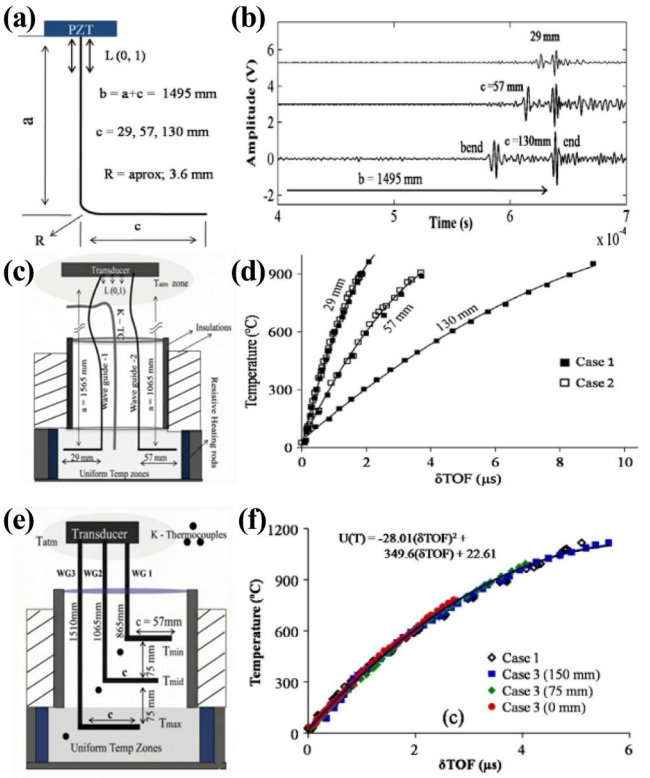


FIG. 23. (a) Single bent waveguide with bend lengths of 29–130 mm; (b) A-scan signals acquired for bend lengths of 29, 57, and 130 mm; (c) experimental setup for two waveguides connected to a single transducer; (d) sensitivity results for cases 1 and 2 under different experimental configurations; (e) multiple-bent-waveguide setup for measurements in a temperature-gradient field; and (f) corresponding calibration curves. Reproduced with permission from Periyannan *et al.*, Ultrasonics **74**, 62–74 (2017). Copyright 2017 Elsevier.[258]

any additional external load, thereby avoiding the disturbance to the bolt's operating condition that is inherent in traditional mechanical measurement methods.[261,262]

Traditionally, two approaches have been used to achieve ultrasonic excitation and reception on bolts: (i) bonding a piezoelectric ceramic plate to the bolt end face, and (ii) acoustically coupling an external ultrasonic probe to the bolt surface via a coupling agent. The former suffers from the risk of debonding, whereas the latter is cumbersome to operate and highly dependent on the quality of the coupling layer.[212] These challenges are further exacerbated in high-temperature environments, where maintaining stable and repeatable acoustic coupling becomes considerably more difficult. In contrast, film-type transducers can be directly deposited onto the bolt surface and provide stable and reliable ultrasonic excitation and signal transmission without the need for an additional coupling layer, making them particularly suitable for high-temperature conditions and long-term online monitoring.[213,214]

Wang *et al.* likewise employed radio frequency magnetron sputtering to deposit an AlN piezoelectric film *in situ* on the end face of a GH4169 nickel-based alloy bolt, thereby constructing an integrated film ultrasonic transducer that requires no coupling agent. The device

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comprises multilayer Mo electrodes (introduced to relieve internal stress and improve film densification) and an AlN piezoelectric layer with a thickness up to 12.90 μm [as illustrated in Fig. 24(a)]. To evaluate its practical performance in preload monitoring, the authors designed and conducted a systematic preload calibration and measurement experiment [schematic shown in Fig. 24(b)]: first, the bolt with the integrated AlN transducer was mounted in a dedicated fixture, and the bottom-echo signal in the stress-free state was recorded, with its arrival time defined as the reference zero. Subsequently, a tensile testing machine was used to apply a series of known axial loads ranging from 0 to 21 kN, during which the system incrementally increased the load and simultaneously recorded the flight-time changes of the echo relative to the zero-load reference. By fitting the measured time-difference data, a calibration curve correlating preload with flight-time variation was established and subsequently used to determine unknown stress states. The experimental results [Fig. 24(c)] demonstrate that the absolute measurement error is below $\pm$0.2 kN and the relative error does not exceed $\pm$2.50%, achieving high-accuracy and high-stability *in situ* monitoring of bolt preload.[263]

Xu *et al.* employed radio frequency magnetron sputtering to directly deposit a ZnO piezoelectric film onto the bolt end face, thereby constructing an integrated film-based ultrasonic transducer. The device consists of a ~13.8 μm ZnO piezoelectric layer, a 2.2 μm SiO_2 protective layer, and Ti electrodes [as illustrated in Fig. 24(d)], and is capable of efficiently exciting and receiving longitudinal ultrasonic echoes at approximately 24 MHz without the need for a coupling agent. Based on this transducer, the authors established a tensile preload measurement system [schematically shown in Fig. 24(e)]: torques of 40–130 N m were applied to the bolt, and the corresponding axial forces were calibrated using a load cell, while the variations in the round trip travel time of longitudinal waves inside the bolt were recorded. The experimental results [Fig. 24(f)] indicate that the echo flight time increases approximately linearly with increasing preload. This trend arises from the combined effects of acoustoelasticity-induced reductions in wave velocity and the elongation of the bolt under axial load, which increases the propagation path length.[264]

For bolt preload and stress monitoring, the central requirement is high-precision and repeatable time-of-flight measurement under coupled thermal and mechanical loading. Directly deposited AlN, ZnO, or $LiNbO_3$ thin-film transducers are suitable because they eliminate the need for an additional couplant and can be integrated on the bolt end face. However, temperature-stress cross-sensitivity, electrode stability, thin-film fatigue, and long-term calibration drift remain key barriers to practical deployment.

E. Density measurement of liquids

Continuous monitoring of high-temperature liquids or molten metals is a critical requirement in fields such as petrochemical processing and nuclear power.[265] Among the relevant thermophysical parameters, density is not only essential for evaluating product quality but also a key factor governing process cost and overall economic efficiency.[266] Conventional density measurements are typically performed offline in laboratory environments using hydrometers, differential-pressure densitometers, and similar instruments. These methods require sampling and therefore cannot capture the real-time

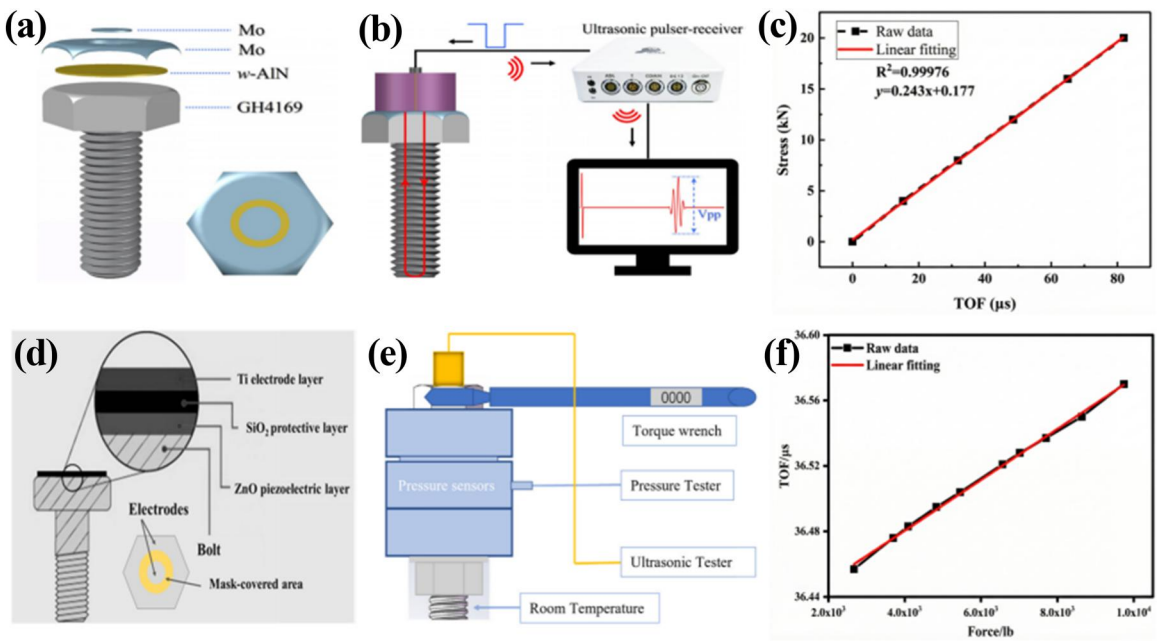


FIG. 24. Schematic of (a) AlN thin film ultrasonic transducer and (b) the experimental device for ultrasonic response measurement. (c) The calibration curve of the AlN film transducer sample. Reproduced with permission from Wang *et al.*, Sensors **24**(17), 5820 (2024). Copyright 2024 Authors, licensed under a Creative Commons Attribution (CC BY) license.[263] The structure diagram of (d) ZnO thin-film ultrasonic transducer and (e) the experimental setup for ultrasonic-response measurement. (f) Preload calibration curve for the ZnO thin-film transducer. Reproduced with permission from Xu *et al.*, Coatings **14**(6), 718 (2024). Copyright 2024 Authors, licensed under a Creative Commons Attribution (CC BY) license.[264]

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state of the process; moreover, if the liquid properties change between sampling and measurement, the resulting data may mislead process control.[267] In contrast, online density measurement aims to obtain real-time density information directly from the process medium without disturbing the main flow. Current technologies suitable for online deployment include near-infrared/spectroscopic analysis, x-ray absorption, electromagnetic techniques, and ultrasonic methods.[268] Among them, ultrasonics offers distinct advantages—high accuracy, strong reliability, and comparatively low cost—making it particularly promising for real-time density monitoring of liquids.[269]

The fundamental principle of ultrasonic density measurement is to determine the acoustic impedance of the target liquid from the ultrasonic reflection coefficient at the solid–liquid interface and, together with the sound speed in the liquid, calculate its density. A schematic of the measurement principle is shown in Fig. 25(a). Specifically, the ultrasonic measurement system consists of transducer 1 and transducer 2, two acoustic matching layers with acoustic impedance Z_2, and the liquid under test with acoustic impedance Z_3. Transducer 1 generates an ultrasonic pulse that propagates sequentially through waveguide 1 and matching layer 1 before reaching the liquid. A portion of the acoustic energy is reflected at the waveguide–liquid interface, and the reflection coefficient is directly related to Z_3. Meanwhile, the portion of the pulse transmitted into the liquid propagates through the medium and is received by transducer 2; the sound speed c_3 in the liquid is obtained from the time of flight between transmission and reception. Finally, the density ρ_3 of the liquid is calculated from the relationship between acoustic impedance and sound speed.[270]

Kazys *et al.* designed an ultrasonic transducer based on a steel waveguide for online density measurement of fluids inside high-pressure pipelines, with the overall structure shown in Fig. 25(c).[270] The waveguide is threaded into the pipeline wall—through which the test fluid flows—using a standard 1/2–20 UNF-2A port. The authors performed finite element simulations to evaluate the temperature distribution along the stainless-steel waveguide. As shown in Fig. 25(b), when the pipeline wall is at 200 °C, a waveguide length of approximately 110 mm is sufficient to reduce the temperature at the transducer end to near ambient conditions. A corresponding experimental setup was constructed based on this design, as illustrated in Fig. 25(d). Using this setup, the ratio of the reflected signal amplitudes before

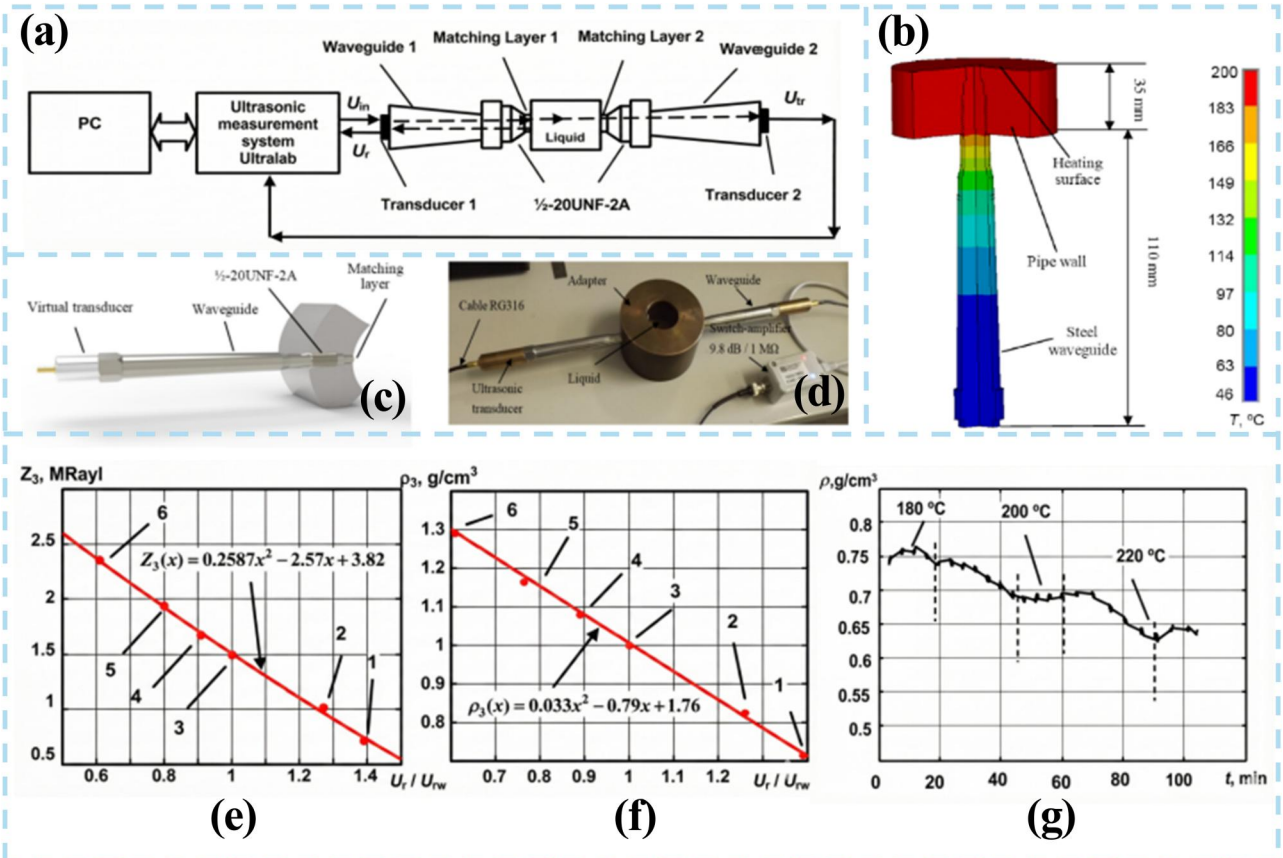


FIG. 25. (a) Schematic illustration of the principle of ultrasonic liquid density measurement. (b) Temperature distribution in the AISI 316 steel waveguide. (c) Graphical representation of the 3D model. (d) Photograph of the experimental setup. (e) Relationship between the normalized amplitude Ur/Urw and the acoustic impedance of the measured liquid; (f) relationship between the normalized amplitude U_t/U_{rw} and the density of the measured liquid. (g) Variations of the density of the virgin polypropylene melt at different temperatures. Reproduced with permission from Kazys *et al.*, Sensors **15**, 19393–19415 (2015). Copyright 2015 Authors, licensed under a Creative Commons Attribution (CC BY) license.[270]

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and after incidence at the liquid–waveguide interface was measured, from which the relationship between the reflection amplitude ratio and the acoustic impedance and density of different liquids was established [Figs. 25(e) and 25(f)]. Based on this, the system was further applied to measure the density of polypropylene (PP) melt at various temperatures. The results [Fig. 25(g)] show that PP melt exhibits higher density at lower temperatures; within the range of 180–220 °C, the PP density varies between ∼0.64 and 0.76 g/cm^3, in good agreement with reported literature values. A complex uncertainty analysis indicates that the relative expanded measurement uncertainty of the polymer melt density is $U_\Sigma = 7.4 \times 10^{-3}$ g/cm^3 (∼1%), thereby validating the accuracy and reliability of this ultrasonic waveguide–based density measurement method.[270]

High-temperature liquid or melt density measurement requires stable reflection-coefficient extraction at the solid–liquid interface and reliable sound-speed measurement in the liquid. Waveguide or buffer-rod structures are useful when the piezoelectric element must be thermally isolated, whereas direct-contact configurations require careful selection of protective layers and high-temperature-stable piezoelectric materials. The major challenges are insufficient wettability, interfacial bubble formation, corrosion, sealing reliability, and uncertainty in acoustic impedance under high-temperature operating conditions.

F. PMUT-based high-temperature applications

With the continuous development of high-temperature equipment condition monitoring toward greater continuity, multi-point deployment, and system integration, miniaturization, array configuration, and on-chip integration capabilities of ultrasonic transducers have received increasing attention. Conventional bulk piezoelectric ultrasonic transducers for high-temperature applications offer advantages in acoustic output and penetration capability; however, their discrete packaging structures still impose certain limitations in the construction of high-density arrays, multichannel interconnections, and integration with front-end electronics. Against this background, piezoelectric micromachined ultrasonic transducers (PMUTs) based on MEMS technology have gradually emerged as a promising device architecture for high-temperature ultrasonic applications. PMUTs feature small device dimensions, low driving voltage, and strong compatibility with semiconductor fabrication processes. Through microfabrication techniques, they enable high-density arrays, low-voltage actuation, large-scale batch manufacturing, and co-integration with CMOS/ASIC front-end circuits.[271,272,274,281] Therefore, PMUTs provide an alternative implementation pathway to traditional bulk transducers for localized detection, near-field monitoring, process parameter measurement, and integrated ultrasonic arrays. They are particularly suitable for high-temperature application scenarios that impose stringent requirements on device size, power consumption, array density, and system integration level.

Regarding PMUTs, extensive reviews have already systematically covered their structural design, performance metrics, fabrication processes, and application progress; therefore, this paper provides only a brief overview of their fundamental operating mechanisms, while focusing on device-level challenges in high-temperature applications.[271–273] Unlike conventional bulk piezoelectric ultrasonic transducers that primarily rely on thickness-mode vibration, PMUTs typically use piezoelectric thin films to drive microscale suspended membranes into flexural vibration, thereby enabling electromechanical energy conversion. As shown in Fig. 26(a), a typical PMUT cell consists of a piezoelectric thin film, top and bottom electrodes, a supporting membrane layer, an acoustic cavity, and a substrate. Its resonant frequency and acoustic response are jointly governed by membrane geometry, multilayer structural stiffness, residual stress, boundary conditions, and acoustic loading.[271,272] For high-temperature applications, this multilayer suspended-membrane structure is further influenced by external operating conditions. As illustrated in Fig. 26(b), temperature elevation, thermal cycling, and prolonged high-temperature exposure may induce structural-level effects such as electrode degradation, residual stress redistribution, changes in multilayer effective stiffness, and thermal deformation of the cavity and substrate. These effects ultimately manifest as membrane warping, resonant frequency drift, variations in output or sensitivity, and reduced consistency among array elements. Therefore, in high-temperature environments, the performance of PMUTs should not be attributed solely to the thermal stability of the piezoelectric layer but should instead be understood as a device-level thermo-mechanical-electrical coupled behavior arising from the collective interaction of the piezoelectric film, electrodes, membrane structure, cavity, and substrate.[275]

In recent years, several studies on high-temperature MEMS/PMUTs and related thin-film micromechanical acoustic devices have provided preliminary device-level validation of this research direction. Esteves *et al.* developed a high-temperature MEMS platform based on an AlN/SiC heterostructure, using PMUTs and microresonators as process demonstration devices. The PMUTs maintained stable resonant responses before and after annealing at 935 °C, with a frequency

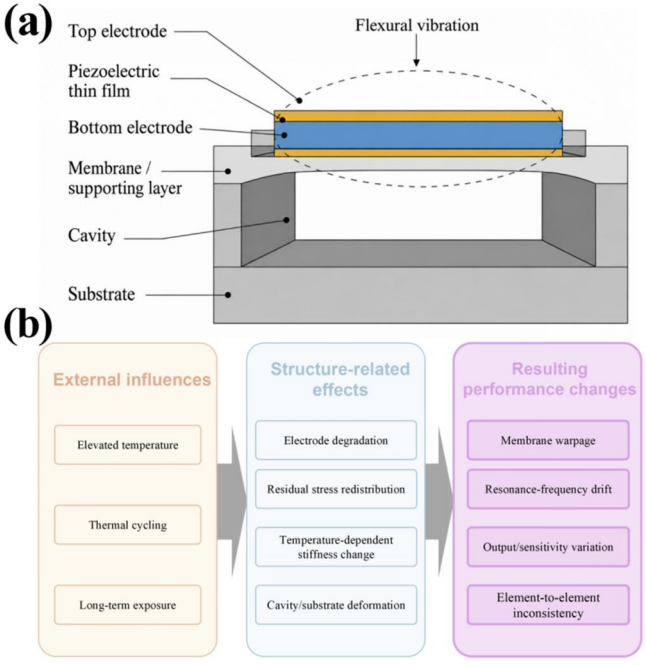


FIG. 26. (a) Basic structure of a representative PMUT and (b) high-temperature effects on PMUTs. The PMUT structure was drawn based on representative PMUT design principles reported in previous reviews.[271,272]

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variation of less than 1%. Meanwhile, the microresonators showed no observable degradation in electromechanical coupling coefficient during *in situ* RF measurements up to 500 °C, indicating that the AlN/SiC multilayer structure can retain a certain level of electromechanical conversion stability at elevated temperatures.[275] Sui *et al.* further reported an AlScN-on-SiC thin-film micromechanical resonant transducer and characterized its macroscopic and microscopic vibration behavior at high temperatures using laser interferometry and Raman spectroscopy. The device was able to maintain resonant operation up to 600 °C, suggesting that the combination of Sc-doped AlN and SiC substrates may offer a favorable balance between high piezoelectric response and thermal structural stability.[251] In more PMUT-specific studies, Lyu and Safari investigated the high-temperature performance of SOI-based Sc-doped AlN PMUTs, comparing the effects of different cavity sizes and top silicon layer thicknesses on resonant frequency, electromechanical coupling, vibration amplitude, and membrane warping as the temperature increased from room temperature to 200 °C.[276] The results show that larger cavity sizes and thinner top silicon layers enhance temperature-induced frequency drift and structural deformation. A related study on AlN/ScAlN PMUTs further demonstrated temperature-dependent frequency drift and their potential for compact contact temperature monitoring.[277] In addition, recent demonstrations of stable high-temperature operation and thermal survivability of LiNbO3 bimorph PMUTs further indicate that high-coupling thin-film platforms may serve as a complementary route for high-temperature microscale piezoelectric ultrasonic devices.[278]

From an application perspective, the primary value of MEMS/PMUT-based high-temperature ultrasonic devices does not lie in directly replacing conventional high-power bulk high-temperature probes, but rather in providing a new device implementation pathway for miniaturized, array-based, and integrated localized sensing. Existing studies have demonstrated that PMUT arrays can be used for defect detection and imaging in solid structures. For example, Yin *et al.* designed a miniaturized nondestructive testing probe based on AlN PMUTs, where defect locations in solid samples were identified using time-of-flight differences in pulse-echo signals, demonstrating the application potential of PMUTs in compact solid-state inspection probes.[279] Ji *et al.* further implemented full-matrix capture (FMC) ultrasonic imaging of solid defects using a PZT PMUT array, providing additional evidence for the feasibility of PMUT arrays in defect localization and image reconstruction.[280] Beyond nondestructive testing, monolithic integration of PMUTs with CMOS front-end circuits also offers new possibilities for process parameter sensing. The PMUT-on-CMOS device reported by Ledesma *et al.* can be used for monitoring fluid properties such as density, viscosity, sound velocity, and compressibility, demonstrating its advantages in highly integrated, multi-parameter, and low-power sensing nodes.[281] When these capabilities are extended to high-temperature applications, PMUTs are more suitable for localized monitoring of thin-walled structures, miniaturized process monitoring probes, monitoring of medium-to-high-temperature manufacturing processes such as injection molding or packaging, liquid property measurements, and distributed multi-point ultrasonic arrays. In particular, in scenarios characterized by limited space, a large number of sensing nodes, power constraints, or tight integration with front-end electronic systems, PMUTs exhibit a more distinct system-level integration advantage compared with conventional discrete high-temperature transducers.

Based on the above high-temperature device validation and application cases, Table VII further provides a cross-comparative summary of representative MEMS/PMUT-based ultrasonic devices in terms of device type, validation conditions, application relevance, and unresolved issues. It can be observed that, although existing studies have separately demonstrated the feasibility of high-temperature material platforms at the device level and the application potential of PMUTs in miniaturized sensing, array imaging, and integrated measurements, a clear gap remains between these two lines of work. The former is mostly limited to high-temperature annealing, *in situ* resonant testing, or unit-level thermal drift analysis, without sufficient validation under practical application conditions. The latter, although demonstrating relatively well-defined functional capabilities, is still largely confined to room-temperature or limited experimental environments, with insufficient verification of thermal cycling stability and temperature tolerance. Therefore, the current critical gap in high-temperature PMUT research does not lie in the realization of individual device concepts or short-term resonant responses, but rather in the lack of system-level validation under realistic service conditions. Future research should move beyond the high-temperature operability of material stacks and single devices toward issues such as high-temperature pulse-echo signal stability, acoustic coupling robustness, array amplitude and phase consistency, thermal drift compensation, high-temperature interconnection and packaging reliability, and co-integration with front-end readout electronics. Only when a complete validation framework is established across these aspects can PMUTs evolve from high-temperature MEMS device demonstrations into integrated high-temperature ultrasonic transducer platforms for engineering inspection and process monitoring.

G. Application-oriented material–structure–application relationships

The three aspects of HTPUTs—high-temperature piezoelectric materials, transducer structural design, and representative engineering applications—have been reviewed in Secs. III, V A–V F, respectively. However, in practical device design, these three aspects are not independent of one another. The selection of piezoelectric materials only becomes practically meaningful when considered in relation to specific operating temperatures, measurement targets, acoustic coupling conditions, and structural configurations. Similarly, a transducer structure that is effective in one application scenario may not necessarily be suitable for another high-temperature environment, because the dominant constraints in different applications may shift from acoustic output and sensitivity to thermal isolation, wettability, oxidation resistance, miniaturization, or long-term stability.

To further highlight the integrated "materials–structures–applications" design logic of HTPUTs, Table VIII summarizes the typical operating temperatures, primary measurement targets, candidate material systems, applicable structural configurations, and unresolved issues for several representative high-temperature ultrasonic applications from the perspective of application requirements. It can be seen that, for applications within a moderate temperature range that require relatively high acoustic output and sensitivity, such as online pipeline wall-thickness monitoring and polymer processing monitoring, PZT-based materials still retain considerable application value,

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TABLE VII. Representative MEMS/PMUT-based ultrasonic devices and their implications for high-temperature ultrasonic applications.

Device/platform	Device type	Testing condition or demonstrated capability	Main contribution	Implications for high-temperature ultrasonic applications	Remaining issues
AlN/SiC MEMS platform[275]	PMUT and micromechanical resonator	The PMUT retained a stable resonance response after 935 °C annealing, while the microresonator was tested *in situ* up to 500 °C.	Demonstrated the thermal robustness of AlN/SiC multilayer MEMS structures.	Provides an important material and structural platform for high-temperature micromachined ultrasonic devices.	Mainly limited to platform- and device-level validation; a mature high-temperature pulse–echo inspection system has not yet been demonstrated.
AlScN-on-SiC thin-film micromachined resonant transducer[251]	Thin-film microchanical resonant transducer	Resonant operation was demonstrated at temperatures up to 600 °C.	Verified the feasibility of AlScN/SiC heterostructures for high-temperature acoustic MEMS devices.	Suggests that Sc-doped AlN combined with SiC substrates may be suitable for harsh-environment miniature piezoelectric acoustic devices.	The device remains closer to a resonant MEMS demonstration than to a practical PMUT probe or array for ultrasonic NDE.
SOI-based ScAlN PMUT[276]	PMUT cell	Resonant frequency, electromechanical coupling, vibration amplitude, and membrane warpage were analyzed from room temperature to approximately 200 °C.	Revealed the influence of cavity size, silicon device-layer thickness, and multilayer thermal stress on the high-temperature response of PMUTs.	Directly illustrates the frequency drift and membrane deformation issues that may arise in high-temperature PMUTs.	Higher-temperature testing, thermal cycling, long-term stability, and array-level consistency still require further validation.
AlN/ScAlN PMUT with temperature-dependent frequency response[277]	PMUT contact sensor	Temperature-dependent resonance-frequency drift was characterized under elevated-temperature conditions.	Demonstrated that the resonance frequency of PMUTs can be used as a temperature-sensitive parameter.	Indicates the possibility of compact high-temperature contact monitoring based on PMUT frequency shifts.	Temperature range, repeatability, calibration stability, and high-temperature packaging reliability remain insufficiently verified.
$LiNbO_3$ bimorph PMUT[278]	Bimorph thin-film PMUT	Stable operation at elevated temperature and thermal survival under more extreme exposure were reported.	Extends the candidate material platforms for high-temperature PMUTs beyond AlN/ScAlN toward high-coupling LiNbO3 thin films.	Provides a complementary route for high-coupling thin-film piezoelectric platforms in miniature high-temperature ultrasonic devices.	Still at an early device-demonstration stage; process compatibility, array integration, and long-term reliability need further investigation.
AlN-based PMUT NDT probe[279]	PMUT probe	Pulse–echo signals were used for defect detection in solid samples.	Demonstrated the feasibility of AlN PMUTs for miniaturized solid-state NDT probes.	Provides an application basis for future local inspection of thin-walled structures and compact high-temperature NDT probes.	Most demonstrations remain at room temperature; high-temperature acoustic coupling, packaging, and acoustic output capability are not yet sufficiently verified.

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TABLE VII. (*Continued.*)

Device/platform	Device type	Testing condition or demonstrated capability	Main contribution	Implications for high-temperature ultrasonic applications	Remaining issues
PZT PMUT array for total-focus imaging[280]	PMUT array	Solid defects were imaged using total-focus ultrasonic reconstruction.	Demonstrated the defect localization and imaging capability of PMUT arrays.	Supports the potential of PMUT arrays for localized ultrasonic imaging and compact inspection systems.	The high-temperature stability of PZT, array packaging, and element-to-element phase/amplitude consistency under high-temperature remain major concerns.
PMUT-on-CMOS fluid-property sensor[281]	Monolithically integrated PMUT-on-CMOS device	Fluid density, viscosity, sound velocity, and compressibility were measured using an integrated PMUT-CMOS system.	Demonstrated integrated multiparameter acoustic measurement using a compact PMUT-on-CMOS platform.	Relevant to high-temperature process monitoring, liquid-property measurement, and integrated sensing nodes.	CMOS thermal tolerance, high-temperature interconnects, thermal isolation packaging, and long-term stability must be addressed before high-temperature deployment.

TABLE VIII. Application-oriented summary of material–structure–application relationships for HTPUT design.

Application scenario	Typical temperature range/operating environment	Main measurement target	Candidate material systems	Applicable structural configurations	Main limitations or unresolved issues
Online pipeline wall-thickness/corrosion monitoring[231–237]	Typically approximately 150–400 °C; high-temperature steel pipes, pressure vessels, petrochemical pipelines, etc.	Wall-thickness variation, corrosion or erosion rate, and long-term thickness evolution.	PZT or BSPT may be considered for moderate-temperature scenarios; $LiNbO_3$, $GaPO_4$, AlN, and other single crystals may be considered for higher-temperature or long-term stable operation, while conventional piezoelectric elements may also be used when waveguide-based thermal isolation is adopted.	Buffer-rod transducers, waveguide transducers, and direct-contact transducers with protective layers.	Degradation of high-temperature coupling layers, temperature-dependent sound velocity in the pipe wall, interfacial failure induced by thermal cycling, and insufficient repeatability of field installation.
Injection molding/polymer-processing monitoring[238–248]	Typically approximately 180–350 °C; polymer melts, molds, nozzles, barrels, etc.	Melt filling, solidification, demolding, shrinkage, and variations in sound velocity and echo amplitude.	Sol-gel PZT/$Bi_4Ti_3O_{12}$ thick films, ZnO thin films, and AlN thin films; bulk PZT may also be used in	Directly deposited film transducers, embedded probes, surface-mounted probes, and delay-line probes.	Film/substrate adhesion stability, residual stress, film cracking, and complex calibration caused by

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TABLE VIII. (*Continued.*)

Application scenario	Typical temperature range/operating environment	Main measurement target	Candidate material systems	Applicable structural configurations	Main limitations or unresolved issues
			lower temperature regions.		variations in the melt state.
High-temperature ultrasonic thermometry[249–258]	Up to approximately 600–1800 °C; furnaces, combustion chambers, strongly oxidizing environments, or environments with large temperature gradients.	Temperature reconstruction based on sound velocity or echo time delay.	The piezoelectric element is usually placed at the low-temperature end, where PZT, BSPT, or $LiNbO_3$ may be used.	Grooved waveguides, bent waveguides, and multipoint waveguide thermometry structures.	Waveguide oxidation, measurement errors caused by temperature gradients, signal attenuation induced by long-distance propagation, and long-term drift of calibration curves.
Bolt preload/stress monitoring[259–264]	From room temperature to several hundred degrees Celsius; high-temperature bolts, aero-engine components, mechanical joints, etc.	Preload, axial stress, and time-of-flight shift.	AlN thin films, ZnO thin films, and $LiNbO_3$ thin films; PZT thin films or bulk materials may be considered for low-to-moderate temperature scenarios.	Directly deposited film transducers on bolt end faces and integrated thin-film ultrasonic sensors.	Cross-sensitivity between temperature and stress, thin-film fatigue, electrode stability, and calibration drift after long-term service.
High-temperature liquid/melt density measurement[265–270]	Polymer melts are commonly measured at approximately 180–250 °C; for higher-temperature liquids or molten media, the usable range should be determined according to medium corrosiveness and interfacial conditions.	Density, sound velocity, acoustic impedance, and changes in liquid state.	Conventional piezoelectric materials may be used when waveguide-based thermal isolation is adopted; $LiNbO_3$, AlN, and related materials may be considered for direct high-temperature contact.	Waveguide-based pitch-catch structures, buffer-rod structures, and direct-contact transducers with protective layers.	Insufficient wettability, interfacial bubbles, corrosion, uncertainty in acoustic impedance, and insufficient sealing and long-term stability.
PMUT/MEMS integrated high-temperature ultrasonic devices[271–281]	Current validations are mostly concentrated at approximately 200–600 °C; some AlN/SiC or ScAlN/SiC platforms show higher thermal-stability potential.	Local monitoring, near-field detection, process-parameter measurement, miniaturized arrays, and distributed sensing.	AlN, ScAlN, ZnO, and $LiNbO_3$ thin films, as well as high-temperature PZT-based systems.	PMUT arrays, MEMS resonators, PMUT-on-CMOS nodes, and miniaturized thin-film ultrasonic probes.	Limited acoustic output, membrane thermal deformation, frequency drift, reliability of packaging and interconnections, and insufficient validation of high-temperature pulse-echo operation.

provided that depolarization, dielectric loss, and coupling stability are properly controlled. For scenarios involving higher temperatures, longer service durations, or strong temperature gradients, thermally stable single-crystal materials such as $LiNbO_3$, AlN, and $GaPO_4$, or thermally isolated structures such as delay lines and waveguides, usually offer greater advantages, because the piezoelectric element can either directly withstand higher temperatures or be isolated within a relatively low-temperature region. In contrast, directly deposited film

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transducers and PMUT/MEMS devices provide new implementation pathways for conformal integration, miniaturization, array configuration, and distributed monitoring, although their acoustic output capability, packaging and interconnection reliability, and array uniformity still require further validation.

From a practical design perspective, the selection of an HTPUT should follow a hierarchical rather than material-first procedure. The operating temperature should first be considered together with exposure duration and thermal-cycling severity to determine whether the active transducer can be directly exposed to the hot region or requires thermal isolation. The acoustic medium and coupling condition then constrain the selection of matching, bonding, and protection strategies, particularly for molten media or chemically aggressive environments. Subsequently, the operating frequency and required acoustic performance determine the relative importance of electromechanical coupling, mechanical quality factor, dielectric loss, acoustic attenuation, and bandwidth. Finally, the required sensitivity, monitoring duration, geometric accessibility, and integration level determine the appropriate transducer architecture. Therefore, HTPUT design should be regarded as an application-driven multi-constraint selection process in which material, structural, interfacial, and packaging choices are progressively narrowed according to the dominant service requirements.

Beyond the above design-selection considerations, the application fields reviewed here also exhibit markedly different levels of technology readiness. High-temperature pipeline wall-thickness monitoring and injection-molding process monitoring have progressed beyond basic proof-of-concept studies, with integrated systems demonstrated on representative pipelines, flow-accelerated-corrosion test facilities, molding machines, or process-monitoring platforms. These studies can generally be regarded as application-oriented prototype or process-level validation, although systematic evidence of multi-year industrial deployment under uncontrolled service conditions remains limited in the open literature. High-temperature ultrasonic thermometry has also been validated over wide temperature ranges using different waveguide configurations, but many reported systems remain laboratory- or prototype-scale demonstrations, with long-term calibration stability and environmental durability still requiring further validation. Bolt preload monitoring and high-temperature liquid-density measurement are predominantly represented by laboratory validation and application-oriented prototype studies. In contrast, high-temperature PMUT/MEMS technologies remain at a comparatively earlier stage, with current work focused mainly on high-temperature material-stack stability, resonant-device operation, thermal drift, and proof-of-concept sensing rather than mature long-term industrial ultrasonic monitoring systems. Overall, the transition from successful high-temperature operation to industrial deployment requires not only functional ultrasonic response, but also reproducible fabrication, long-duration stability, thermal-cycling resistance, calibration retention, packaging reliability, and maintainability under actual service conditions.

VI. CONCLUSIONS AND OUTLOOK

In this review, we summarize and highlight the latest advancements in the field of HTPUTs. First, starting with the fundamental electroacoustic energy conversion mechanism of PUTs, we systematically review common electrical and acoustic characterization methods, key performance indicators, and typical high-temperature effects. We then provide a comprehensive overview of widely used high-temperature piezoelectric materials in current research and engineering practice—covering ceramics, single crystals, and thin/thick film materials—including their main characteristics, applicable temperature ranges, performance degradation, and failure mechanisms. More importantly, we explore typical structural design strategies, such as direct and nondirect contact designs, focusing on the design requirements and material selection for key components under high-temperature conditions (e.g., matching layers and backing layers), and we discuss effective high-temperature interconnection and acoustic coupling methods. As these technologies have progressed, new application scenarios have also appeared. Finally, we systematically introduce some of the newer applications, including pipeline wall thickness monitoring, injection molding process monitoring, ultrasonic temperature measurement, stress monitoring, high-temperature liquid density measurement and PMUT-based MEMS.

Overall, as high-temperature industrial equipment continues to develop and manufacturing technologies progress, the requirements for HTPUTs have become increasingly stringent. Although significant progress has been made in areas such as maximum operating temperature, measurement accuracy, long-term stability, and system reliability, several major challenges still persist in the practical application of HTPUTs. First, the issue of performance degradation of piezoelectric materials in high-temperature environments has not been fully resolved, and higher-performance piezoelectric materials are still in need of further development. Second, the thermal shock resistance and acoustic coupling efficiency of the transducer structure, as well as how to maintain long-term stability under thermal cycling, require further improvement for better solutions. Third, the design of multilayer structures and interfaces in high-temperature environments still presents technical difficulties. The compatibility between materials, thermal expansion matching, and the impact of long-term thermal cycling on the transducer's performance remain key challenges in the design and optimization process.

Given the latest advancements in HTPUTs and the challenges that still persist, Table IX summarizes a temperature-regime-based technology roadmap to clarify the current material and architectural options across different service-temperature ranges. These temperature regimes should be regarded as application-oriented design references rather than rigid operating boundaries, because the practical operating limit of an HTPUT also depends on exposure duration, thermal cycling, device architecture, interfaces, and environmental conditions.

Based on this roadmap, the future development directions of HTPUTs are further discussed below from the materials and transducer perspectives.

A. Material

The future development of high-temperature piezoelectric materials should not be limited to purely metric-driven optimization but should further move toward device demand-driven optimization, thereby enabling the collaborative design of materials and devices, as shown in Fig. 27(a). In recent years, with the rapid advancement of materials science, high-performance high-temperature piezoelectric materials have been continuously developed through the

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TABLE IX. Temperature-regime-based technology roadmap for HTPUT materials and transducer architectures.

Temperature regime	Candidate material systems	Preferred transducer architecture	Dominant limitations
<300 °C	PZT and modified PZT; BSPT; selected thin-film systems	Direct-contact transducers; directly deposited films and integrated devices where appropriate	Depolarization near the upper limit; polymer/couplant aging; interface stability during long-term exposure
300–600 °C	BSPT; $LiNbO_3$; $GaPO_4$; AlN/ZnO-based films; selected BLSF systems	High-temperature direct-contact structures with stable joining; delay-line structures where thermal isolation is required	Bonding/coupling degradation; CTE mismatch; electrode and interconnection stability; thermal cycling
600–1000 °C	$LiNbO_3$; AlN; $GaPO_4$; ReCOB; high-temperature BLSF systems	Direct-contact structures with ceramic/metallic interfaces under suitable conditions; delay-line or waveguide architectures for stronger thermal isolation and long-duration operation	Oxidation; electrical resistivity/loss degradation; interfacial stress; coupling reliability; insufficient long-term validation
>1000 °C	AlN and ReCOB as potential material platforms; direct transducer-level evidence remains very limited	Thermally isolated waveguide architectures are generally more realistic; direct-contact systems remain primarily a research target	Limited validation of complete transducer systems; electrode/interconnect stability; oxidation; insulation; coupling and packaging reliability

multi-objective optimization of parameters such as d_{33}, k_t, T_c, dielectric loss, resistivity, and thermal stability. However, the engineering applicability of these materials ultimately depends on their contribution to system-level transducer performance under specific working modes and structural constraints. Therefore, while the enhancement of individual material parameters remains important, their influence on practical device performance is more critical, especially in terms of energy conversion efficiency near resonance, bandwidth maintenance, center-frequency drift, sensitivity retention, acoustic output stability, and repeatability under thermal cycling.

From this perspective, future materials research should place greater emphasis on establishing comprehensive high-temperature property databases covering both temperature and frequency domains. At a minimum, such databases should include temperature- and frequency-dependent piezoelectric coefficients d_{ij}, dielectric permittivity ε, elastic constants, electromechanical coupling factors,

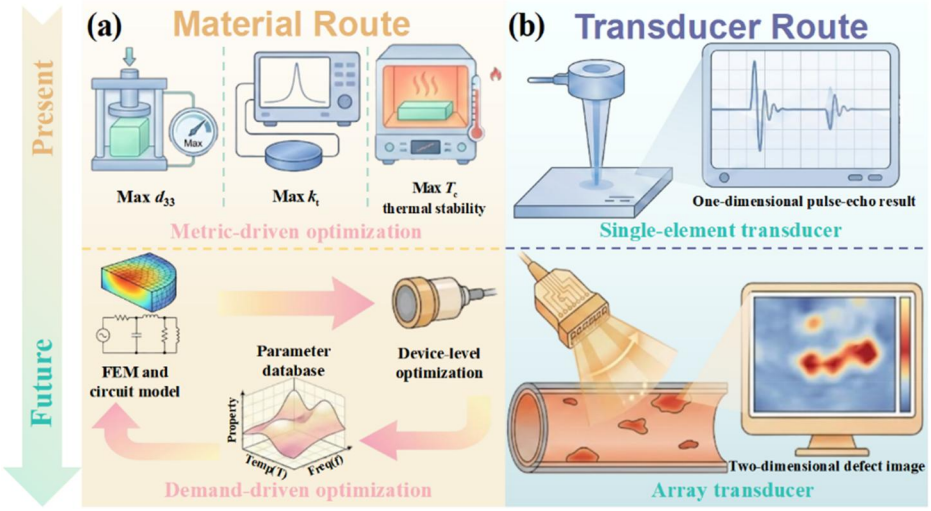


FIG. 27. Future development directions of HTPUTs in terms of (a) materials and (b) transducers.

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dielectric and mechanical losses, electrical resistivity, density, acoustic velocity, acoustic impedance and attenuation, together with CTE. For ferroelectric materials, depolarization behavior and property-retention kinetics during prolonged thermal exposure should also be recorded, while thermal-cycle retention is required to distinguish reversible temperature drift from irreversible degradation. These parameters should be accompanied by the relevant material state and measurement conditions, including composition, crystal cut or orientation, poling state, atmosphere, frequency range, dwell time, and thermal-cycling history, so that datasets obtained from different studies can be meaningfully compared and incorporated into equivalent-circuit and finite-element transducer models. Such a database would enable a closed-loop linkage among material design, parameter optimization, device simulation, and transducer performance evaluation. In particular, the lack of high-temperature acoustic impedance, sound velocity, attenuation, dielectric loss, resistivity, and long-term thermal-aging data remains a major obstacle to translating newly developed materials into practical HTPUTs. This issue is especially important for emerging high-temperature material systems, because many of them show promising intrinsic properties but still lack sufficient device-oriented parameter sets for reliable transducer design.

Within this device-demand-driven framework, lead-free high-temperature piezoelectric materials should be further explored as important candidates for future HTPUTs. These include bismuth-layered, tungsten-bronze, and wurtzite-structured systems, as well as lead-free perovskite systems such as KNN-based, $BiFeO_3$-based, and $BaTiO_3$-based materials. For example, KNN-based ceramics may provide relatively high piezoelectric activity and lead-free compatibility, while $BiFeO_3$-based systems are attractive because of their high Curie temperature and potential high-temperature stability. However, their engineering value should not be judged only by high T_C, room temperature piezoelectric coefficients, or lead-free composition. Instead, these materials need to be evaluated using the same device-oriented criteria discussed above, including high-temperature electromechanical coupling, dielectric loss, resistivity, acoustic impedance, sound velocity, attenuation, electrode compatibility, and long-term retention after thermal exposure and thermal cycling. Therefore, future material screening should integrate intrinsic material optimization with device-level requirements such as stable acoustic output, low-loss operation, reliable electrical insulation, manufacturability, electrode stability, packaging compatibility, and matching/backing layer design. Only through this material–device co-design strategy can new high-temperature piezoelectric materials, including lead-free systems, be effectively transformed from promising material candidates into practical sensing elements for HTPUTs.

B. Transducer

The future development of HTPUTs is expected to evolve from single-element, point-wise measurement toward array-based, integrated, and intelligent ultrasonic sensing systems, as shown in Fig. 27(b). Conventional single-element HTPUTs remain valuable for simple thickness measurement, time-of-flight monitoring, and localized inspection because of their structural simplicity and relatively high acoustic output. However, as high-temperature monitoring tasks become more complex, especially in aerospace structures, energy equipment, petrochemical pipelines, and manufacturing processes, single-point measurement is often insufficient to describe the spatial distribution and evolution of defects, temperature fields, or material states. Therefore, array-based HTPUTs and high-temperature ultrasonic imaging systems will be an important future direction.

Compared with single-element transducers, high-temperature piezoelectric ultrasonic arrays can provide electronic focusing, beam steering, multichannel signal synthesis, and imaging capability. These functions can significantly improve inspection efficiency, spatial resolution, and defect localization accuracy. In high-temperature pipeline inspection, array transducers may extend conventional wall-thickness monitoring toward inner-wall defect imaging and corrosion mapping. In aerospace and power-generation components, they may enable real-time tracking of crack initiation, propagation, and structural degradation. In process monitoring, arrayed or distributed transducers can provide spatially resolved information rather than only a single averaged measurement. Therefore, the value of high-temperature arrays is not only higher imaging resolution, but also the ability to transform HTPUTs from local sensing probes into spatially informative monitoring systems.

However, this evolution also introduces array-specific thermomechanical and signal-coherence challenges. The key issue is no longer only whether individual piezoelectric elements can survive high-temperature, but whether the complete multichannel array can preserve the geometric and phase relationships required for coherent beamforming during prolonged thermal exposure and cycling. Differential thermal expansion among the piezoelectric elements, matching/backing layers, substrate, and packaging can alter the effective element pitch and aperture geometry, while temperature-dependent changes in material properties and inter-element coupling may produce nonuniform variations in resonance frequency, amplitude, and phase among individual elements. In addition, temperature-dependent propagation delay and phase changes in cables and interconnections can introduce further channel-to-channel errors. These effects may cause the actual transmit–receive delay relationships to deviate from the nominal beamforming laws, resulting in focal shift, steering errors, increased sidelobes, reduced image contrast, and degraded defect localization. Repeated thermal cycling can further introduce calibration drift through irreversible interfacial degradation, oxidation, and residual-stress accumulation. Therefore, future high-temperature array systems should combine thermally stable packaging and multichannel interconnections with temperature-aware element calibration, pitch and phase compensation, crosstalk characterization, and periodic recalibration to maintain coherent imaging performance over the required service life.

To support these complex transducer architectures, advanced manufacturing technologies will become increasingly important. Additive manufacturing, microfabrication, and other precision fabrication methods may provide new routes for preparing customized matching layers, gradient backing materials, integrated housings, conformal transducers, and complex array layouts that are difficult to realize using conventional machining and bonding methods. In particular, additive manufacturing may allow acoustic, thermal, and mechanical functions to be co-designed within the same transducer structure, thereby improving impedance matching, heat dissipation, thermal-stress management, and structural integration. Nevertheless, the acoustic attenuation, porosity control, interfacial bonding strength, high-temperature stability, and batch-to-batch reproducibility of printed or microfabricated functional structures still require systematic investigation before they can be widely used in HTPUTs.

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Beyond the transducer head itself, future HTPUTs should be developed as complete sensing systems. In practical high-temperature monitoring, signal quality and system reliability are strongly affected by high-temperature cables, connectors, impedance-matching circuits, preamplifiers, data-acquisition units, thermal protection structures, and communication modules. Therefore, the integration of HTPUTs with front-end electronics, wireless transmission, and distributed sensor networks is a necessary step toward field deployment. Compact integrated sensing nodes would be particularly useful in scenarios where wired access is difficult, sensor density is high, or long-term online monitoring is required, such as rotating machinery, aero-engines, nuclear facilities, furnaces, and petrochemical pipelines. For PMUT/MEMS-based devices and thin-film transducers, such integration is especially important because their advantages in miniaturization and array density can only be fully exploited when packaging, interconnection, and readout circuits are co-designed.

Finally, as HTPUTs evolve toward arrayed, distributed, and long-term monitoring systems, data interpretation will become as important as device fabrication. Digital twins and artificial-intelligence-assisted NDE provide promising tools for this stage. Digital-twin models can combine temperature fields, structural mechanics, acoustic propagation, transducer response, and degradation behavior to predict the performance of HTPUT systems under complex service conditions. Meanwhile, AI-based methods can assist in signal denoising, feature extraction, defect recognition, temperature compensation, coupling-state evaluation, adaptive calibration, and lifetime prediction. However, these methods should not be used as purely data-driven black boxes. For high-temperature ultrasonic applications, reliable intelligent monitoring will require physically meaningful features, high-quality experimental datasets, uncertainty quantification, and validation under realistic thermal, mechanical, and environmental conditions. Therefore, the future development of HTPUTs should move toward an integrated framework combining transducer arrays, advanced manufacturing, high-temperature electronics, distributed sensing networks, digital twins, and physics-informed data analysis.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Shaotong Gong: Data curation (lead); Investigation (lead); Writing – original draft (lead); Writing – review & editing (equal). **Xinhao Sun:** Supervision (equal); Writing – review & editing (equal). **Yi Quan:** Data curation (equal); Funding acquisition (equal). **Tianlong Zhao:** Data curation (equal); Funding acquisition (equal). **Guangzhi Dong:** Resources (equal); Supervision (equal). **Chunlong Fei:** Data curation (lead); Funding acquisition (lead); Supervision (lead); Writing – review & editing (lead). **Yintang Yang:** Data curation (equal); Funding acquisition (equal).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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